FICTION

Eureka: A Prose Poem

Edgar Allan Poe

FAHAD.COM — Free Book Library

Public Domain Edition

EUREKA: A PROSE POEM.

BY

EDGAR A. POE.

NEW-YORK: GEO. P. PUTNAM, OF LATE FIRM OF “WILEY & PUTNAM,” 155 BROADWAY.

MDCCCXLVIII.

ENTERED, according to Act of Congress, in the year 1848, BY EDGAR A. POE, In the Clerk’s Office of the District Court for the Southern District of New-York.

LEAVITT, TROW & CO Prs., 33 Ann-street.

WITH VERY PROFOUND RESPECT, This Work is Dedicated TO ALEXANDER VON HUMBOLDT.

PREFACE.

To the few who love me and whom I love—to those who feel rather than to those who think—to the dreamers and those who put faith in dreams as in the only realities—I offer this Book of Truths, not in its character of Truth-Teller, but for the Beauty that abounds in its Truth; constituting it true. To these I present the composition as an Art-Product alone:—let us say as a Romance; or, if I be not urging too lofty a claim, as a Poem.

What I here propound is true:—therefore it cannot die:—or if by any means it be now trodden down so that it die, it will “rise again to the Life Everlasting.”

Nevertheless it is as a Poem only that I wish this work to be judged after I am dead.

E. A. P.

EUREKA:

AN ESSAY ON THE MATERIAL AND SPIRITUAL UNIVERSE.

It is with humility really unassumed—it is with a sentiment even of awe—that I pen the opening sentence of this work: for of all conceivable subjects I approach the reader with the most solemn—the most comprehensive—the most difficult—the most august.

What terms shall I find sufficiently simple in their sublimity—sufficiently sublime in their simplicity—for the mere enunciation of my theme?

I design to speak of the _Physical, Metaphysical and Mathematical—of the Material and Spiritual Universe:—of its Essence, its Origin, its Creation, its Present Condition and its Destiny_. I shall be so rash, moreover, as to challenge the conclusions, and thus, in effect, to question the sagacity, of many of the greatest and most justly

reverenced of men.

In the beginning, let me as distinctly as possible announce—not the theorem which I hope to demonstrate—for, whatever the mathematicians may assert, there is, in this world at least, _no such thing_ as demonstration—but the ruling idea which, throughout this volume, I shall be continually endeavoring to suggest.

My general proposition, then, is this:—_In the Original Unity of the First Thing lies the Secondary Cause of All Things, with the Germ of their Inevitable Annihilation_.

In illustration of this idea, I propose to take such a survey of the Universe that the mind may be able really to receive and to perceive an individual impression.

He who from the top of Ætna casts his eyes leisurely around, is affected chiefly by the _extent_ and _diversity_ of the scene. Only by a rapid whirling on his heel could he hope to comprehend the panorama in the sublimity of its _oneness_. But as, on the summit of Ætna, _no_ man has thought of whirling on his heel, so no man has ever taken into his brain the full uniqueness of the prospect; and so, again, whatever considerations lie involved in this uniqueness, have as yet no practical existence for mankind.

I do not know a treatise in which a survey of the _Universe_—using the word in its most comprehensive and only legitimate acceptation—is taken at all:—and it may be as well here to mention that by the term “Universe,” wherever employed without qualification in this essay, I mean to designate _the utmost conceivable expanse of space, with all things, spiritual and material, that can be imagined to exist within the compass of that expanse_. In speaking of what is ordinarily implied by the expression, “Universe,” I shall take a phrase of limitation—“the Universe of stars.” Why this distinction is considered necessary, will be seen in the sequel.

But even of treatises on the really limited, although always assumed as the _un_limited, Universe of _stars_, I know none in which a survey, even of this limited Universe, is so taken as to warrant deductions from its _individuality_. The nearest approach to such a work is made in the “Cosmos” of Alexander Von Humboldt. He presents the subject, however, _not_ in its individuality but in its generality. His theme, in its last result, is the law of _each_ portion of the merely physical Universe, as this law is related to the laws of _every other_ portion of this merely physical Universe. His design is simply synœretical. In a word, he discusses the universality of material relation, and discloses to the eye of Philosophy whatever inferences have hitherto lain hidden _behind_ this universality. But however admirable be the succinctness with which he has treated each particular point of his topic, the mere multiplicity of these points occasions, necessarily, an amount of detail, and thus an involution of idea, which precludes all _individuality_ of impression.

It seems to me that, in aiming at this latter effect, and, through it, at the consequences—the conclusions—the suggestions—the speculations—or, if nothing better offer itself the mere guesses which may result from it—we require something like a mental gyration on the heel. We need so rapid a revolution of all things about the central point of sight that, while the minutiæ vanish altogether, even the more conspicuous objects become blended into one. Among the vanishing minutiæ, in a

survey of this kind, would be all exclusively terrestrial matters. The Earth would be considered in its planetary relations alone. A man, in this view, becomes mankind; mankind a member of the cosmical family of Intelligences.

And now, before proceeding to our subject proper, let me beg the reader's attention to an extract or two from a somewhat remarkable letter, which appears to have been found corked in a bottle and floating on the _Mare Tenebrarum_—an ocean well described by the Nubian geographer, Ptolemy Hephestion, but little frequented in modern days unless by the Transcendentalists and some other divers for crotchets. The date of this letter, I confess, surprises me even more particularly than its contents; for it seems to have been written in the year _two_ thousand eight hundred and forty-eight. As for the passages I am about to transcribe, they, I fancy, will speak for themselves.

"Do you know, my dear friend," says the writer, addressing, no doubt, a contemporary—"Do you know that it is scarcely more than eight or nine hundred years ago since the metaphysicians first consented to relieve the people of the singular fancy that there exist _but two practicable roads to Truth_? Believe it if you can! It appears, however, that long, long ago, in the night of Time, there lived a Turkish philosopher called Aries and surnamed Tottle." [Here, possibly, the letter-writer means Aristotle; the best names are wretchedly corrupted in two or three thousand years.] "The fame of this great man depended mainly upon his demonstration that sneezing is a natural provision, by means of which over-profound thinkers are enabled to expel superfluous ideas through the nose; but he obtained a scarcely less valuable celebrity as the founder, or at all events as the principal propagator, of what was termed the _de_ductive or _à priori_ philosophy. He started with what he maintained to be axioms, or self-evident truths:—and the now well understood fact that _no_ truths are _self_-evident, really does not make in the slightest degree against his speculations:—it was sufficient for his purpose that the truths in question were evident at all. From axioms he proceeded, logically, to results. His most illustrious disciples were one Tuclid, a geometrician," [meaning Euclid] "and one Kant, a Dutchman, the originator of that species of Transcendentalism which, with the change merely of a C for a K, now bears his peculiar name.

"Well, Aries Tottle flourished supreme, until the advent of one Hog, surnamed 'the Ettrick shepherd,' who preached an entirely different system, which he called the _à posteriori_ or _in_ductive. His plan referred altogether to sensation. He proceeded by observing, analyzing, and classifying facts—_instantiæ Naturæ_, as they were somewhat affectedly called—and arranging them into general laws. In a word, while the mode of Aries rested on _noumena_, that of Hog depended on _phenomena_; and so great was the admiration excited by this latter system that, at its first introduction, Aries fell into general disrepute. Finally, however, he recovered ground, and was permitted to divide the empire of Philosophy with his more modern rival:—the savans contenting themselves with proscribing all _other_ competitors, past, present, and to come; putting an end to all controversy on the topic by the promulgation of a Median law, to the effect that the Aristotelian and Baconian roads are, and of right ought to be, the solo possible avenues to knowledge:—'Baconian,' you must know, my dear

friend," adds the letter-writer at this point, "was an adjective invented as equivalent to Hog-ian, and at the same time more dignified and euphonious.

"Now I do assure you most positively"—proceeds the epistle—"that I represent these matters fairly; and you can easily understand how restrictions so absurd on their very face must have operated, in those days, to retard the progress of true Science, which makes its most important advances—as all History will show—by seemingly intuitive _leaps_. These ancient ideas confined investigation to crawling; and I need not suggest to you that crawling, among varieties of locomotion, is a very capital thing of its kind;—but because the tortoise is sure of foot, for this reason must we clip the wings of the eagles? For many centuries, so great was the infatuation, about Hog especially, that a virtual stop was put to all thinking, properly so called. No man dared utter a truth for which he felt himself indebted to his soul alone. It mattered not whether the truth was even demonstrably such; for the dogmatizing philosophers of that epoch regarded only _the road_ by which it professed to have been attained. The end, with them, was a point of no moment, whatever:—'the means!' they vociferated—'let us look at the means!'—and if, on scrutiny of the means, it was found to come neither under the category Hog, nor under the category Aries (which means ram), why then the savans went no farther, but, calling the thinker a fool and branding him a 'theorist,' would never, thenceforward, have any thing to do either with _him_ or with his truths.

"Now, my dear friend," continues the letter-writer, "it cannot be maintained that by the crawling system, exclusively adopted, men would arrive at the maximum amount of truth, even in any long series of ages; for the repression of imagination was an evil not to be counterbalanced even by _absolute_ certainty in the snail processes. But their certainty was very far from absolute. The error of our progenitors was quite analogous with that of the wiseacre who fancies he must necessarily see an object the more distinctly, the more closely he holds it to his eyes. They blinded themselves, too, with the impalpable, titillating Scotch snuff of _detail_; and thus the boasted facts of the Hog-ites were by no means always facts—a point of little importance but for the assumption that they always _were_. The vital taint, however, in Baconianism—its most lamentable fount of error—lay in its tendency to throw power and consideration into the hands of merely perceptive men—of those inter-Tritonic minnows, the microscopical savans—the diggers and pedlers of minute _facts_, for the most part in physical science—facts all of which they retailed at the same price upon the highway; their value depending, it was supposed, simply upon the _fact of their fact_, without reference to their applicability or inapplicability in the development of those ultimate and only legitimate facts, called Law.

"Than the persons"—the letter goes on to say—"Than the persons thus suddenly elevated by the Hog-ian philosophy into a station for which they were unfitted—thus transferred from the sculleries into the parlors of Science—from its pantries into its pulpits—than these individuals a more intolerant—a more intolerable set of bigots and tyrants never existed on the face of the earth. Their creed, their text and their sermon were, alike, the one word '_fact_'—but, for the most part, even of this one word, they knew not even the meaning. On those who ventured to _disturb_ their

facts with the view of putting them in order and to use, the disciples of Hog had no mercy whatever. All attempts at generalization were met at once by the words ‘theoretical,’ ‘theory,’ ‘theorist’—all _thought_, to be brief, was very properly resented as a personal affront to themselves. Cultivating the natural sciences to the exclusion of Metaphysics, the Mathematics, and Logic, many of these Bacon-engendered philosophers—one-idead, one-sided and lame of a leg—were more wretchedly helpless—more miserably ignorant, in view of all the comprehensible objects of knowledge, than the veriest unlettered hind who proves that he knows something at least, in admitting that he knows absolutely nothing.

“Nor had our forefathers any better right to talk about _certainty_, when pursuing, in blind confidence, the _à priori_ path of axioms, or of the Ram. At innumerable points this path was scarcely as straight as a ram’s-horn. The simple truth is, that the Aristotelians erected their castles upon a basis far less reliable than air; _for no such things as axioms ever existed or can possibly exist at all_. This they must have been very blind, indeed, not to see, or at least to suspect; for, even in their own day, many of their long-admitted ‘axioms’ had been abandoned:—‘_ex nihilo nihil fit_,’ for example, and a ‘thing cannot act where it is not,’ and ‘there cannot be antipodes,’ and ‘darkness cannot proceed from light.’ These and numerous similar propositions formerly accepted, without hesitation, as axioms, or undeniable truths, were, even at the period of which I speak, seen to be altogether untenable:—how absurd in these people, then, to persist in relying upon a basis, as immutable, whose mutability had become so repeatedly manifest!

“But, even through evidence afforded by themselves against themselves, it is easy to convict these _à priori_ reasoners of the grossest unreason—it is easy to show the futility—the impalpability of their axioms in general. I have now lying before me”—it will be observed that we still proceed with the letter—“I have now lying before me a book printed about a thousand years ago. Pundit assures me that it is decidedly the cleverest ancient work on its topic, which is ‘Logic.’ The author, who was much esteemed in his day, was one Miller, or Mill; and we find it recorded of him, as a point of some importance, that he rode a mill-horse whom he called Jeremy Bentham:—but let us glance at the volume itself!

“Ah!—‘Ability or inability to conceive,’ says Mr. Mill very properly, ‘is _in no case_ to be received as a criterion of axiomatic truth.’ Now, that this is a palpable truism no one in his senses will deny. _Not_ to admit the proposition, is to insinuate a charge of variability in Truth itself, whose very title is a synonym of the Steadfast. If ability to conceive be taken as a criterion of Truth, then a truth to _David_ Hume would very seldom be a truth to _Joe_; and ninety-nine hundredths of what is undeniable in Heaven would be demonstrable falsity upon Earth. The proposition of Mr. Mill, then, is sustained. I will not grant it to be an _axiom_; and this merely because I am showing that _no_ axioms exist; but, with a distinction which could not have been cavilled at even by Mr. Mill himself, I am ready to grant that, _if_ an axiom _there be_, then the proposition of which we speak has the fullest right to be considered an axiom—that no _more_ absolute axiom _is_—and, consequently, that any subsequent proposition which shall conflict with this one primarily advanced, must be either a

falsity in itself—that is to say no axiom—or, if admitted axiomatic, must at once neutralize both itself and its predecessor.

"And now, by the logic of their own propounder, let us proceed to test any one of the axioms propounded. Let us give Mr. Mill the fairest of play. We will bring the point to no ordinary issue. We will select for investigation no common-place axiom—no axiom of what, not the less preposterously because only impliedly, he terms his secondary class—as if a positive truth by definition could be either more or less positively a truth:—we will select, I say, no axiom of an unquestionability so questionable as is to be found in Euclid. We will not talk, for example, about such propositions as that two straight lines cannot enclose a space, or that the whole is greater than any one of its parts. We will afford the logician _every_ advantage. We will come at once to a proposition which he regards as the acme of the unquestionable—as the quintessence of axiomatic undeniability. Here it is:—'Contradictions cannot _both_ be true—that is, cannot cöexist in nature.' Here Mr. Mill means, for instance,—and I give the most forcible instance conceivable—that a tree must be either a tree or _not_ a tree—that it cannot be at the same time a tree _and_ not a tree:—all which is quite reasonable of itself and will answer remarkably well as an axiom, until we bring it into collation with an axiom insisted upon a few pages before—in other words—words which I have previously employed—until we test it by the logic of its own propounder. 'A tree,' Mr. Mill asserts, 'must be either a tree or _not_ a tree.' Very well:—and now let me ask him, _why_. To this little query there is but one response:—I defy any man living to invent a second. The sole answer is this:—'Because we find it _impossible to conceive_ that a tree can be any thing else than a tree or not a tree.' This, I repeat, is Mr. Mill's sole answer:—he will not _pretend_ to suggest another:—and yet, by his own showing, his answer is clearly no answer at all; for has he not already required us to admit, _as an axiom_, that ability or inability to conceive is _in no case_ to be taken as a criterion of axiomatic truth? Thus all—absolutely _all_ his argumentation is at sea without a rudder. Let it not be urged that an exception from the general rule is to be made, in cases where the 'impossibility to conceive' is so peculiarly great as when we are called upon to conceive a tree _both_ a tree and _not_ a tree. Let no attempt, I say, be made at urging this sotticism; for, in the first place, there are no _degrees_ of 'impossibility,' and thus no one impossible conception can be _more_ peculiarly impossible than another impossible conception:—in the second place, Mr. Mill himself, no doubt after thorough deliberation, has most distinctly, and most rationally, excluded all opportunity for exception, by the emphasis of his proposition, that, _in no case_, is ability or inability to conceive, to be taken as a criterion of axiomatic truth:—in the third place, even were exceptions admissible at all, it remains to be shown how any exception is admissible _here_. That a tree can be both a tree and not a tree, is an idea which the angels, or the devils, _may_ entertain, and which no doubt many an earthly Bedlamite, or Transcendentalist, _does_.

"Now I do not quarrel with these ancients," continues the letter-writer, "_so much_ on account of the transparent frivolity of their logic—which, to be plain, was baseless, worthless and fantastic altogether—as on account of their pompous and infatuate proscription of all _other_ roads to Truth than the two narrow and crooked paths—the

one of creeping and the other of crawling—to which, in their ignorant perversity, they have dared to confine the Soul—the Soul which loves nothing so well as to soar in those regions of illimitable intuition which are utterly incognizant of '_path_.'

"By the bye, my dear friend, is it not an evidence of the mental slavery entailed upon those bigoted people by their Hogs and Rams, that in spite of the eternal prating of their savans about _roads_ to Truth, none of them fell, even by accident, into what we now so distinctly perceive to be the broadest, the straightest and most available of all mere roads—the great thoroughfare—the majestic highway of the _Consistent_? Is it not wonderful that they should have failed to deduce from the works of God the vitally momentous consideration that _a perfect consistency can be nothing but an absolute truth_? How plain—how rapid our progress since the late announcement of this proposition! By its means, investigation has been taken out of the hands of the ground-moles, and given as a duty, rather than as a task, to the true—to the _only_ true thinkers—to the generally-educated men of ardent imagination. These latter—our Keplers—our Laplaces—'speculate'—'theorize'—these are the terms—can you not fancy the shout of scorn with which they would be received by our progenitors, were it possible for them to be looking over my shoulders as I write? The Keplers, I repeat, speculate—theorize—and their theories are merely corrected—reduced—sifted—cleared, little by little, of their chaff of inconsistency—until at length there stands apparent an unencumbered _Consistency_—a consistency which the most stolid admit—because it _is_ a consistency—to be an absolute and an unquestionable _Truth_.

"I have often thought, my friend, that it must have puzzled these dogmaticians of a thousand years ago, to determine, even, by which of their two boasted roads it is that the cryptographist attains the solution of the more complicate cyphers—or by which of them Champollion guided mankind to those important and innumerable truths which, for so many centuries, have lain entombed amid the phonetical hieroglyphics of Egypt. In especial, would it not have given these bigots some trouble to determine by which of their two roads was reached the most momentous and sublime of _all_ their truths—the truth—the fact of _gravitation_? Newton deduced it from the laws of Kepler. Kepler admitted that these laws he _guessed_—these laws whose investigation disclosed to the greatest of British astronomers that principle, the basis of all (existing) physical principle, in going behind which we enter at once the nebulous kingdom of Metaphysics. Yes!—these vital laws Kepler _guessed_—that is to say, he _imagined_ them. Had he been asked to point out either the _de_ductive or _in_ductive route by which he attained them, his reply might have been—'I know nothing about _routes_—but I _do_ know the machinery of the Universe. Here it is. I grasped it with _my soul_—I reached it through mere dint of _intuition_.' Alas, poor ignorant old man! Could not any metaphysician have told him that what he called 'intuition' was but the conviction resulting from _de_ductions or _in_ductions of which the processes were so shadowy as to have escaped his consciousness, eluded his reason, or bidden defiance to his capacity of expression? How great a pity it is that some 'moral philosopher' had not enlightened him about all this! How it would have comforted him on his death-bed to know that, instead of having gone intuitively and thus unbecomingly, he had, in fact, proceeded decorously and legitimately—that is to

say Hog-ishly, or at least Ram-ishly—into the vast halls where lay gleaming, untended, and hitherto untouched by mortal hand—unseen by mortal eye—the imperishable and priceless secrets of the Universe!

"Yes, Kepler was essentially a _theorist_; but this title, _now_ of so much sanctity, was, in those ancient days, a designation of supreme contempt. It is only _now_ that men begin to appreciate that divine old man—to sympathize with the prophetical and poetical rhapsody of his ever-memorable words. For _my_ part," continues the unknown correspondent, "I glow with a sacred fire when I even think of them, and feel that I shall never grow weary of their repetition:—in concluding this letter, let me have the real pleasure of transcribing them once again:—'_I care not whether my work be read now or by posterity. I can afford to wait a century for readers when God himself has waited six thousand years for an observer. I triumph. I have stolen the golden secret of the Egyptians. I will indulge my sacred fury._'"

Here end my quotations from this very unaccountable and, perhaps, somewhat impertinent epistle; and perhaps it would be folly to comment, in any respect, upon the chimerical, not to say revolutionary, fancies of the writer—whoever he is—fancies so radically at war with the well-considered and well-settled opinions of this age. Let us proceed, then, to our legitimate thesis, _The Universe_.

This thesis admits a choice between two modes of discussion:—We may _as_cend or _de_scend. Beginning at our own point of view—at the Earth on which we stand—we may pass to the other planets of our system—thence to the Sun—thence to our system considered collectively—and thence, through other systems, indefinitely outwards; or, commencing on high at some point as definite as we can make it or conceive it, we may come down to the habitation of Man. Usually—that is to say, in ordinary essays on Astronomy—the first of these two modes is, with certain reservation, adopted:—this for the obvious reason that astronomical _facts_, merely, and principles, being the object, that object is best fulfilled in stepping from the known because proximate, gradually onward to the point where all certitude becomes lost in the remote. For my present purpose, however,—that of enabling the mind to take in, as if from afar and at one glance, a distinct conception of the _individual_ Universe—it is clear that a descent to small from great—to the outskirts from the centre (if we could establish a centre)—to the end from the beginning (if we could fancy a beginning) would be the preferable course, but for the difficulty, if not impossibility, of presenting, in this course, to the unastronomical, a picture at all comprehensible in regard to such considerations as are involved in _quantity_—that is to say, in number, magnitude and distance.

Now, distinctness—intelligibility, at all points, is a primary feature in my general design. On important topics it is better to be a good deal prolix than even a very little obscure. But abstruseness is a quality appertaining to no subject _per se_. All are alike, in facility of comprehension, to him who approaches them by properly graduated steps. It is merely because a stepping-stone, here and there, is heedlessly left unsupplied in our road to the Differential Calculus, that this latter is not altogether as simple a thing as a sonnet by Mr. Solomon Seesaw.

By way of admitting, then, no _chance_ for misapprehension, I think it advisable to proceed as if even the more obvious facts of Astronomy were unknown to the reader. In combining the two modes of discussion to which I have referred, I propose to avail myself of the advantages peculiar to each—and very especially of the _iteration in detail_ which will be unavoidable as a consequence of the plan. Commencing with a descent, I shall reserve for the return upwards those indispensable considerations of _quantity_ to which allusion has already been made.

Let us begin, then, at once, with that merest of words, “Infinity.” This, like “God,” “spirit,” and some other expressions of which the equivalents exist in all languages, is by no means the expression of an idea—but of an effort at one. It stands for the possible attempt at an impossible conception. Man needed a term by which to point out the _direction_ of this effort—the cloud behind which lay, forever invisible, the _object_ of this attempt. A word, in fine, was demanded, by means of which one human being might put himself in relation at once with another human being and with a certain _tendency_ of the human intellect. Out of this demand arose the word, “Infinity;” which is thus the representative but of the _thought of a thought_.

As regards _that_ infinity now considered—the infinity of space—we often hear it said that “its idea is admitted by the mind—is acquiesced in—is entertained—on account of the greater difficulty which attends the conception of a limit.” But this is merely one of those _phrases_ by which even profound thinkers, time out of mind, have occasionally taken pleasure in deceiving _themselves_. The quibble lies concealed in the word “difficulty.” “The mind,” we are told, “entertains the idea of _limitless_, through the greater _difficulty_ which it finds in entertaining that of _limited_, space.” Now, were the proposition but fairly _put_, its absurdity would become transparent at once. Clearly, there is no mere _difficulty_ in the case. The assertion intended, if presented _according_ to its intention and without sophistry, would run thus:—“The mind admits the idea of limitless, through the greater _impossibility_ of entertaining that of limited, space.”

It must be immediately seen that this is not a question of two statements between whose respective credibilities—or of two arguments between whose respective validities—the _reason_ is called upon to decide:—it is a matter of two conceptions, directly conflicting, and each avowedly impossible, one of which the _intellect_ is supposed to be capable of entertaining, on account of the greater _impossibility_ of entertaining the other. The choice is _not_ made between two difficulties;—it is merely _fancied_ to be made between two impossibilities. Now of the former, there _are_ degrees—but of the latter, none:—just as our impertinent letter-writer has already suggested. A task _may_ be more or less difficult; but it is either possible or not possible:—there are no gradations. It _might_ be more _difficult_ to overthrow the Andes than an ant-hill; but it _can_ be no more _impossible_ to annihilate the matter of the one than the matter of the other. A man may jump ten feet with less _difficulty_ than he can jump twenty, but the _impossibility_ of his leaping to the moon is not a whit less than that of his leaping to the dog-star.

Since all this is undeniable: since the choice of the mind is to be made between _impossibilities_ of conception: since one impossibility cannot be greater than

another: and since, thus, one cannot be preferred to another: the philosophers who not only maintain, on the grounds mentioned, man's _idea_ of infinity but, on account of such supposititious idea, _infinity itself_—are plainly engaged in demonstrating one impossible thing to be possible by showing how it is that some one other thing—is impossible too. This, it will be said, is nonsense; and perhaps it is:—indeed I think it very capital nonsense—but forego all claim to it as nonsense of mine.

The readiest mode, however, of displaying the fallacy of the philosophical argument on this question, is by simply adverting to a _fact_ respecting it which has been hitherto quite overlooked—the fact that the argument alluded to both proves and disproves its own proposition. "The mind is impelled," say the theologians and others, "to admit a _First Cause_, by the superior difficulty it experiences in conceiving cause beyond cause without end." The quibble, as before, lies in the word "difficulty"—but _here_ what is it employed to sustain? A First Cause. And what is a First Cause? An ultimate termination of causes. And what is an ultimate termination of causes? Finity—the Finite. Thus the one quibble, in two processes, by God knows how many philosophers, is made to support now Finity and now Infinity—could it not be brought to support something besides? As for the quibblers—_they_, at least, are insupportable. But—to dismiss them:—what they prove in the one case is the identical nothing which they demonstrate in the other.

Of course, no one will suppose that I here contend for the absolute impossibility of _that_ which we attempt to convey in the word "Infinity." My purpose is but to show the folly of endeavoring to prove Infinity itself or even our conception of it, by any such blundering ratiocination as that which is ordinarily employed.

Nevertheless, as an individual, I may be permitted to say that _I cannot_ conceive Infinity, and am convinced that no human being can. A mind not thoroughly self-conscious—not accustomed to the introspective analysis of its own operations—will, it is true, often deceive itself by supposing that it _has_ entertained the conception of which we speak. In the effort to entertain it, we proceed step beyond step—we fancy point still beyond point; and so long as we _continue_ the effort, it may be said, in fact, that we are _tending_ to the formation of the idea designed; while the strength of the impression that we actually form or have formed it, is in the ratio of the period during which we keep up the mental endeavor. But it is in the act of discontinuing the endeavor—of fulfilling (as we think) the idea—of putting the finishing stroke (as we suppose) to the conception—that we overthrow at once the whole fabric of our fancy by resting upon some one ultimate and therefore definite point. This fact, however, we fail to perceive, on account of the absolute coincidence, in time, between the settling down upon the ultimate point and the act of cessation in thinking.—In attempting, on the other hand, to frame the idea of a _limited_ space, we merely converse the processes which involve the impossibility.

We _believe_ in a God. We may or may not _believe_ in finite or in infinite space; but our belief, in such cases, is more properly designated as _faith_, and is a matter quite distinct from that belief proper—from that _intellectual_ belief—which presupposes the mental conception.

The fact is, that, upon the enunciation of any one of that class of terms to which “Infinity” belongs—the class representing _thoughts of thought_—he who has a right to say that he thinks _at all_, feels himself called upon, _not_ to entertain a conception, but simply to direct his mental vision toward some given point, in the intellectual firmament, where lies a nebula never to be resolved. To solve it, indeed, he makes no effort; for with a rapid instinct he comprehends, not only the impossibility, but, as regards all human purposes, the _inessentiality_, of its solution. He perceives that the Deity has not _designed_ it to be solved. He sees, at once, that it lies _out_ of the brain of man, and even _how_, if not exactly _why_, it lies out of it. There _are_ people, I am aware, who, busying themselves in attempts at the unattainable, acquire very easily, by dint of the jargon they emit, among those thinkers-that-they-think with whom darkness and depth are synonymous, a kind of cuttle-fish reputation for profundity; but the finest quality of Thought is its self-cognizance; and, with some little equivocation, it may be said that no fog of the mind can well be greater than that which, extending to the very boundaries of the mental domain, shuts out even these boundaries themselves from comprehension.

It will now be understood that, in using the phrase, “Infinity of Space,” I make no call upon the reader to entertain the impossible conception of an _absolute_ infinity. I refer simply to the “_utmost conceivable expanse_” of space—a shadowy and fluctuating domain, now shrinking, now swelling, in accordance with the vacillating energies of the imagination.

Hitherto, the Universe of stars has always been considered as coincident with the Universe proper, as I have defined it in the commencement of this Discourse. It has been always either directly or indirectly assumed—at least since the dawn of intelligible Astronomy—that, were it possible for us to attain any given point in space, we should still find, on all sides of us, an interminable succession of stars. This was the untenable idea of Pascal when making perhaps the most successful attempt ever made, at periphrasing the conception for which we struggle in the word “Universe.” “It is a sphere,” he says, “of which the centre is everywhere, the circumference, nowhere.” But although this intended definition is, in fact, _no_ definition of the Universe of _stars_, we may accept it, with some mental reservation, as a definition (rigorous enough for all practical purposes) of the Universe _proper_—that is to say, of the Universe of _space_. This latter, then, let us regard as “_a sphere of which the centre is everywhere, the circumference nowhere_.” In fact, while we find it impossible to fancy an _end_ to space, we have no difficulty in picturing to ourselves any one of an infinity of _beginnings_.

As our starting-point, then, let us adopt the _Godhead_. Of this Godhead, _in itself_, he alone is not imbecile—he alone is not impious who propounds—nothing. “_Nous ne connaissons rien_,” says the Baron de Bielfeld—“_Nous ne connaissons rien de la nature ou de l’essence de Dieu:—pour savoir ce qu’il est, il faut être Dieu même._”—“We know absolutely _nothing_ of the nature or essence of God:—in order to comprehend what he is, we should have to be God ourselves.”

“_We should have to be God ourselves!_”—With a phrase so startling as this yet ringing in my ears, I nevertheless venture to demand if this our present ignorance of

the Deity is an ignorance to which the soul is _everlastingly_ condemned.

By _Him_, however—_now_, at least, the Incomprehensible—by Him—assuming him as _Spirit_—that is to say, as _not Matter_—a distinction which, for all intelligible purposes, will stand well instead of a definition—by Him, then, existing as Spirit, let us content ourselves, to-night, with supposing to have been _created_, or made out of Nothing, by dint of his Volition—at some point of Space which we will take as a centre—at some period into which we do not pretend to inquire, but at all events immensely remote—by Him, then again, let us suppose to have been created——_what_? This is a vitally momentous epoch in our considerations. _What_ is it that we are justified—that alone we are justified in supposing to have been, primarily and solely, _created_?

We have attained a point where only _Intuition_ can aid us:—but now let me recur to the idea which I have already suggested as that alone which we can properly entertain of intuition. It is but _the conviction arising from those inductions or deductions of which the processes are so shadowy as to escape our consciousness, elude our reason, or defy our capacity of expression_. With this understanding, I now assert—that an intuition altogether irresistible, although inexpressible, forces me to the conclusion that what God originally created—that that Matter which, by dint of his Volition, he first made from his Spirit, or from Nihility, _could_ have been nothing but Matter in its utmost conceivable state of——what?—of _Simplicity_?

This will be found the sole absolute _assumption_ of my Discourse. I use the word “assumption” in its ordinary sense; yet I maintain that even this my primary proposition, is very, very far indeed, from being really a mere assumption. Nothing was ever more certainly—no human conclusion was ever, in fact, more regularly—more rigorously _de_duced:—but, alas! the processes lie out of the human analysis—at all events are beyond the utterance of the human tongue.

Let us now endeavor to conceive what Matter must be, when, or if, in its absolute extreme of _Simplicity_. Here the Reason flies at once to Imparticularity—to a particle—to _one_ particle—a particle of _one_ kind—of _one_ character—of _one_ nature—of _one size_—of one form—a particle, therefore, “_without_ form and void”—a particle positively a particle at all points—a particle absolutely unique, individual, undivided, and not indivisible only because He who _created_ it, by dint of his Will, can by an infinitely less energetic exercise of the same Will, as a matter of course, divide it.

Oneness, then, is all that I predicate of the originally created Matter; but I propose to show that this _Oneness is a principle abundantly sufficient to account for the constitution, the existing phænomena and the plainly inevitable annihilation of at least the material Universe_.

The willing into being the primordial particle, has completed the act, or more properly the _conception_, of Creation. We now proceed to the ultimate purpose for which we are to suppose the Particle created—that is to say, the ultimate purpose so far as our considerations _yet_ enable us to see it—the constitution of the Universe from it, the Particle.

This constitution has been effected by _forcing_ the originally and therefore normally _One_ into the abnormal condition of _Many_. An action of this character implies rëaction. A diffusion from Unity, under the conditions, involves a tendency to return into Unity—a tendency ineradicable until satisfied. But on these points I will speak more fully hereafter.

The assumption of absolute Unity in the primordial Particle includes that of infinite divisibility. Let us conceive the Particle, then, to be only not totally exhausted by diffusion into Space. From the one Particle, as a centre, let us suppose to be irradiated spherically—in all directions—to immeasurable but still to definite distances in the previously vacant space—a certain inexpressibly great yet limited number of unimaginably yet not infinitely minute atoms.

Now, of these atoms, thus diffused, or upon diffusion, what conditions are we permitted—not to assume, but to infer, from consideration as well of their source as of the character of the design apparent in their diffusion? _Unity_ being their source, and _difference from Unity_ the character of the design manifested in their diffusion, we are warranted in supposing this character to be at least _generally_ preserved throughout the design, and to form a portion of the design itself:—that is to say, we shall be warranted in conceiving continual differences at all points from the uniquity and simplicity of the origin. But, for these reasons, shall we be justified in imagining the atoms heterogeneous, dissimilar, unequal, and inequidistant? More explicitly—are we to consider no two atoms as, at their diffusion, of the same nature, or of the same form, or of the same size?—and, after fulfilment of their diffusion into Space, is absolute inequidistance, each from each, to be understood of all of them? In such arrangement, under such conditions, we most easily and immediately comprehend the subsequent most feasible carrying out to completion of any such design as that which I have suggested—the design of variety out of unity—diversity out of sameness—heterogeneity out of homogeneity—complexity out of simplicity—in a word, the utmost possible multiplicity of _relation_ out of the emphatically irrelative _One_. Undoubtedly, therefore, we _should_ be warranted in assuming all that has been mentioned, but for the reflection, first, that supererogation is not presumable of any Divine Act; and, secondly, that the object supposed in view, appears as feasible when some of the conditions in question are dispensed with, in the beginning, as when all are understood immediately to exist. I mean to say that some are involved in the rest, or so instantaneous a consequence of them as to make the distinction inappreciable. Difference of _size_, for example, will at once be brought about through the tendency of one atom to a second, in preference to a third, on account of particular inequidistance; which is to be comprehended as _particular inequidistances between centres of quantity, in neighboring atoms of different form_—a matter not at all interfering with the generally-equable distribution of the atoms. Difference of _kind_, too, is easily conceived to be merely a result of differences in size and form, taken more or less conjointly:—in fact, since the _Unity_ of the Particle Proper implies absolute homogeneity, we cannot imagine the atoms, at their diffusion, differing in kind, without imagining, at the same time, a special exercise of the Divine Will, at the emission of each atom, for the purpose of effecting, in each, a change of its essential nature:—so fantastic an idea is the less to be

indulged, as the object proposed is seen to be thoroughly attainable without such minute and elaborate interposition. We perceive, therefore, upon the whole, that it would be supererogatory, and consequently unphilosophical, to predicate of the atoms, in view of their purposes, any thing more than _difference of form_ at their dispersion, with particular inequidistance after it—all other differences arising at once out of these, in the very first processes of mass-constitution:—We thus establish the Universe on a purely _geometrical_ basis. Of course, it is by no means necessary to assume absolute difference, even of form, among _all_ the atoms irradiated—any more than absolute particular inequidistance of each from each. We are required to conceive merely that no _neighboring_ atoms are of similar form—no atoms which can ever approximate, until their inevitable rëunition at the end.

Although the immediate and perpetual _tendency_ of the disunited atoms to return into their normal Unity, is implied, as I have said, in their abnormal diffusion; still it is clear that this tendency will be without consequence—a tendency and no more—until the diffusive energy, in ceasing to be exerted, shall leave _it_, the tendency, free to seek its satisfaction. The Divine Act, however, being considered as determinate, and discontinued on fulfilment of the diffusion, we understand, at once, a _rëaction_—in other words, a _satisfiable_ tendency of the disunited atoms to return into _One_.

But the diffusive energy being withdrawn, and the rëaction having commenced in furtherance of the ultimate design—_that of the utmost possible Relation_—this design is now in danger of being frustrated, in detail, by reason of that very tendency to return which is to effect its accomplishment in general. _Multiplicity_ is the object; but there is nothing to prevent proximate atoms, from lapsing _at once_, through the now satisfiable tendency—_before_ the fulfilment of any ends proposed in multiplicity—into absolute oneness among themselves:—there is nothing to impede the aggregation of various _unique_ masses, at various points of space:—in other words, nothing to interfere with the accumulation of various masses, each absolutely One.

For the effectual and thorough completion of the general design, we thus see the necessity for a repulsion of limited capacity—a separative _something_ which, on withdrawal of the diffusive Volition, shall at the same time allow the approach, and forbid the junction, of the atoms; suffering them infinitely to approximate, while denying them positive contact; in a word, having the power—_up to a certain epoch_—of preventing their _coalition_, but no ability to interfere with their _coalescence_ in any respect _or degree_. The repulsion, already considered as so peculiarly limited in other regards, must be understood, let me repeat, as having power to prevent absolute coalition, _only up to a certain epoch_. Unless we are to conceive that the appetite for Unity among the atoms is doomed to be satisfied _never_;—unless we are to conceive that what had a beginning is to have no end—a conception which cannot _really_ be entertained, however much we may talk or dream of entertaining it—we are forced to conclude that the repulsive influence imagined, will, finally—under pressure of the _Unitendency collectively_ applied, but never and in no degree _until_, on fulfilment of the Divine purposes, such collective application shall be naturally made—yield to a force which, at that ultimate epoch,

shall be the superior force precisely to the extent required, and thus permit the universal subsidence into the inevitable, because original and therefore normal, _One_.—The conditions here to be reconciled are difficult indeed:—we cannot even comprehend the possibility of their conciliation;—nevertheless, the apparent impossibility is brilliantly suggestive.

That the repulsive something actually exists, _we see_. Man neither employs, nor knows, a force sufficient to bring two atoms into contact. This is but the well-established proposition of the impenetrability of matter. All Experiment proves—all Philosophy admits it. The _design_ of the repulsion—the necessity for its existence—I have endeavored to show; but from all attempt at investigating its nature have religiously abstained; this on account of an intuitive conviction that the principle at issue is strictly spiritual—lies in a recess impervious to our present understanding—lies involved in a consideration of what now—in our human state—is _not_ to be considered—in a consideration of _Spirit in itself_. I feel, in a word, that here the God has interposed, and here only, because here and here only the knot demanded the interposition of the God.

In fact, while the tendency of the diffused atoms to return into Unity, will be recognized, at once, as the principle of the Newtonian Gravity, what I have spoken of as a repulsive influence prescribing limits to the (immediate) satisfaction of the tendency, will be understood as _that_ which we have been in the practice of designating now as heat, now as magnetism, now as _electricity_; displaying our ignorance of its awful character in the vacillation of the phraseology with which we endeavor to circumscribe it.

Calling it, merely for the moment, electricity, we know that all experimental analysis of electricity has given, as an ultimate result, the principle, or seeming principle, _heterogeneity_. _Only_ where things differ is electricity apparent; and it is presumable that they _never_ differ where it is not developed at least, if not apparent. Now, this result is in the fullest keeping with that which I have reached unempirically. The design of the repulsive influence I have maintained to be that of preventing immediate Unity among the diffused atoms; and these atoms are represented as different each from each. _Difference_ is their character—their essentiality—just as _no-difference_ was the essentiality of their source. When we say, then, that an attempt to bring any two of these atoms together would induce an effort, on the part of the repulsive influence, to prevent the contact, we may as well use the strictly convertible sentence that an attempt to bring together any two differences will result in a development of electricity. All existing bodies, of course, are composed of these atoms in proximate contact, and are therefore to be considered as mere assemblages of more or fewer differences; and the resistance made by the repulsive spirit, on bringing together any two such assemblages, would be in the ratio of the two sums of the differences in each:—an expression which, when reduced, is equivalent to this:—_The amount of electricity developed on the approximation of two bodies, is proportional to the difference between the respective sums of the atoms of which the bodies are composed._ That _no_ two bodies are absolutely alike, is a simple corollary from all that has been here said. Electricity,

therefore, existing always, is _developed_ whenever _any_ bodies, but _manifested_ only when bodies of appreciable difference, are brought into approximation.

To electricity—so, for the present, continuing to call it—we _may_ not be wrong in referring the various physical appearances of light, heat and magnetism; but far less shall we be liable to err in attributing to this strictly spiritual principle the more important phænomena of vitality, consciousness and _Thought_. On this topic, however, I need pause _here_ merely to suggest that these phænomena, whether observed generally or in detail, seem to proceed _at least in the ratio of the heterogeneous_.

Discarding now the two equivocal terms, “gravitation” and “electricity,” let us adopt the more definite expressions, “_attraction_” and “_repulsion_.” The former is the body; the latter the soul: the one is the material; the other the spiritual, principle of the Universe. _No other principles exist._ _All_ phænomena are referable to one, or to the other, or to both combined. So rigorously is this the case—so thoroughly demonstrable is it that attraction and repulsion are the _sole_ properties through which we perceive the Universe—in other words, by which Matter is manifested to Mind—that, for all merely argumentative purposes, we are fully justified in assuming that matter _exists_ only as attraction and repulsion—that attraction and repulsion _are_ matter:—there being no conceivable case in which we may not employ the term “matter” and the terms “attraction” and “repulsion,” taken together, as equivalent, and therefore convertible, expressions in Logic.

I said, just now, that what I have described as the tendency of the diffused atoms to return into their original unity, would be understood as the principle of the Newtonian law of gravity: and, in fact, there can be little difficulty in such an understanding, if we look at the Newtonian gravity in a merely general view, as a force impelling matter to seek matter; that is to say, when we pay no attention to the known _modus operandi_ of the Newtonian force. The general coincidence satisfies us; but, upon looking closely, we see, in detail, much that appears _in_coincident, and much in regard to which no coincidence, at least, is established. For example; the Newtonian gravity, when we think of it in certain moods, does _not_ seem to be a tendency to _oneness_ at all, but rather a tendency of all bodies in all directions—a phrase apparently expressive of a tendency to diffusion. Here, then, is an _in_coincidence. Again; when we reflect on the mathematical _law_ governing the Newtonian tendency, we see clearly that no coincidence has been made good, in respect of the _modus operandi_, at least, between gravitation as known to exist and that seemingly simple and direct tendency which I have assumed.

In fact, I have attained a point at which it will be advisable to strengthen my position by reversing my processes. So far, we have gone on _à priori_, from an abstract consideration of _Simplicity_, as that quality most likely to have characterized the original action of God. Let us now see whether the established facts of the Newtonian Gravitation may not afford us, _à posteriori_, some legitimate inductions.

What does the Newtonian law declare?—That all bodies attract each other with forces proportional to their quantities of matter and inversely proportional to the squares of their distances. Purposely, I have here given, in the first place, the vulgar version of the law; and I confess that in this, as in most other vulgar versions of great truths, we find little of a suggestive character. Let us now adopt a more philosophical phraseology:—_Every atom, of every body, attracts every other atom, both of its own and of every other body, with a force which varies inversely as the squares of the distances between the attracting and attracted atom._—Here, indeed, a flood of suggestion bursts upon the mind.

But let us see distinctly what it was that Newton _proved_—according to the grossly irrational definitions of _proof_ prescribed by the metaphysical schools. He was forced to content himself with showing how thoroughly the motions of an imaginary Universe, composed of attracting and attracted atoms obedient to the law he announced, coincide with those of the actually existing Universe so far as it comes under our observation. This was the amount of his _demonstration_—that is to say, this was the amount of it, according to the conventional cant of the “philosophies.” His successes added proof multiplied by proof—such proof as a sound intellect admits—but the _demonstration_ of the law itself, persist the metaphysicians, had not been strengthened in any degree. “_Ocular_, _physical_ proof,” however, of attraction, here upon Earth, in accordance with the Newtonian theory, was, at length, much to the satisfaction of some intellectual grovellers, afforded. This proof arose collaterally and incidentally (as nearly all important truths have arisen) out of an attempt to ascertain the mean density of the Earth. In the famous Maskelyne, Cavendish and Bailly experiments for this purpose, the attraction of the mass of a mountain was seen, felt, measured, and found to be mathematically consistent with the immortal theory of the British astronomer.

But in spite of this confirmation of that which needed none—in spite of the so-called corroboration of the “theory” by the so-called “ocular and physical proof”—in spite of the _character_ of this corroboration—the ideas which even really philosophical men cannot help imbibing of gravity—and, especially, the ideas of it which ordinary men get and contentedly maintain, are _seen_ to have been derived, for the most part, from a consideration of the principle as they find it developed—_merely in the planet upon which they stand_.

Now, to what does so partial a consideration tend—to what species of error does it give rise? On the Earth we _see_ and _feel_, only that gravity impels all bodies towards the _centre_ of the Earth. No man in the common walks of life could be _made_ to see or to feel anything else—could be made to perceive that anything, anywhere, has a perpetual, gravitating tendency in any _other_ direction than to the centre of the Earth; yet (with an exception hereafter to be specified) it is a fact that every earthly thing (not to speak now of every heavenly thing) has a tendency not _only_ to the Earth’s centre but in every conceivable direction besides.

Now, although the philosophic cannot be said to _err with_ the vulgar in this matter, they nevertheless permit themselves to be influenced, without knowing it, by the _sentiment_ of the vulgar idea. “Although the Pagan fables are not believed,” says

Bryant, in his very erudite “Mythology,” “yet we forget ourselves continually and make inferences from them as from existing realities.” I mean to assert that the merely _sensitive perception_ of gravity as we experience it on Earth, beguiles mankind into the fancy of _concentralization_ or _especiality_ respecting it—has been continually biasing towards this fancy even the mightiest intellects—perpetually, although imperceptibly, leading them away from the real characteristics of the principle; thus preventing them, up to this date, from ever getting a glimpse of that vital truth which lies in a diametrically opposite direction—behind the principle’s _essential_ characteristics—those, _not_ of concentralization or especiality—but of _universality_ and _diffusion_. This “vital truth” is _Unity_ as the _source_ of the phænomenon.

Let me now repeat the definition of gravity:—_Every atom, of every body, attracts every other atom, both of its own and of every other body_, with a force which varies inversely as the squares of the distances of the attracting and attracted atom.

Here let the reader pause with me, for a moment, in contemplation of the miraculous—of the ineffable—of the altogether unimaginable complexity of relation involved in the fact that _each atom attracts every other atom_—involved merely in this fact of the attraction, without reference to the law or mode in which the attraction is manifested—involved _merely_ in the fact that each atom attracts every other atom _at all_, in a wilderness of atoms so numerous that those which go to the composition of a cannon-ball, exceed, probably, in mere point of number, all the stars which go to the constitution of the Universe.

Had we discovered, simply, that each atom tended to some one favorite point—to some especially attractive atom—we should still have fallen upon a discovery which, in itself, would have sufficed to overwhelm the mind:—but what is it that we are actually called upon to comprehend? That each atom attracts—sympathizes with the most delicate movements of every other atom, and with each and with all at the same time, and forever, and according to a determinate law of which the complexity, even considered by itself solely, is utterly beyond the grasp of the imagination of man. If I propose to ascertain the influence of one mote in a sunbeam upon its neighboring mote, I cannot accomplish my purpose without first counting and weighing all the atoms in the Universe and defining the precise positions of all at one particular moment. If I venture to displace, by even the billionth part of an inch, the microscopical speck of dust which lies now upon the point of my finger, what is the character of that act upon which I have adventured? I have done a deed which shakes the Moon in her path, which causes the Sun to be no longer the Sun, and which alters forever the destiny of the multitudinous myriads of stars that roll and glow in the majestic presence of their Creator.

These ideas—conceptions such as _these_—unthoughtlike thoughts—soul-reveries rather than conclusions or even considerations of the intellect:—ideas, I repeat, such as these, are such as we can alone hope profitably to entertain in any effort at grasping the great principle, _Attraction_.

But now,—_with_ such ideas—with such a _vision_ of the marvellous complexity of Attraction fairly in his mind—let any person competent of thought on such topics as these, set himself to the task of imagining a _principle_ for the phænomena observed—a condition from which they sprang.

Does not so evident a brotherhood among the atoms point to a common parentage? Does not a sympathy so omniprevalent, so ineradicable, and so thoroughly irrespective, suggest a common paternity as its source? Does not one extreme impel the reason to the other? Does not the infinitude of division refer to the utterness of individuality? Does not the entireness of the complex hint at the perfection of the simple? It is _not_ that the atoms, as we see them, are divided or that they are complex in their relations—but that they are inconceivably divided and unutterably complex:—it is the extremeness of the conditions to which I now allude, rather than to the conditions themselves. In a word, is it not because the atoms were, at some remote epoch of time, even _more than together_—is it not because originally, and therefore normally, they were _One_—that now, in all circumstances—at all points—in all directions—by all modes of approach—in all relations and through all conditions—they struggle _back_ to this absolutely, this irrelatively, this unconditionally _one_?

Some person may here demand:—“Why—since it is to the _One_ that the atoms struggle back—do we not find and define Attraction ‘a merely general tendency to a centre?’—why, in especial, do not _your_ atoms—the atoms which you describe as having been irradiated from a centre—proceed at once, rectilinearly, back to the central point of their origin?”

I reply that _they do_; as will be distinctly shown; but that the cause of their so doing is quite irrespective of the centre _as such_. They all tend rectilinearly towards a centre, because of the sphereicity with which they have been irradiated into space. Each atom, forming one of a generally uniform globe of atoms, finds more atoms in the direction of the centre, of course, than in any other, and in that direction, therefore, is impelled—but is _not_ thus impelled because the centre is _the point of its origin_. It is not to any _point_ that the atoms are allied. It is not any _locality_, either in the concrete or in the abstract, to which I suppose them bound. Nothing like _location_ was conceived as their origin. Their source lies in the principle, _Unity_. _This_ is their lost parent. _This_ they seek always—immediately—in all directions—wherever it is even partially to be found; thus appeasing, in some measure, the ineradicable tendency, while on the way to its absolute satisfaction in the end. It follows from all this, that any principle which shall be adequate to account for the _law_, or _modus operandi_, of the attractive force in general, will account for this law in particular:—that is to say, any principle which will show why the atoms should tend to their _general centre of irradiation_ with forces inversely proportional to the squares of the distances, will be admitted as satisfactorily accounting, at the same time, for the tendency, according to the same law, of these atoms each to each:—_for_ the tendency to the centre _is_ merely the tendency each to each, and not any tendency to a centre as such.—Thus it will be seen, also, that the establishment of my propositions would involve no _necessity_ of modification in the

terms of the Newtonian definition of Gravity, which declares that each atom attracts each other atom and so forth, and declares this merely; but (always under the supposition that what I propose be, in the end, admitted) it seems clear that some error might occasionally be avoided, in the future processes of Science, were a more ample phraseology adopted:—for instance:—“Each atom tends to every other atom &c. with a force &c.: _the general result being a tendency of all, with a similar force, to a general centre_.”

The reversal of our processes has thus brought us to an identical result; but, while in the one process _intuition_ was the starting-point, in the other it was the goal. In commencing the former journey I could only say that, with an irresistible intuition, I _felt_ Simplicity to have been the characteristic of the original action of God:—in ending the latter I can only declare that, with an irresistible intuition, I perceive Unity to have been the source of the observed phænomena of the Newtonian gravitation. Thus, according to the schools, I _prove_ nothing. So be it:—I design but to suggest—and to _convince_ through the suggestion. I am proudly aware that there exist many of the most profound and cautiously discriminative human intellects which cannot _help_ being abundantly content with my—suggestions. To these intellects—as to my own—there is no mathematical demonstration which _could_ bring the least additional _true proof_ of the great _Truth_ which I have advanced—_the truth of Original Unity as the source—as the principle of the Universal Phænomena_. For my part, I am not so sure that I speak and see—I am not so sure that my heart beats and that my soul lives:—of the rising of to-morrow’s sun—a probability that as yet lies in the Future—I do not pretend to be one thousandth part as sure—as I am of the irretrievably by-gone _Fact_ that All Things and All Thoughts of Things, with all their ineffable Multiplicity of Relation, sprang at once into being from the primordial and irrelative _One_.

Referring to the Newtonian Gravity, Dr. Nichol, the eloquent author of “The Architecture of the Heavens,” says:—“In truth we have no reason to suppose this great Law, as now revealed, to be the ultimate or simplest, and therefore the universal and all-comprehensive, form of a great Ordinance. The mode in which its intensity diminishes with the element of distance, has not the aspect of an ultimate _principle_; which always assumes the simplicity and self-evidence of those axioms which constitute the basis of Geometry.”

Now, it is quite true that “ultimate principles,” in the common understanding of the words, always assume the simplicity of geometrical axioms—(as for “self-evidence,” there is no such thing)—but these principles are clearly _not_ “ultimate;” in other terms what we are in the habit of calling principles are no principles, properly speaking—since there can be but one _principle_, the Volition of God. We have no right to assume, then, from what we observe in rules that we choose foolishly to name “principles,” anything at all in respect to the characteristics of a principle proper. The “ultimate principles” of which Dr. Nichol speaks as having geometrical simplicity, may and do have this geometrical turn, as being part and parcel of a vast geometrical system, and thus a system of simplicity itself—in which, nevertheless, the _truly_ ultimate principle is, _as we know_, the consummation of the

complex—that is to say, of the unintelligible—for is it not the Spiritual Capacity of God?

I quoted Dr. Nichol's remark, however, not so much to question its philosophy, as by way of calling attention to the fact that, while all men have admitted _some_ principle as existing behind the Law of Gravity, no attempt has been yet made to point out what this principle in particular _is_:—if we except, perhaps, occasional fantastic efforts at referring it to Magnetism, or Mesmerism, or Swedenborgianism, or Transcendentalism, or some other equally delicious _ism_ of the same species, and invariably patronized by one and the same species of people. The great mind of Newton, while boldly grasping the Law itself, shrank from the principle of the Law. The more fluent and comprehensive at least, if not the more patient and profound, sagacity of Laplace, had not the courage to attack it. But hesitation on the part of these two astronomers it is, perhaps, not so very difficult to understand. They, as well as all the first class of mathematicians, were mathematicians _solely_:—their intellect, at least, had a firmly-pronounced mathematico-physical tone. What lay not distinctly within the domain of Physics, or of Mathematics, seemed to them either Non-Entity or Shadow. Nevertheless, we may well wonder that Leibnitz, who was a marked exception to the general rule in these respects, and whose mental temperament was a singular admixture of the mathematical with the physico-metaphysical, did not at once investigate and establish the point at issue. Either Newton or Laplace, seeking a principle and discovering none _physical_, would have rested contentedly in the conclusion that there was absolutely none; but it is almost impossible to fancy, of Leibnitz, that, having exhausted in his search the physical dominions, he would not have stepped at once, boldly and hopefully, amid his old familiar haunts in the kingdom of Metaphysics. Here, indeed, it is clear that he _must_ have adventured in search of the treasure:—that he did not find it after all, was, perhaps, because his fairy guide, Imagination, was not sufficiently well-grown, or well-educated, to direct him aright.

I observed, just now, that, in fact, there had been certain vague attempts at referring Gravity to some very uncertain _isms_. These attempts, however, although considered bold and justly so considered, looked no farther than to the generality—the merest generality—of the Newtonian Law. Its _modus operandi_ has never, to my knowledge, been approached in the way of an effort at explanation. It is, therefore, with no unwarranted fear of being taken for a madman at the outset, and before I can bring my propositions fairly to the eye of those who alone are competent to decide upon them, that I here declare the _modus operandi_ of the Law of Gravity to be an exceedingly simple and perfectly explicable thing—that is to say, when we make our advances towards it in just gradations and in the true direction—when we regard it from the proper point of view.

Whether we reach the idea of absolute _Unity_ as the source of All Things, from a consideration of Simplicity as the most probable characteristic of the original action of God;—whether we arrive at it from an inspection of the universality of relation in the gravitating phænomena;—or whether we attain it as a result of the mutual corroboration afforded by both processes;—still, the idea itself, if entertained at all, is

entertained in inseparable connection with another idea—that of the condition of the Universe of stars as we _now_ perceive it—that is to say, a condition of immeasurable _diffusion_ through space. Now a connection between these two ideas—unity and diffusion—cannot be established unless through the entertainment of a third idea—that of _irradiation_. Absolute Unity being taken as a centre, then the existing Universe of stars is the result of _irradiation_ from that centre.

Now, the laws of irradiation are _known_. They are part and parcel of the _sphere_. They belong to the class of _indisputable geometrical properties_. We say of them, “they are true—they are evident.” To demand _why_ they are true, would be to demand why the axioms are true upon which their demonstration is based. _Nothing_ is demonstrable, strictly speaking; but _if_ anything _be_, then the properties—the laws in question are demonstrated.

But these laws—what do they declare? Irradiation—how—by what steps does it proceed outwardly from a centre?

From a _luminous_ centre, _Light_ issues by irradiation; and the quantities of light received upon any given plane, supposed to be shifting its position so as to be now nearer the centre and now farther from it, will be diminished in the same proportion as the squares of the distances of the plane from the luminous body, are increased; and will be increased in the same proportion as these squares are diminished.

The expression of the law may be thus generalized:—the number of light-particles (or, if the phrase be preferred, the number of light-impressions) received upon the shifting plane, will be _inversely_ proportional with the squares of the distances of the plane. Generalizing yet again, we may say that the diffusion—the scattering—the irradiation, in a word—is _directly_ proportional with the squares of the distances.

[Illustration]

For example: at the distance B, from the luminous centre A, a certain number of particles are so diffused as to occupy the surface B. Then at double the distance—that is to say at C—they will be so much farther diffused as to occupy four such surfaces:—at treble the distance, or at D, they will be so much farther separated as to occupy nine such surfaces:—while, at quadruple the distance, or at E, they will have become so scattered as to spread themselves over sixteen such surfaces—and so on forever.

In saying, generally, that the irradiation proceeds in direct proportion with the squares of the distances, we use the term irradiation to express _the degree of the diffusion_ as we proceed outwardly from the centre. Conversing the idea, and employing the word “concentralization” to express _the degree of the drawing together_ as we come back toward the centre from an outward position, we may say that concentralization proceeds _inversely_ as the squares of the distances. In other words, we have reached the conclusion that, on the hypothesis that matter was originally irradiated from a centre and is now returning to it, the concentralization, in the return, proceeds _exactly as we know the force of gravitation to proceed_.

Now here, if we could be permitted to assume that concentralization exactly represented the _force of the tendency to the centre_—that the one was exactly proportional to the other, and that the two proceeded together—we should have shown all that is required. The sole difficulty existing, then, is to establish a direct proportion between “concentralization” and the _force_ of concentralization; and this is done, of course, if we establish such proportion between “irradiation” and the _force_ of irradiation.

A very slight inspection of the Heavens assures us that the stars have a certain general uniformity, equability, or equidistance, of distribution through that region of space in which, collectively, and in a roughly globular form, they are situated:—this species of very general, rather than absolute, equability, being in full keeping with my deduction of inequidistance, within certain limits, among the originally diffused atoms, as a corollary from the evident design of infinite complexity of relation out of irrelation. I started, it will be remembered, with the idea of a generally uniform but particularly _un_uniform distribution of the atoms;—an idea, I repeat, which an inspection of the stars, as they exist, confirms.

But even in the merely general equability of distribution, as regards the atoms, there appears a difficulty which, no doubt, has already suggested itself to those among my readers who have borne in mind that I suppose this equability of distribution effected through _irradiation from a centre_. The very first glance at the idea, irradiation, forces us to the entertainment of the hitherto unseparated and seemingly inseparable idea of agglomeration about a centre, with dispersion as we recede from it—the idea, in a word, of _in_equability of distribution in respect to the matter irradiated.

Now, I have elsewhere[1] observed that it is by just such difficulties as the one now in question—such roughnesses—such peculiarities—such protuberances above the plane of the ordinary—that Reason feels her way, if at all, in her search for the True. By the difficulty—the “peculiarity”—now presented, I leap at once to _the_ secret—a secret which I might never have attained _but_ for the peculiarity and the inferences which, _in its mere character of peculiarity_, it affords me.

[1] “_Murders in the Rue Morgue_”—p. 133.

The process of thought, at this point, may be thus roughly sketched:—I say to myself—“Unity, as I have explained it, is a truth—I feel it. Diffusion is a truth—I see it. Irradiation, by which alone these two truths are reconciled, is a consequent truth—I perceive it. _Equability_ of diffusion, first deduced _à priori_ and then corroborated by the inspection of phænomena, is also a truth—I fully admit it. So far all is clear around me:—there are no clouds behind which _the_ secret—the great secret of the gravitating _modus operandi_—can possibly lie hidden;—but this secret lies _hereabouts_, most assuredly; and _were_ there but a cloud in view, I should be driven to suspicion of that cloud.” And now, just as I say this, there actually comes a cloud into view. This cloud is the seeming impossibility of reconciling my truth, _irradiation_, with my truth, _equability of diffusion_. I say now:—“Behind this _seeming_ impossibility is to be found what I desire.” I do not say “_real_

impossibility;” for invincible faith in my truths assures me that it is a mere difficulty after all—but I go on to say, with unflinching confidence, that, _when_ this _difficulty_ shall be solved, we shall find, _wrapped up in the process of solution_, the key to the secret at which we aim. Moreover—I _feel_ that we shall discover _but one_ possible solution of the difficulty; this for the reason that, were there two, one would be supererogatory—would be fruitless—would be empty—would contain no key—since no duplicate key can be needed to any secret of Nature.

And now, let us see:—Our usual notions of irradiation—in fact _all_ our distinct notions of it—are caught merely from the process as we see it exemplified in _Light_. Here there is a _continuous_ outpouring of _ray-streams_, and _with a force which we have at least no right to suppose varies at all_. Now, in any such irradiation _as this_—continuous and of unvarying force—the regions nearer the centre must _inevitably_ be always more crowded with the irradiated matter than the regions more remote. But I have assumed _no_ such irradiation _as this_. I assumed no _continuous_ irradiation; and for the simple reason that such an assumption would have involved, first, the necessity of entertaining a conception which I have shown no man _can_ entertain, and which (as I will more fully explain hereafter) all observation of the firmament refutes—the conception of the absolute infinity of the Universe of stars—and would have involved, secondly, the impossibility of understanding a rëaction—that is, gravitation—as existing now—since, while an act is continued, no rëaction, of course, can take place. My assumption, then, or rather my inevitable deduction from just premises—was that of a _determinate_ irradiation—one finally _dis_continued.

Let me now describe the sole possible mode in which it is conceivable that matter could have been diffused through space, so as to fulfil the conditions at once of irradiation and of generally equable distribution.

For convenience of illustration, let us imagine, in the first place, a hollow sphere of glass, or of anything else, occupying the space throughout which the universal matter is to be thus equally diffused, by means of irradiation, from the absolute, irrelative, unconditional particle, placed in the centre of the sphere.

Now, a certain exertion of the diffusive power (presumed to be the Divine Volition)—in other words, a certain _force_—whose measure is the quantity of matter—that is to say, the number of atoms—emitted; emits, by irradiation, this certain number of atoms; forcing them in all directions outwardly from the centre—their proximity to each other diminishing as they proceed—until, finally, they are distributed, loosely, over the interior surface of the sphere.

When these atoms have attained this position, or while proceeding to attain it, a second and inferior exercise of the same force—or a second and inferior force of the same character—emits, in the same manner—that is to say, by irradiation as before—a second stratum of atoms which proceeds to deposit itself upon the first; the number of atoms, in this case as in the former, being of course the measure of the force which emitted them; in other words the force being precisely adapted to the purpose it effects—the force and the number of atoms sent out by the force, being

directly proportional.

When this second stratum has reached its destined position—or while approaching it—a third still inferior exertion of the force, or a third inferior force of a similar character—the number of atoms emitted being in _all_ cases the measure of the force—proceeds to deposit a third stratum upon the second:—and so on, until these concentric strata, growing gradually less and less, come down at length to the central point; and the diffusive matter, simultaneously with the diffusive force, is exhausted.

We have now the sphere filled, through means of irradiation, with atoms equably diffused. The two necessary conditions—those of irradiation and of equable diffusion—are satisfied; and by the _sole_ process in which the possibility of their simultaneous satisfaction is conceivable. For this reason, I confidently expect to find, lurking in the present condition of the atoms as distributed throughout the sphere, the secret of which I am in search—the all-important principle of the _modus operandi_ of the Newtonian law. Let us examine, then, the actual condition of the atoms.

They lie in a series of concentric strata. They are equably diffused throughout the sphere. They have been irradiated into these states.

The atoms being _equably_ distributed, the greater the superficial extent of any of these concentric strata, or spheres, the more atoms will lie upon it. In other words, the number of atoms lying upon the surface of any one of the concentric spheres, is directly proportional with the extent of that surface.

But, in any series of concentric spheres, the surfaces are directly proportional with the squares of the distances from the centre.[2]

[2] Succinctly—The surfaces of spheres are as the squares of their radii.

Therefore the number of atoms in any stratum is directly proportional with the square of that stratum’s distance from the centre.

But the number of atoms in any stratum is the measure of the force which emitted that stratum—that is to say, is _directly proportional_ with the force.

Therefore the force which irradiated any stratum is directly proportional with the square of that stratum’s distance from the centre:—or, generally,

The force of the irradiation has been directly proportional with the squares of the distances.

Now, Rëaction, as far as we know anything of it, is Action conversed. The _general_ principle of Gravity being, in the first place, understood as the rëaction of an act—as the expression of a desire on the part of Matter, while existing in a state of diffusion, to return into the Unity whence it was diffused; and, in the second place, the mind being called upon to determine the _character_ of the desire—the manner in which it would, naturally, be manifested; in other words, being called upon to conceive a probable law, or _modus operandi_, for the return; could not well help arriving at the conclusion that this law of return would be precisely the converse of the law of departure. That such would be the case, any one, at least, would be abundantly

justified in taking for granted, until such time as some person should suggest something like a plausible reason why it should _not_ be the case—until such period as a law of return shall be imagined which the intellect can consider as preferable.

Matter, then, irradiated into space with a force varying as the squares of the distances, might, _à priori_, be supposed to return towards its centre of irradiation with a force varying _inversely_ as the squares of the distances: and I have already shown[3] that any principle which will explain why the atoms should tend, according to any law, to the general centre, must be admitted as satisfactorily explaining, at the same time, why, according to the same law, they should tend each to each. For, in fact, the tendency to the general centre is not to a centre as such, but because of its being a point in tending towards which each atom tends most directly to its real and essential centre, _Unity_—the absolute and final Union of all.

[3] Page 44.

The consideration here involved presents to my own mind no embarrassment whatever—but this fact does not blind me to the possibility of its being obscure to those who may have been less in the habit of dealing with abstractions:—and, upon the whole, it may be as well to look at the matter from one or two other points of view.

The absolute, irrelative particle primarily created by the Volition of God, must have been in a condition of positive _normality_, or rightfulness—for wrongfulness implies _relation_. Right is positive; wrong is negative—is merely the negation of right; as cold is the negation of heat—darkness of light. That a thing may be wrong, it is necessary that there be some other thing in _relation_ to which it _is_ wrong—some condition which it fails to satisfy; some law which it violates; some being whom it aggrieves. If there be no such being, law, or condition, in respect to which the thing is wrong—and, still more especially, if no beings, laws, or conditions exist at all—then the thing can_not_ be wrong and consequently must be _right_. Any deviation from normality involves a tendency to return into it. A difference from the normal—from the right—from the just—can be understood as effected only by the overcoming a difficulty; and if the force which overcomes the difficulty be not infinitely continued, the ineradicable tendency to return will at length be permitted to act for its own satisfaction. Upon withdrawal of the force, the tendency acts. This is the principle of rëaction as the inevitable consequence of finite action. Employing a phraseology of which the seeming affectation will be pardoned for its expressiveness, we may say that Rëaction is the return from the condition of _as it is and ought not to be_ into the condition of _as it was, originally, and therefore ought to be_:—and let me add here that the _absolute_ force of Rëaction would no doubt be always found in direct proportion with the reality—the truth—the absoluteness—of the _originality_—if ever it were possible to measure this latter:—and, consequently, the greatest of all conceivable reactions must be that produced by the tendency which we now discuss—the tendency to return into the _absolutely original_—into the _supremely_ primitive. Gravity, then, _must be the strongest of forces_—an idea reached _à priori_ and abundantly confirmed by induction. What use I make of the idea, will be seen in the sequel.

The atoms, now, having been diffused from their normal condition of Unity, seek to return to——what? Not to any particular _point_, certainly; for it is clear that if, upon the diffusion, the whole Universe of matter had been projected, collectively, to a distance from the point of irradiation, the atomic tendency to the general centre of the sphere would not have been disturbed in the least:—the atoms would not have sought the point _in absolute space_ from which they were originally impelled. It is merely the _condition_, and not the point or locality at which this condition took its rise, that these atoms seek to re-establish;—it is merely _that condition which is their normality_, that they desire. “But they seek a centre,” it will be said, “and a centre is a point.” True; but they seek this point not in its character of point—(for, were the whole sphere moved from its position, they would seek, equally, the centre; and the centre _then_ would be a _new_ point)—but because it so happens, on account of the form in which they collectively exist—(that of the sphere)—that only _through_ the point in question—the sphere’s centre—they can attain their true object, Unity. In the direction of the centre each atom perceives more atoms than in any other direction. Each atom is impelled towards the centre because along the straight line joining it and the centre and passing on to the circumference beyond, there lie a greater number of atoms than along any other straight line—a greater number of objects that seek it, the individual atom—a greater number of tendencies to Unity—a greater number of satisfactions for its own tendency to Unity—in a word, because in the direction of the centre lies the utmost possibility of satisfaction, generally, for its own individual appetite. To be brief, the _condition_, Unity, is all that is really sought; and if the atoms _seem_ to seek the centre of the sphere, it is only impliedly, through implication—because such centre happens to imply, to include, or to involve, the only essential centre, Unity. But _on account of_ this implication or involution, there is no possibility of practically separating the tendency to Unity in the abstract, from the tendency to the concrete centre. Thus the tendency of the atoms to the general centre _is_, to all practical intents and for all logical purposes, the tendency each to each; and the tendency each to each _is_ the tendency to the centre; and the one tendency may be assumed _as_ the other; whatever will apply to the one must be thoroughly applicable to the other; and, in conclusion, whatever principle will satisfactorily explain the one, cannot be questioned as an explanation of the other.

In looking carefully around me for rational objection to what I have advanced, I am able to discover _nothing_;—but of that class of objections usually urged by the doubters for Doubt’s sake, I very readily perceive _three_; and proceed to dispose of them in order.

It may be said, first: “The proof that the force of irradiation (in the case described) is directly proportional to the squares of the distances, depends upon an unwarranted assumption—that of the number of atoms in each stratum being the measure of the force with which they are emitted.”

I reply, not only that I am warranted in such assumption, but that I should be utterly _un_warranted in any other. What I assume is, simply, that an effect is the measure of its cause—that every exercise of the Divine Will will be proportional to that which demands the exertion—that the means of Omnipotence, or of

Omniscience, will be exactly adapted to its purposes. Neither can a deficiency nor an excess of cause bring to pass any effect. Had the force which irradiated any stratum to its position, been either more or less than was needed for the purpose—that is to say, not _directly proportional_ to the purpose—then to its position that stratum could not have been irradiated. Had the force which, with a view to general equability of distribution, emitted the proper number of atoms for each stratum, been not _directly proportional_ to the number, then the number would _not_ have been the number demanded for the equable distribution.

The second supposable objection is somewhat better entitled to an answer.

It is an admitted principle in Dynamics that every body, on receiving an impulse, or disposition to move, will move onward in a straight line, in the direction imparted by the impelling force, until deflected, or stopped, by some other force. How then, it may be asked, is my first or external stratum of atoms to be understood as discontinuing their movement at the circumference of the imaginary glass sphere, when no second force, of more than an imaginary character, appears, to account for the discontinuance?

I reply that the objection, in this case, actually does arise out of “an unwarranted assumption”—on the part of the objector—the assumption of a principle, in Dynamics, at an epoch when _no_ “principles,” in _anything_, exist:—I use the word “principle,” of course, in the objector’s understanding of the word.

“In the beginning” we can admit—indeed we can comprehend—but one _First Cause_—the truly ultimate _Principle_—the Volition of God. The primary _act_—that of Irradiation from Unity—must have been independent of all that which the world now calls “principle”—because all that we so designate is but a consequence of the rëaction of that primary act:—I say “_primary_” act; for the creation of the absolute material particle is more properly to be regarded as a _conception_ than as an “_act_” in the ordinary meaning of the term. Thus, we must regard the primary act as an act for the establishment of what we now call “principles.” But this primary act itself is to be considered as _continuous Volition_. The Thought of God is to be understood as originating the Diffusion—as proceeding with it—as regulating it—and, finally, as being withdrawn from it upon its completion. _Then_ commences Rëaction, and through Rëaction, “Principle,” as we employ the word. It will be advisable, however, to limit the application of this word to the two _immediate_ results of the discontinuance of the Divine Volition—that is, to the two agents, _Attraction_ and _Repulsion_. Every other Natural agent depends, either more or less immediately, upon these two, and therefore would be more conveniently designated as _sub_-principle.

It may be objected, thirdly, that, in general, the peculiar mode of distribution which I have suggested for the atoms, is “an hypothesis and nothing more.”

Now, I am aware that the word hypothesis is a ponderous sledge-hammer, grasped immediately, if not lifted, by all very diminutive thinkers, upon the first appearance of any proposition wearing, in any particular, the garb of _a theory_. But “hypothesis” cannot be wielded _here_ to any good purpose, even by those who succeed in lifting

it—little men or great.

I maintain, first, that _only_ in the mode described is it conceivable that Matter could have been diffused so as to fulfil at once the conditions of irradiation and of generally equable distribution. I maintain, secondly, that these conditions themselves have been imposed upon me, as necessities, in a train of ratiocination _as rigorously logical as that which establishes any demonstration in Euclid_; and I maintain, thirdly, that even if the charge of “hypothesis” were as fully sustained as it is, in fact, unsustained and untenable, still the validity and indisputability of my result would not, even in the slightest particular, be disturbed.

To explain:—The Newtonian Gravity—a law of Nature—a law whose existence as such no one out of Bedlam questions—a law whose admission as such enables us to account for nine-tenths of the Universal phænomena—a law which, merely because it does so enable us to account for these phænomena, we are perfectly willing, without reference to any other considerations, to admit, and cannot help admitting, as a law—a law, nevertheless, of which neither the principle nor the _modus operandi_ of the principle, has ever yet been traced by the human analysis—a law, in short, which, neither in its detail nor in its generality, has been found susceptible of explanation _at all_—is at length seen to be at every point thoroughly explicable, provided only we yield our assent to——what? To an hypothesis? Why _if_ an hypothesis—if the merest hypothesis—if an hypothesis for whose assumption—as in the case of that _pure_ hypothesis the Newtonian law itself—no shadow of _à priori_ reason could be assigned—if an hypothesis, even so absolute as all this implies, would enable us to perceive a principle for the Newtonian law—would enable us to understand as satisfied, conditions so miraculously—so ineffably complex and seemingly irreconcileable as those involved in the relations of which Gravity tells us,—what rational being _could_ so expose his fatuity as to call even this absolute hypothesis an hypothesis any longer—unless, indeed, he were to persist in so calling it, with the understanding that he did so, simply for the sake of consistency _in words_?

But what is the true state of our present case? What is _the fact_? Not only that it is _not_ an hypothesis which we are required _to adopt_, in order to admit the principle at issue explained, but that it _is_ a logical conclusion which we are requested _not_ to adopt if we can avoid it—which we are simply invited to _deny if we can_:—a conclusion of so accurate a logicality that to dispute it would be the effort—to doubt its validity beyond our power:—a conclusion from which we see no mode of escape, turn as we will; a result which confronts us either at the end of an _in_ductive journey from the phænomena of the very Law discussed, or at the close of a _de_ductive career from the most rigorously simple of all conceivable assumptions—_the assumption, in a word, of Simplicity itself_.

And if here, for the mere sake of cavilling, it be urged, that although my starting-point is, as I assert, the assumption of absolute Simplicity, yet Simplicity, considered merely in itself, is no axiom; and that only deductions from axioms are indisputable—it is thus that I reply:—

Every other science than Logic is the science of certain concrete relations. Arithmetic, for example, is the science of the relations of number—Geometry, of the relations of form—Mathematics in general, of the relations of quantity in general—of whatever can be increased or diminished. Logic, however, is the science of Relation in the abstract—of absolute Relation—of Relation considered solely in itself. An axiom in any particular science other than Logic is, thus, merely a proposition announcing certain concrete relations which seem to be too obvious for dispute—as when we say, for instance, that the whole is greater than its part:—and, thus again, the principle of the _Logical_ axiom—in other words, of an axiom in the abstract—is, simply, _obviousness of relation_. Now, it is clear, not only that what is obvious to one mind may not be obvious to another, but that what is obvious to one mind at one epoch, may be anything but obvious, at another epoch, to the same mind. It is clear, moreover, that what, to-day, is obvious even to the majority of mankind, or to the majority of the best intellects of mankind, may to-morrow be, to either majority, more or less obvious, or in no respect obvious at all. It is seen, then, that the _axiomatic principle_ itself is susceptible of variation, and of course that axioms are susceptible of similar change. Being mutable, the “truths” which grow out of them are necessarily mutable too; or, in other words, are never to be positively depended upon as truths at all—since Truth and Immutability are one.

It will now be readily understood that no axiomatic idea—no idea founded in the fluctuating principle, obviousness of relation—can possibly be so secure—so reliable a basis for any structure erected by the Reason, as _that_ idea—(whatever it is, wherever we can find it, or _if_ it be practicable to find it anywhere)—which is _ir_relative altogether—which not only presents to the understanding _no obviousness_ of relation, either greater or less, to be considered, but subjects the intellect, not in the slightest degree, to the necessity of even looking at _any relation at all_. If such an idea be not what we too heedlessly term “an axiom,” it is at least preferable, as a Logical basis, to any axiom ever propounded, or to all imaginable axioms combined:—and such, precisely, is the idea with which my deductive process, so thoroughly corroborated by induction, commences. My _particle proper_ is but _absolute Irrelation_. To sum up what has been here advanced:—As a starting point I have taken it for granted, simply, that the Beginning had nothing behind it or before it—that it was a Beginning in fact—that it was a beginning and nothing different from a beginning—in short that this Beginning was——_that which it was_. If this be a “mere assumption” then a “mere assumption” let it be.

To conclude this branch of the subject:—I am fully warranted in announcing that _the Law which we have been in the habit of calling Gravity exists on account of Matter’s having been irradiated, at its origin, atomically, into a limited[4] sphere of Space, from one, individual, unconditional, irrelative, and absolute Particle Proper, by the sole process in which it was possible to satisfy, at the same time, the two conditions, irradiation, and generally-equable distribution throughout the sphere—that is to say, by a force varying in direct proportion with the squares of the distances between the irradiated atoms, respectively, and the Particular centre of Irradiation_.

[4] Limited sphere—A sphere is _necessarily_ limited. I prefer tautology to a chance of misconception.

I have already given my reasons for presuming Matter to have been diffused by a determinate rather than by a continuous or infinitely continued force. Supposing a continuous force, we should be unable, in the first place, to comprehend a rëaction at all; and we should be required, in the second place, to entertain the impossible conception of an infinite extension of Matter. Not to dwell upon the impossibility of the conception, the infinite extension of Matter is an idea which, if not positively disproved, is at least not in any respect warranted by telescopic observation of the stars—a point to be explained more fully hereafter; and this empirical reason for believing in the original finity of Matter is unempirically confirmed. For example:—Admitting, for the moment, the possibility of understanding Space _filled_ with the irradiated atoms—that is to say, admitting, as well as we can, for argument's sake, that the succession of the irradiated atoms had absolutely _no end_—then it is abundantly clear that, even when the Volition of God had been withdrawn from them, and thus the tendency to return into Unity permitted (abstractly) to be satisfied, this permission would have been nugatory and invalid—practically valueless and of no effect whatever. No Rëaction could have taken place; no movement toward Unity could have been made; no Law of Gravity could have obtained.

To explain:—Grant the _abstract_ tendency of any one atom to any one other as the inevitable result of diffusion from the normal Unity:—or, what is the same thing, admit any given atom as _proposing_ to move in any given direction—it is clear that, since there is an _infinity_ of atoms on all sides of the atom proposing to move, it never can actually move toward the satisfaction of its tendency in the direction given, on account of a precisely equal and counterbalancing tendency in the direction diametrically opposite. In other words, exactly as many tendencies to Unity are behind the hesitating atom as before it; for it is a mere sotticism to say that one infinite line is longer or shorter than another infinite line, or that one infinite number is greater or less than another number that is infinite. Thus the atom in question must remain stationary forever. Under the impossible circumstances which we have been merely endeavoring to conceive for argument's sake, there could have been no aggregation of Matter—no stars—no worlds—nothing but a perpetually atomic and inconsequential Universe. In fact, view it as we will, the whole idea of unlimited Matter is not only untenable, but impossible and preposterous.

With the understanding of a _sphere_ of atoms, however, we perceive, at once, a _satisfiable_ tendency to union. The general result of the tendency each to each, being a tendency of all to the centre, the _general_ process of condensation, or approximation, commences immediately, by a common and simultaneous movement, on withdrawal of the Divine Volition; the _individual_ approximations, or coalescences—_not_ cöalitions—of atom with atom, being subject to almost infinite variations of time, degree, and condition, on account of the excessive multiplicity of relation, arising from the differences of form assumed as characterizing the atoms at the moment of their quitting the Particle Proper; as well as from the subsequent particular inequidistance, each from each.

What I wish to impress upon the reader is the certainty of there arising, at once, (on withdrawal of the diffusive force, or Divine Volition,) out of the condition of the atoms as described, at innumerable points throughout the Universal sphere, innumerable agglomerations, characterized by innumerable specific differences of form, size, essential nature, and distance each from each. The development of Repulsion (Electricity) must have commenced, of course, with the very earliest particular efforts at Unity, and must have proceeded constantly in the ratio of Coalescence—that is to say, _in that of Condensation_, or, again, of Heterogeneity.

Thus the two Principles Proper, _Attraction_ and _Repulsion_—the Material and the Spiritual—accompany each other, in the strictest fellowship, forever. Thus _The Body and The Soul walk hand in hand_.

If now, in fancy, we select _any one_ of the agglomerations considered as in their primary stages throughout the Universal sphere, and suppose this incipient agglomeration to be taking place at that point where the centre of our Sun exists—or rather where it _did_ exist originally; for the Sun is perpetually shifting his position—we shall find ourselves met, and borne onward for a time at least, by the most magnificent of theories—by the Nebular Cosmogony of Laplace:—although “Cosmogony” is far too comprehensive a term for what he really discusses—which is the constitution of our solar system alone—of one among the myriad of similar systems which make up the Universe Proper—that Universal sphere—that all-inclusive and absolute _Kosmos_ which forms the subject of my present Discourse.

Confining himself to an _obviously limited_ region—that of our solar system with its comparatively immediate vicinity—and _merely_ assuming—that is to say, assuming without any basis whatever, either deductive or inductive—_much_ of what I have been just endeavoring to place upon a more stable basis than assumption; assuming, for example, matter as diffused (without pretending to account for the diffusion) throughout, and somewhat beyond, the space occupied by our system—diffused in a state of heterogeneous nebulosity and obedient to that omniprevalent law of Gravity at whose principle he ventured to make no guess;—assuming all this (which is quite true, although he had no logical right to its assumption) Laplace has shown, dynamically and mathematically, that the results in such case necessarily ensuing, are those and those alone which we find manifested in the actually existing condition of the system itself.

To explain:—Let us conceive _that_ particular agglomeration of which we have just spoken—the one at the point designated by our Sun’s centre—to have so far proceeded that a vast quantity of nebulous matter has here assumed a roughly globular form; its centre being, of course, coincident with what is now, or rather was originally, the centre of our Sun; and its periphery extending out beyond the orbit of Neptune, the most remote of our planets:—in other words, let us suppose the diameter of this rough sphere to be some 6000 millions of miles. For ages, this mass of matter has been undergoing condensation, until at length it has become reduced into the bulk we imagine; having proceeded gradually, of course, from its atomic and imperceptible state, into what we understand of visible, palpable, or otherwise appreciable nebulosity.

Now, the condition of this mass implies a rotation about an imaginary axis—a rotation which, commencing with the absolute incipiency of the aggregation, has been ever since acquiring velocity. The very first two atoms which met, approaching each other from points not diametrically opposite, would, in rushing partially past each other, form a nucleus for the rotary movement described. How this would increase in velocity, is readily seen. The two atoms are joined by others:—an aggregation is formed. The mass continues to rotate while condensing. But any atom at the circumference has, of course, a more rapid motion than one nearer the centre. The outer atom, however, with its superior velocity, approaches the centre; carrying this superior velocity with it as it goes. Thus every atom, proceeding inwardly, and finally attaching itself to the condensed centre, adds something to the original velocity of that centre—that is to say, increases the rotary movement of the mass.

Let us now suppose this mass so far condensed that it occupies _precisely_ the space circumscribed by the orbit of Neptune, and that the velocity with which the surface of the mass moves, in the general rotation, is precisely that velocity with which Neptune now revolves about the Sun. At this epoch, then, we are to understand that the constantly increasing centrifugal force, having gotten the better of the non-increasing centripetal, loosened and separated the exterior and least condensed stratum, or a few of the exterior and least condensed strata, at the equator of the sphere, where the tangential velocity predominated; so that these strata formed about the main body an independent ring encircling the equatorial regions:—just as the exterior portion thrown off, by excessive velocity of rotation, from a grindstone, would form a ring about the grindstone, but for the solidity of the superficial material: were this caoutchouc, or anything similar in consistency, precisely the phænomenon I describe would be presented.

The ring thus whirled from the nebulous mass, _revolved_, of course, _as_ a separate ring, with just that velocity with which, while the surface of the mass, it _rotated_. In the meantime, condensation still proceeding, the interval between the discharged ring and the main body continued to increase, until the former was left at a vast distance from the latter.

Now, admitting the ring to have possessed, by some seemingly accidental arrangement of its heterogeneous materials, a constitution nearly uniform, then this ring, _as_ such, would never have ceased revolving about its primary; but, as might have been anticipated, there appears to have been enough irregularity in the disposition of the materials, to make them cluster about centres of superior solidity; and thus the annular form was destroyed.[5] No doubt, the band was soon broken up into several portions, and one of these portions, predominating in mass, absorbed the others into itself; the whole settling, spherically, into a planet. That this latter, _as_ a planet, continued the revolutionary movement which characterized it while a ring, is sufficiently clear; and that it took upon itself also, an additional movement in its new condition of sphere, is readily explained. The ring being understood as yet unbroken, we see that its exterior, while the whole revolves about the parent body, moves more rapidly than its interior. When the rupture occurred, then, some portion in each fragment must have been moving with greater velocity than the others. The superior

movement prevailing, must have whirled each fragment round—that is to say, have caused it to rotate; and the direction of the rotation must, of course, have been the direction of the revolution whence it arose. _All_ the fragments having become subject to the rotation described, must, in coalescing, have imparted it to the one planet constituted by their coalescence.—This planet was Neptune. Its material continuing to undergo condensation, and the centrifugal force generated in its rotation getting, at length, the better of the centripetal, as before in the case of the parent orb, a ring was whirled also from the equatorial surface of this planet: this ring, having been ununiform in its constitution, was broken up, and its several fragments, being absorbed by the most massive, were collectively spherified into a moon. Subsequently, the operation was repeated, and a second moon was the result. We thus account for the planet Neptune, with the two satellites which accompany him.

[5] Laplace assumed his nebulosity heterogeneous, merely that he might be thus enabled to account for the breaking up of the rings; for had the nebulosity been homogeneous, they would not have broken. I reach the same result—heterogeneity of the secondary masses immediately resulting from the atoms—purely from an _à priori_ consideration of their general design—_Relation_.

In throwing off a ring from its equator, the Sun re-established that equilibrium between its centripetal and centrifugal forces which had been disturbed in the process of condensation; but, as this condensation still proceeded, the equilibrium was again immediately disturbed, through the increase of rotation. By the time the mass had so far shrunk that it occupied a spherical space just that circumscribed by the orbit of Uranus, we are to understand that the centrifugal force had so far obtained the ascendency that new relief was needed: a second equatorial band was, consequently, thrown off, which, proving ununiform, was broken up, as before in the case of Neptune; the fragments settling into the planet Uranus; the velocity of whose actual revolution about the Sun indicates, of course, the rotary speed of that Sun’s equatorial surface at the moment of the separation. Uranus, adopting a rotation from the collective rotations of the fragments composing it, as previously explained, now threw off ring after ring; each of which, becoming broken up, settled into a moon:—three moons, at different epochs, having been formed, in this manner, by the rupture and general spherification of as many distinct ununiform rings.

By the time the Sun had shrunk until it occupied a space just that circumscribed by the orbit of Saturn, the balance, we are to suppose, between its centripetal and centrifugal forces had again become so far disturbed, through increase of rotary velocity, the result of condensation, that a third effort at equilibrium became necessary; and an annular band was therefore whirled off as twice before; which, on rupture through ununiformity, became consolidated into the planet Saturn. This latter threw off, in the first place, seven uniform bands, which, on rupture, were spherified respectively into as many moons; but, subsequently, it appears to have discharged, at three distinct but not very distant epochs, three rings whose equability of constitution was, by apparent accident, so considerable as to present no occasion for their rupture; thus they continue to revolve as rings. I use the phrase “_apparent_

accident;” for of accident in the ordinary sense there was, of course, nothing:—the term is properly applied only to the result of indistinguishable or not immediately traceable _law_.

Shrinking still farther, until it occupied just the space circumscribed by the orbit of Jupiter, the Sun now found need of farther effort to restore the counterbalance of its two forces, continually disarranged in the still continued increase of rotation. Jupiter, accordingly, was now thrown off; passing from the annular to the planetary condition; and, on attaining this latter, threw off in its turn, at four different epochs, four rings, which finally resolved themselves into so many moons.

Still shrinking, until its sphere occupied just the space defined by the orbit of the Asteroids, the Sun now discarded a ring which appears to have had _eight_ centres of superior solidity, and, on breaking up, to have separated into eight fragments no one of which so far predominated in mass as to absorb the others. All therefore, as distinct although comparatively small planets, proceeded to revolve in orbits whose distances, each from each, may be considered as in some degree the measure of the force which drove them asunder:—all the orbits, nevertheless, being so closely coincident as to admit of our calling them _one_, in view of the other planetary orbits.

Continuing to shrink, the Sun, on becoming so small as just to fill the orbit of Mars, now discharged this planet—of course by the process repeatedly described. Having no moon, however, Mars could have thrown off no ring. In fact, an epoch had now arrived in the career of the parent body, the centre of the system. The _de_crease of its nebulosity, which is the _in_crease of its density, and which again is the _de_crease of its condensation, out of which latter arose the constant disturbance of equilibrium—must, by this period, have attained a point at which the efforts for restoration would have been more and more ineffectual just in proportion as they were less frequently needed. Thus the processes of which we have been speaking would everywhere show signs of exhaustion—in the planets, first, and secondly, in the original mass. We must not fall into the error of supposing the decrease of interval observed among the planets as we approach the Sun, to be in any respect indicative of an increase of frequency in the periods at which they were discarded. Exactly the converse is to be understood. The longest interval of time must have occurred between the discharges of the two interior; the shortest, between those of the two exterior, planets. The decrease of the interval of space is, nevertheless, the measure of the density, and thus inversely of the condensation, of the Sun, throughout the processes detailed.

Having shrunk, however, so far as to fill only the orbit of our Earth, the parent sphere whirled from itself still one other body—the Earth—in a condition so nebulous as to admit of this body’s discarding, in its turn, yet another, which is our Moon;—but here terminated the lunar formations.

Finally, subsiding to the orbits first of Venus and then of Mercury, the Sun discarded these two interior planets; neither of which has given birth to any moon.

Thus from his original bulk—or, to speak more accurately, from the condition in which we first considered him—from a partially spherified nebular mass, _certainly_

much more than 5,600 millions of miles in diameter—the great central orb and origin of our solar-planetary-lunar system, has gradually descended, by condensation, in obedience to the law of Gravity, to a globe only 882,000 miles in diameter; but it by no means follows, either that its condensation is yet complete, or that it may not still possess the capacity of whirling from itself another planet.

I have here given—in outline of course, but still with all the detail necessary for distinctness—a view of the Nebular Theory as its author himself conceived it. From whatever point we regard it, we shall find it _beautifully true_. It is by far too beautiful, indeed, _not_ to possess Truth as its essentiality—and here I am very profoundly serious in what I say. In the revolution of the satellites of Uranus, there does appear something seemingly inconsistent with the assumptions of Laplace; but that _one_ inconsistency can invalidate a theory constructed from a million of intricate consistencies, is a fancy fit only for the fantastic. In prophecying, confidently, that the apparent anomaly to which I refer, will, sooner or later, be found one of the strongest possible corroborations of the general hypothesis, I pretend to no especial spirit of divination. It is a matter which the only difficulty seems _not_ to foresee.[6]

[6] I am prepared to show that the anomalous revolution of the satellites of Uranus is a simply perspective anomaly arising from the inclination of the axis of the planet.

The bodies whirled off in the processes described, would exchange, it has been seen, the superficial _rotation_ of the orbs whence they originated, for a _revolution_ of equal velocity about these orbs as distant centres; and the revolution thus engendered must proceed, so long as the centripetal force, or that with which the discarded body gravitates toward its parent, is neither greater nor less than that by which it was discarded; that is, than the centrifugal, or, far more properly, than the tangential, velocity. From the unity, however, of the origin of these two forces, we might have expected to find them as they are found—the one accurately counterbalancing the other. It has been shown, indeed, that the act of whirling-off is, in every case, merely an act for the preservation of the counterbalance.

After referring, however, the centripetal force to the omniprevalent law of Gravity, it has been the fashion with astronomical treatises, to seek beyond the limits of mere Nature—that is to say, of _Secondary_ Cause—a solution of the phænomenon of tangential velocity. This latter they attribute directly to a _First_ Cause—to God. The force which carries a stellar body around its primary they assert to have originated in an impulse given immediately by the finger—this is the childish phraseology employed—by the finger of Deity itself. In this view, the planets, fully formed, are conceived to have been hurled from the Divine hand, to a position in the vicinity of the suns, with an impetus mathematically adapted to the masses, or attractive capacities, of the suns themselves. An idea so grossly unphilosophical, although so supinely adopted, could have arisen only from the difficulty of otherwise accounting for the absolutely accurate adaptation, each to each, of two forces so seemingly independent, one of the other, as are the gravitating and tangential. But it should be remembered that, for a long time, the coincidence between the moon’s rotation and her sidereal revolution—two matters seemingly far more independent than those now

considered—was looked upon as positively miraculous; and there was a strong disposition, even among astronomers, to attribute the marvel to the direct and continual agency of God—who, in this case, it was said, had found it necessary to interpose, specially, among his general laws, a set of subsidiary regulations, for the purpose of forever concealing from mortal eyes the glories, or perhaps the horrors, of the other side of the Moon—of that mysterious hemisphere which has always avoided, and must perpetually avoid, the telescopic scrutiny of mankind. The advance of Science, however, soon demonstrated—what to the philosophical instinct needed _no_ demonstration—that the one movement is but a portion—something more, even, than a consequence—of the other.

For my part, I have no patience with fantasies at once so timorous, so idle, and so awkward. They belong to the veriest _cowardice_ of thought. That Nature and the God of Nature are distinct, no thinking being can long doubt. By the former we imply merely the laws of the latter. But with the very idea of God, omnipotent, omniscient, we entertain, also, the idea of _the infallibility_ of his laws. With Him there being neither Past nor Future—with Him all being _Now_—do we not insult him in supposing his laws so contrived as not to provide for every possible contingency?—or, rather, what idea _can_ we have of _any_ possible contingency, except that it is at once a result and a manifestation of his laws? He who, divesting himself of prejudice, shall have the rare courage to think absolutely for himself, cannot fail to arrive, in the end, at the condensation of _laws_ into _Law_—cannot fail of reaching the conclusion that _each law of Nature is dependent at all points upon all other laws_, and that all are but consequences of one primary exercise of the Divine Volition. Such is the principle of the Cosmogony which, with all necessary deference, I here venture to suggest and to maintain.

In this view, it will be seen that, dismissing as frivolous, and even impious, the fancy of the tangential force having been imparted to the planets immediately by “the finger of God,” I consider this force as originating in the rotation of the stars:—this rotation as brought about by the in-rushing of the primary atoms, towards their respective centres of aggregation:—this in-rushing as the consequence of the law of Gravity:—this law as but the mode in which is necessarily manifested the tendency of the atoms to return into imparticularity:—this tendency to return as but the inevitable rëaction of the first and most sublime of Acts—that act by which a God, self-existing and alone existing, became all things at once, through dint of his volition, while all things were thus constituted a portion of God.

The radical assumptions of this Discourse suggest to me, and in fact imply, certain important _modifications_ of the Nebular Theory as given by Laplace. The efforts of the repulsive power I have considered as made for the purpose of preventing contact among the atoms, and thus as made in the ratio of the approach to contact—that is to say, in the ratio of condensation.[7] In other words, _Electricity_, with its involute phænomena, heat, light and magnetism, is to be understood as proceeding as condensation proceeds, and, of course, inversely as density proceeds, or the _cessation to condense_. Thus the Sun, in the process of its aggregation, must soon, in developing repulsion, have become excessively heated—perhaps incandescent: and

we can perceive how the operation of discarding its rings must have been materially assisted by the slight incrustation of its surface consequent on cooling. Any common experiment shows us how readily a crust of the character suggested, is separated, through heterogeneity, from the interior mass. But, on every successive rejection of the crust, the new surface would appear incandescent as before; and the period at which it would again become so far encrusted as to be readily loosened and discharged, may well be imagined as exactly coincident with that at which a new effort would be needed, by the whole mass, to restore the equilibrium of its two forces, disarranged through condensation. In other words:—by the time the electric influence (Repulsion) has prepared the surface for rejection, we are to understand that the gravitating influence (Attraction) is precisely ready to reject it. Here, then, as everywhere, _the Body and the Soul walk hand in hand_.

[7] See page 70.

These ideas are empirically confirmed at all points. Since condensation can never, in any body, be considered as absolutely at an end, we are warranted in anticipating that, whenever we have an opportunity of testing the matter, we shall find indications of resident luminosity in _all_ the stellar bodies—moons and planets as well as suns. That our Moon is strongly self-luminous, we see at her every total eclipse, when, if not so, she would disappear. On the dark part of the satellite, too, during her phases, we often observe flashes like our own Auroras; and that these latter, with our various other so-called electrical phænomena, without reference to any more steady radiance, must give our Earth a certain appearance of luminosity to an inhabitant of the Moon, is quite evident. In fact, we should regard all the phænomena referred to, as mere manifestations, in different moods and degrees, of the Earth’s feebly-continued condensation.

If my views are tenable, we should be prepared to find the newer planets—that is to say, those nearer the Sun—more luminous than those older and more remote:—and the extreme brilliancy of Venus (on whose dark portions, during her phases, the Auroras are frequently visible) does not seem to be altogether accounted for by her mere proximity to the central orb. She is no doubt vividly self-luminous, although less so than Mercury: while the luminosity of Neptune may be comparatively nothing.

Admitting what I have urged, it is clear that, from the moment of the Sun’s discarding a ring, there must be a continuous diminution both of his heat and light, on account of the continuous encrustation of his surface; and that a period would arrive—the period immediately previous to a new discharge—when a _very material_ decrease of both light and heat, must become apparent. Now, we know that tokens of such changes are distinctly recognizable. On the Melville islands—to adduce merely one out of a hundred examples—we find traces of _ultra-tropical_ vegetation—of plants that never could have flourished without immensely more light and heat than are at present afforded by our Sun to any portion of the surface of the Earth. Is such vegetation referable to an epoch immediately subsequent to the whirling-off of Venus? At this epoch must have occurred to us our greatest access of solar influence; and, in fact, this influence must then have attained its maximum:—leaving out of view, of course, the period when the Earth itself was discarded—the period of its

mere organization.

Again:—we know that there exist _non-luminous suns_—that is to say, suns whose existence we determine through the movements of others, but whose luminosity is not sufficient to impress us. Are these suns invisible merely on account of the length of time elapsed since their discharge of a planet? And yet again:—may we not—at least in certain cases—account for the sudden appearances of suns where none had been previously suspected, by the hypothesis that, having rolled with encrusted surfaces throughout the few thousand years of our astronomical history, each of these suns, in whirling off a new secondary, has at length been enabled to display the glories of its still incandescent interior?—To the well-ascertained fact of the proportional increase of heat as we descend into the Earth, I need of course, do nothing more than refer:—it comes in the strongest possible corroboration of all that I have said on the topic now at issue.

In speaking, not long ago, of the repulsive or electrical influence, I remarked that “the important phænomena of vitality, consciousness, and thought, whether we observe them generally or in detail, seem to proceed _at least in the ratio of the heterogeneous_.”[8] I mentioned, too, that I would recur to the suggestion:—and this is the proper point at which to do so. Looking at the matter, first, in detail, we perceive that not merely the _manifestation_ of vitality, but its importance, consequence, and elevation of character, keep pace, very closely, with the heterogeneity, or complexity, of the animal structure. Looking at the question, now, in its generality, and referring to the first movements of the atoms towards mass-constitution, we find that heterogeneousness, brought about directly through condensation, is proportional with it forever. We thus reach the proposition that _the importance of the development of the terrestrial vitality proceeds equably with the terrestrial condensation_.

[8] Page 36.

Now this is in precise accordance with what we know of the succession of animals on the Earth. As it has proceeded in its condensation, superior and still superior races have appeared. Is it impossible that the successive geological revolutions which have attended, at least, if not immediately caused, these successive elevations of vitalic character—is it improbable that these revolutions have themselves been produced by the successive planetary discharges from the Sun—in other words, by the successive variations in the solar influence on the Earth? Were this idea tenable, we should not be unwarranted in the fancy that the discharge of yet a new planet, interior to Mercury, may give rise to yet a new modification of the terrestrial surface—a modification from which may spring a race both materially and spiritually superior to Man. These thoughts impress me with all the force of truth—but I throw them out, of course, merely in their obvious character of suggestion.

The Nebular Theory of Laplace has lately received far more confirmation than it needed, at the hands of the philosopher, Compte. These two have thus together shown—_not_, to be sure, that Matter at any period actually existed as described, in a state of nebular diffusion, but that, admitting it so to have existed throughout the

space and much beyond the space now occupied by our solar system, _and to have commenced a movement towards a centre_—it must gradually have assumed the various forms and motions which are now seen, in that system, to obtain. A demonstration such as this—a dynamical and mathematical demonstration, as far as demonstration can be—unquestionable and unquestioned—unless, indeed, by that unprofitable and disreputable tribe, the professional questioners—the mere madmen who deny the Newtonian law of Gravity on which the results of the French mathematicians are based—a demonstration, I say, such as this, would to most intellects be conclusive—and I confess that it is so to mine—of the validity of the nebular hypothesis upon which the demonstration depends.

That the demonstration does not _prove_ the hypothesis, according to the common understanding of the word “proof,” I admit, of course. To show that certain existing results—that certain established facts—may be, even mathematically, accounted for by the assumption of a certain hypothesis, is by no means to establish the hypothesis itself. In other words:—to show that, certain data being given, a certain existing result might, or even _must_, have ensued, will fail to prove that this result _did_ ensue, _from the data_, until such time as it shall be also shown that there are, _and can be_, no other data from which the result in question might _equally_ have ensued. But, in the case now discussed, although all must admit the deficiency of what we are in the habit of terming “proof,” still there are many intellects, and those of the loftiest order, to which _no_ proof could bring one iota of additional _conviction_. Without going into details which might impinge upon the Cloud-Land of Metaphysics, I may as well here observe that the force of conviction, in cases such as this, will always, with the right-thinking, be proportional to the amount of _complexity_ intervening between the hypothesis and the result. To be less abstract:—The greatness of the complexity found existing among cosmical conditions, by rendering great in the same proportion the difficulty of accounting for all these conditions _at once_, strengthens, also in the same proportion, our faith in that hypothesis which does, in such manner, satisfactorily account for them:—and as _no_ complexity can well be conceived greater than that of the astronomical conditions, so no conviction can be stronger—to _my_ mind at least—than that with which I am impressed by an hypothesis that not only reconciles these conditions, with mathematical accuracy, and reduces them into a consistent and intelligible whole, but is, at the same time, the _sole_ hypothesis by means of which the human intellect has been ever enabled to account for them _at all_.

A most unfounded opinion has become latterly current in gossiping and even in scientific circles—the opinion that the so-called Nebular Cosmogony has been overthrown. This fancy has arisen from the report of late observations made, among what hitherto have been termed the “nebulæ,” through the large telescope of Cincinnati, and the world-renowned instrument of Lord Rosse. Certain spots in the firmament which presented, even to the most powerful of the old telescopes, the appearance of nebulosity, or haze, had been regarded for a long time as confirming the theory of Laplace. They were looked upon as stars in that very process of condensation which I have been attempting to describe. Thus it was supposed that we “had ocular evidence”—an evidence, by the way, which has always been found very

questionable—of the truth of the hypothesis; and, although certain telescopic improvements, every now and then, enabled us to perceive that a spot, here and there, which we had been classing among the nebulæ, was, in fact, but a cluster of stars deriving its nebular character only from its immensity of distance—still it was thought that no doubt could exist as to the actual nebulosity of numerous other masses, the strong-holds of the nebulists, bidding defiance to every effort at segregation. Of these latter the most interesting was the great “nebulæ” in the constellation Orion:—but this, with innumerable other mis-called “nebulæ,” when viewed through the magnificent modern telescopes, has become resolved into a simple collection of stars. Now this fact has been very generally understood as conclusive against the Nebular Hypothesis of Laplace; and, on announcement of the discoveries in question, the most enthusiastic defender and most eloquent popularizer of the theory, Dr. Nichol, went so far as to “admit the necessity of abandoning” an idea which had formed the material of his most praiseworthy book.[9]

[9] “_Views of the Architecture of the Heavens._” A letter, purporting to be from Dr. Nichol to a friend in America, went the rounds of our newspapers, about two years ago, I think, admitting “the necessity” to which I refer. In a subsequent Lecture, however, Dr. N. appears in some manner to have gotten the better of the necessity, and does not quite _renounce_ the theory, although he seems to wish that he could sneer at it as “a purely hypothetical one.” What else was the Law of Gravity before the Maskelyne experiments? and who questioned the Law of Gravity, even then?

Many of my readers will no doubt be inclined to say that the result of these new investigations _has_ at least a strong _tendency_ to overthrow the hypothesis; while some of them, more thoughtful, will suggest that, although the theory is by no means disproved through the segregation of the particular “nebulæ,” alluded to, still a _failure_ to segregate them, with such telescopes, might well have been understood as a triumphant _corroboration_ of the theory:—and this latter class will be surprised, perhaps, to hear me say that even with _them_ I disagree. If the propositions of this Discourse have been comprehended, it will be seen that, in my view, a failure to segregate the “nebulæ” would have tended to the refutation, rather than to the confirmation, of the Nebular Hypothesis.

Let me explain:—The Newtonian Law of Gravity we may, of course, assume as demonstrated. This law, it will be remembered, I have referred to the rëaction of the first Divine Act—to the rëaction of an exercise of the Divine Volition temporarily overcoming a difficulty. This difficulty is that of forcing the normal into the abnormal—of impelling that whose originality, and therefore whose rightful condition, was _One_, to take upon itself the wrongful condition of _Many_. It is only by conceiving this difficulty as _temporarily_ overcome, that we can comprehend a rëaction. There could have been no rëaction had the act been infinitely continued. So long as the act _lasted_, no rëaction, of course, could commence; in other words, no _gravitation_ could take place—for we have considered the one as but the manifestation of the other. But gravitation _has_ taken place; therefore the act of

Creation has ceased: and gravitation has long ago taken place; therefore the act of Creation has long ago ceased. We can no more expect, then, to observe _the primary processes_ of Creation; and to these primary processes the condition of nebulosity has already been explained to belong.

Through what we know of the propagation of light, we have direct proof that the more remote of the stars have existed, under the forms in which we now see them, for an inconceivable number of years. So far back _at least_, then, as the period when these stars underwent condensation, must have been the epoch at which the mass-constitutive processes began. That we may conceive these processes, then, as still going on in the case of certain “nebulæ,” while in all other cases we find them thoroughly at an end, we are forced into assumptions for which we have really _no_ basis whatever—we have to thrust in, again, upon the revolting Reason, the blasphemous idea of special interposition—we have to suppose that, in the particular instances of these “nebulæ,” an unerring God found it necessary to introduce certain supplementary regulations—certain improvements of the general law—certain retouchings and emendations, in a word, which had the effect of deferring the completion of these individual stars for centuries of centuries beyond the æra during which all the other stellar bodies had time, not only to be fully constituted, but to grow hoary with an unspeakable old age.

Of course, it will be immediately objected that since the light by which we recognize the nebulæ now, must be merely that which left their surfaces a vast number of years ago, the processes at present observed, or supposed to be observed, are, in fact, _not_ processes now actually going on, but the phantoms of processes completed long in the Past—just as I maintain all these mass-constitutive processes _must_ have been.

To this I reply that neither is the now-observed condition of the condensed stars their actual condition, but a condition completed long in the Past; so that my argument drawn from the _relative_ condition of the stars and the “nebulæ,” is in no manner disturbed. Moreover, those who maintain the existence of nebulæ, do _not_ refer the nebulosity to extreme distance; they declare it a real and not merely a perspective nebulosity. That we may conceive, indeed, a nebular mass as visible at all, we must conceive it as _very near us_ in comparison with the condensed stars brought into view by the modern telescopes. In maintaining the appearances in question, then, to be really nebulous, we maintain their comparative vicinity to our point of view. Thus, their condition, as we see them now, must be referred to an epoch _far less remote_ than that to which we may refer the now-observed condition of at least the majority of the stars.—In a word, should Astronomy ever demonstrate a “nebula,” in the sense at present intended, I should consider the Nebular Cosmogony—_not_, indeed, as corroborated by the demonstration—but as thereby irretrievably overthrown.

By way, however, of rendering unto Cæsar _no more_ than the things that are Cæsar’s, let me here remark that the assumption of the hypothesis which led him to so glorious a result, seems to have been suggested to Laplace in great measure by a misconception—by the very misconception of which we have just been speaking—by

the generally prevalent misunderstanding of the character of the nebulæ, so mis-named. These he supposed to be, in reality, what their designation implies. The fact is, this great man had, very properly, an inferior faith in his own merely _perceptive_ powers. In respect, therefore, to the actual existence of nebulæ—an existence so confidently maintained by his telescopic contemporaries—he depended less upon what he saw than upon what he heard.

It will be seen that the only valid objections to his theory, are those made to its hypothesis _as_ such—to what suggested it—not to what it suggests; to its propositions rather than to its results. His most unwarranted assumption was that of giving the atoms a movement towards a centre, in the very face of his evident understanding that these atoms, in unlimited succession, extended throughout the Universal space. I have already shown that, under such circumstances, there could have occurred no movement at all; and Laplace, consequently, assumed one on no more philosophical ground than that something of the kind was necessary for the establishment of what he intended to establish.

His original idea seems to have been a compound of the true Epicurean atoms with the false nebulæ of his contemporaries; and thus his theory presents us with the singular anomaly of absolute truth deduced, as a mathematical result, from a hybrid datum of ancient imagination intertangled with modern inacumen. Laplace's real strength lay, in fact, in an almost miraculous mathematical instinct:—on this he relied; and in no instance did it fail or deceive him:—in the case of the Nebular Cosmogony, it led him, blindfolded, through a labyrinth of Error, into one of the most luminous and stupendous temples of Truth.

Let us now fancy, for the moment, that the ring first thrown off by the Sun—that is to say, the ring whose breaking-up constituted Neptune—did not, in fact, break up until the throwing-off of the ring out of which Uranus arose; that this latter ring, again, remained perfect until the discharge of that out of which sprang Saturn; that this latter, again, remained entire until the discharge of that from which originated Jupiter—and so on. Let us imagine, in a word, that no dissolution occurred among the rings until the final rejection of that which gave birth to Mercury. We thus paint to the eye of the mind a series of cöexistent concentric circles; and looking as well at _them_ as at the processes by which, according to Laplace's hypothesis, they were constructed, we perceive at once a very singular analogy with the atomic strata and the process of the original irradiation as I have described it. Is it impossible that, on measuring the _forces_, respectively, by which each successive planetary circle was thrown off—that is to say, on measuring the successive excesses of rotation over gravitation which occasioned the successive discharges—we should find the analogy in question more decidedly confirmed? _Is it improbable that we should discover these forces to have varied—as in the original radiation—proportionally to the squares of the distances?_

Our solar system, consisting, in chief, of one sun, with sixteen planets certainly, and possibly a few more, revolving about it at various distances, and attended by seventeen moons assuredly, but _very_ probably by several others—is now to be considered as _an example_ of the innumerable agglomerations which proceeded to

take place throughout the Universal Sphere of atoms on withdrawal of the Divine Volition. I mean to say that our solar system is to be understood as affording a _generic instance_ of these agglomerations, or, more correctly, of the ulterior conditions at which they arrived. If we keep our attention fixed on the idea of _the utmost possible Relation_ as the Omnipotent design, and on the precautions taken to accomplish it through difference of form, among the original atoms, and particular inequidistance, we shall find it impossible to suppose for a moment that even any two of the incipient agglomerations reached precisely the same result in the end. We shall rather be inclined to think that _no two_ stellar bodies in the Universe—whether suns, planets or moons—are particularly, while _all_ are generally, similar. Still less, then, can we imagine any two _assemblages_ of such bodies—any two "systems"—as having more than a general resemblance.[10] Our telescopes, at this point, thoroughly confirm our deductions. Taking our own solar system, then, as merely a loose or general type of all, we have so far proceeded in our subject as to survey the Universe under the aspect of a spherical space, throughout which, dispersed with merely general equability, exist a number of but generally similar _systems_.

[10] It is not _impossible_ that some unlooked-for optical improvement may disclose to us, among innumerable varieties of systems, a luminous sun, encircled by luminous and non-luminous rings, within and without and between which, revolve luminous and non-luminous planets, attended by moons having moons—and even these latter again having moons.

Let us now, expanding our conceptions, look upon each of these systems as in itself an atom; which in fact it is, when we consider it as but one of the countless myriads of systems which constitute the Universe. Regarding all, then, as but colossal atoms, each with the same ineradicable tendency to Unity which characterizes the actual atoms of which it consists—we enter at once upon a new order of aggregations. The smaller systems, in the vicinity of a larger one, would, inevitably, be drawn into still closer vicinity. A thousand would assemble here; a million there—perhaps here, again, even a billion—leaving, thus, immeasurable vacancies in space. And if now, it be demanded why, in the case of these systems—of these merely Titanic atoms—I speak, simply, of an "assemblage," and not, as in the case of the actual atoms, of a more or less consolidated agglomeration:—if it be asked, for instance, why I do not carry what I suggest to its legitimate conclusion, and describe, at once, these assemblages of system-atoms as rushing to consolidation in spheres—as each becoming condensed into one magnificent sun—my reply is that μελλοντα ταυτα—I am but pausing, for a moment, on the awful threshold of _the Future_. For the present, calling these assemblages "clusters," we see them in the incipient stages of their consolidation. Their _absolute_ consolidation is _to come_.

We have now reached a point from which we behold the Universe as a spherical space, interspersed, _unequably_, with _clusters_. It will be noticed that I here prefer the adverb "unequably" to the phrase "with a merely general equability," employed before. It is evident, in fact, that the equability of distribution will diminish in the ratio of the agglomerative processes—that is to say, as the things distributed diminish in number. Thus the increase of _in_-equability—an increase which must continue

until, sooner or later, an epoch will arrive at which the largest agglomeration will absorb all the others—should be viewed as, simply, a corroborative indication of the _tendency to One_.

And here, at length, it seems proper to inquire whether the ascertained _facts_ of Astronomy confirm the general arrangement which I have thus, deductively, assigned to the Heavens. Thoroughly, they _do_. Telescopic observation, guided by the laws of perspective, enables us to understand that the perceptible Universe exists as _a cluster of clusters, irregularly disposed_.

The “clusters” of which this Universal “_cluster of clusters_” consists, are merely what we have been in the practice of designating “nebulæ”—and, of these “nebulæ,” _one_ is of paramount interest to mankind. I allude to the Galaxy, or Milky Way. This interests us, first and most obviously, on account of its great superiority in apparent size, not only to any one other cluster in the firmament, but to all the other clusters taken together. The largest of these latter occupies a mere point, comparatively, and is distinctly seen only with the aid of a telescope. The Galaxy sweeps throughout the Heaven and is brilliantly visible to the naked eye. But it interests man chiefly, although less immediately, on account of its being his home; the home of the Earth on which he exists; the home of the Sun about which this Earth revolves; the home of that “system” of orbs of which the Sun is the centre and primary—the Earth one of sixteen secondaries, or planets—the Moon one of seventeen tertiaries, or satellites. The Galaxy, let me repeat, is but one of the _clusters_ which I have been describing—but one of the mis-called “nebulæ” revealed to us—by the telescope alone, sometimes—as faint hazy spots in various quarters of the sky. We have no reason to suppose the Milky Way _really_ more extensive than the least of these “nebulæ.” Its vast superiority in size is but an apparent superiority arising from our position in regard to it—that is to say, from our position in its midst. However strange the assertion may at first appear to those unversed in Astronomy, still the astronomer himself has no hesitation in asserting that we are _in the midst_ of that inconceivable host of stars—of suns—of systems—which constitute the Galaxy. Moreover, not only have _we_—not only has _our_ Sun a right to claim the Galaxy as its own especial cluster, but, with slight reservation, it may be said that all the distinctly visible stars of the firmament—all the stars Visible to the naked eye—have equally a right to claim it as _their_ own.

There has been a great deal of misconception in respect to the _shape_ of the Galaxy; which, in nearly all our astronomical treatises, is said to resemble that of a capital Y. The cluster in question has, in reality, a certain general—_very_ general resemblance to the planet Saturn, with its encompassing triple ring. Instead of the solid orb of that planet, however, we must picture to ourselves a lenticular star-island, or collection of stars; our Sun lying excentrically—near the shore of the island—on that side of it which is nearest the constellation of the Cross and farthest from that of Cassiopeia. The surrounding ring, where it approaches our position, has in it a longitudinal _gash_, which does, in fact, cause _the ring, in our vicinity_, to assume, loosely, the appearance of a capital Y.

We must not fall into the error, however, of conceiving the somewhat indefinite girdle as at all _remote_, comparatively speaking, from the also indefinite lenticular cluster which it surrounds; and thus, for mere purpose of explanation, we may speak of our Sun as actually situated at that point of the Y where its three component lines unite; and, conceiving this letter to be of a certain solidity—of a certain thickness, very trivial in comparison with its length—we may even speak of our position as _in the middle_ of this thickness. Fancying ourselves thus placed, we shall no longer find difficulty in accounting for the phænomena presented—which are perspective altogether. When we look upward or downward—that is to say, when we cast our eyes in the direction of the letter's _thickness_—we look through fewer stars than when we cast them in the direction of its _length_, or _along_ either of the three component lines. Of course, in the former case, the stars appear scattered—in the latter, crowded.—To reverse this explanation:—An inhabitant of the Earth, when looking, as we commonly express ourselves, _at_ the Galaxy, is then beholding it in some of the directions of its length—is looking _along_ the lines of the Y—but when, looking out into the general Heaven, he turns his eyes _from_ the Galaxy, he is then surveying it in the direction of the letter's thickness; and on this account the stars seem to him scattered; while, in fact, they are as close together, on an average, as in the mass of the cluster. _No_ consideration could be better adapted to convey an idea of this cluster's stupendous extent.

If, with a telescope of high space-penetrating power, we carefully inspect the firmament, we shall become aware of _a belt of clusters_—of what we have hitherto called "nebulæ"—a _band_, of varying breadth, stretching from horizon to horizon, at right angles to the general course of the Milky Way. This band is the ultimate _cluster of clusters_. This belt is _The Universe_. Our Galaxy is but one, and perhaps one of the most inconsiderable, of the clusters which go to the constitution of this ultimate, Universal _belt_ or _band_. The appearance of this cluster of clusters, to our eyes, _as_ a belt or band, is altogether a perspective phænomenon of the same character as that which causes us to behold our own individual and roughly-spherical cluster, the Galaxy, under guise also of a belt, traversing the Heavens at right angles to the Universal one. The shape of the all-inclusive cluster is, of course _generally_, that of each individual cluster which it includes. Just as the scattered stars which, on looking _from_ the Galaxy, we see in the general sky, are, in fact, but a portion of that Galaxy itself, and as closely intermingled with it as any of the telescopic points in what seems the densest portion of its mass—so are the scattered "nebulæ" which, on casting our eyes _from_ the Universal _belt_, we perceive at all points of the firmament—so, I say, are these scattered "nebulæ" to be understood as only perspectively scattered, and as part and parcel of the one supreme and Universal _sphere_.

No astronomical fallacy is more untenable, and none has been more pertinaciously adhered to, than that of the absolute _illimitation_ of the Universe of Stars. The reasons for limitation, as I have already assigned them, _à priori_, seem to me unanswerable; but, not to speak of these, _observation_ assures us that there is, in numerous directions around us, certainly, if not in all, a positive limit—or, at the very least, affords us no basis whatever for thinking otherwise. Were the succession of

stars endless, then the background of the sky would present us an uniform luminosity, like that displayed by the Galaxy—_since there could be absolutely no point, in all that background, at which would not exist a star._ The only mode, therefore, in which, under such a state of affairs, we could comprehend the _voids_ which our telescopes find in innumerable directions, would be by supposing the distance of the invisible background so immense that no ray from it has yet been able to reach us at all. That this _may_ be so, who shall venture to deny? I maintain, simply, that we have not even the shadow of a reason for believing that it _is_ so.

When speaking of the vulgar propensity to regard all bodies on the Earth as tending merely to the Earth's centre, I observed that, "with certain exceptions to be specified hereafter, every body on the Earth tended not only to the Earth's centre, but in every conceivable direction besides."[11] The "exceptions" refer to those frequent gaps in the Heavens, where our utmost scrutiny can detect not only no stellar bodies, but no indications of their existence:—where yawning chasms, blacker than Erebus, seem to afford us glimpses, through the boundary walls of the Universe of Stars, into the illimitable Universe of Vacancy, beyond. Now as any body, existing on the Earth, chances to pass, either through its own movement or the Earth's, into a line with any one of these voids, or cosmical abysses, it clearly is no longer attracted _in the direction of that void_, and for the moment, consequently, is "heavier" than at any period, either after or before. Independently of the consideration of these voids, however, and looking only at the generally unequable distribution of the stars, we see that the absolute tendency of bodies on the Earth to the Earth's centre, is in a state of perpetual variation.

[11] Page 62.

We comprehend, then, the insulation of our Universe. We perceive the isolation of _that_—of _all_ that which we grasp with the senses. We know that there exists one _cluster of clusters_—a collection around which, on all sides, extend the immeasurable wildernesses of a Space _to all human perception_ untenanted. But _because_ upon the confines of this Universe of Stars we are compelled to pause, through want of farther evidence from the senses, is it right to conclude that, in fact, there _is_ no material point beyond that which we have thus been permitted to attain? Have we, or have we not, an analogical right to the inference that this perceptible Universe—that this cluster of clusters—is but one of _a series_ of clusters of clusters, the rest of which are invisible through distance—through the diffusion of their light being so excessive, ere it reaches us, as not to produce upon our retinas a light-impression—or from there being no such emanation as light at all, in these unspeakably distant worlds—or, lastly, from the mere interval being so vast, that the electric tidings of their presence in Space, have not yet—through the lapsing myriads of years—been enabled to traverse that interval?

Have we any right to inferences—have we any ground whatever for visions such as these? If we have a right to them in _any_ degree, we have a right to their infinite extension.

The human brain has obviously a leaning to the "_Infinite_," and fondles the phantom of the idea. It seems to long with a passionate fervor for this impossible conception, with the hope of intellectually believing it when conceived. What is general among the whole race of Man, of course no individual of that race can be warranted in considering abnormal; nevertheless, there _may_ be a class of superior intelligences, to whom the human bias alluded to may wear all the character of monomania.

My question, however, remains unanswered:—Have we any right to infer—let us say, rather, to imagine—an interminable succession of the "clusters of clusters," or of "Universes" more or less similar?

I reply that the "right," in a case such as this, depends absolutely upon the hardihood of that imagination which ventures to claim the right. Let me declare, only, that, as an individual, I myself feel impelled to the _fancy_—without daring to call it more—that there _does_ exist a _limitless_ succession of Universes, more or less similar to that of which we have cognizance—to that of which _alone_ we shall ever have cognizance—at the very least until the return of our own particular Universe into Unity. _If_ such clusters of clusters exist, however—_and they do_—it is abundantly clear that, having had no part in our origin, they have no portion in our laws. They neither attract us, nor we them. Their material—their spirit is not ours—is not that which obtains in any part of our Universe. They could not impress our senses or our souls. Among them and us—considering all, for the moment, collectively—there are no influences in common. Each exists, apart and independently, _in the bosom of its proper and particular God_.

In the conduct of this Discourse, I am aiming less at physical than at metaphysical order. The clearness with which even material phænomena are presented to the understanding, depends very little, I have long since learned to perceive, upon a merely natural, and almost altogether upon a moral, arrangement. If then I seem to step somewhat too discursively from point to point of my topic, let me suggest that I do so in the hope of thus the better keeping unbroken that chain of _graduated impression_ by which alone the intellect of Man can expect to encompass the grandeurs of which I speak, and, in their majestic totality, to comprehend them.

So far, our attention has been directed, almost exclusively, to a general and relative grouping of the stellar bodies in space. Of specification there has been little; and whatever ideas of _quantity_ have been conveyed—that is to say, of number, magnitude, and distance—have been conveyed incidentally and by way of preparation for more definitive conceptions. These latter let us now attempt to entertain.

Our solar system, as has been already mentioned, consists, in chief, of one sun and sixteen planets certainly, but in all probability a few others, revolving around it as a centre, and attended by seventeen moons of which we know, with possibly several more of which as yet we know nothing. These various bodies are not true spheres, but oblate spheroids—spheres flattened at the poles of the imaginary axes about which they rotate:—the flattening being a consequence of the rotation. Neither is the Sun absolutely the centre of the system; for this Sun itself, with all the planets, revolves

about a perpetually shifting point of space, which is the system's general centre of gravity. Neither are we to consider the paths through which these different spheroids move—the moons about the planets, the planets about the Sun, or the Sun about the common centre—as circles in an accurate sense. They are, in fact, _ellipses—one of the foci being the point about which the revolution is made_. An ellipse is a curve, returning into itself, one of whose diameters is longer than the other. In the longer diameter are two points, equidistant from the middle of the line, and so situated otherwise that if, from each of them a straight line be drawn to any one point of the curve, the two lines, taken together, will be equal to the longer diameter itself. Now let us conceive such an ellipse. At one of the points mentioned, which are the _foci_, let us fasten an orange. By an elastic thread let us connect this orange with a pea; and let us place this latter on the circumference of the ellipse. Let us now move the pea continuously around the orange—keeping always on the circumference of the ellipse. The elastic thread, which, of course, varies in length as we move the pea, will form what in geometry is called a _radius vector_. Now, if the orange be understood as the Sun, and the pea as a planet revolving about it, then the revolution should be made at such a rate—with a velocity so varying—that the _radius vector_ may pass over _equal areas of space in equal times_. The progress of the pea _should be_—in other words, the progress of the planet _is_, of course,—slow in proportion to its distance from the Sun—swift in proportion to its proximity. Those planets, moreover, move the more slowly which are the farther from the Sun; _the squares of their periods of revolution having the same proportion to each other, as have to each other the cubes of their mean distances from the Sun_.

The wonderfully complex laws of revolution here described, however, are not to be understood as obtaining in our system alone. They _everywhere_ prevail where Attraction prevails. They control _the Universe_. Every shining speck in the firmament is, no doubt, a luminous sun, resembling our own, at least in its general features, and having in attendance upon it a greater or less number of planets, greater or less, whose still lingering luminosity is not sufficient to render them visible to us at so vast a distance, but which, nevertheless, revolve, moon-attended, about their starry centres, in obedience to the principles just detailed—in obedience to the three omniprevalent laws of revolution—the three immortal laws _guessed_ by the imaginative Kepler, and but subsequently demonstrated and accounted for by the patient and mathematical Newton. Among a tribe of philosophers who pride themselves excessively upon matter-of-fact, it is far too fashionable to sneer at all speculation under the comprehensive _sobriquet_, "guess-work." The point to be considered is, _who_ guesses. In guessing with Plato, we spend our time to better purpose, now and then, than in hearkening to a demonstration by Alcmæon.

In many works on Astronomy I find it distinctly stated that the laws of Kepler are _the basis_ of the great principle, Gravitation. This idea must have arisen from the fact that the suggestion of these laws by Kepler, and his proving them _à posteriori_ to have an actual existence, led Newton to account for them by the hypothesis of Gravitation, and, finally, to demonstrate them _à priori_, as necessary consequences of the hypothetical principle. Thus so far from the laws of Kepler being the basis of Gravity, Gravity is the basis of these laws—as it is, indeed, of all the laws of the

material Universe which are not referable to Repulsion alone.

The mean distance of the Earth from the Moon—that is to say, from the heavenly body in our closest vicinity—is 237,000 miles. Mercury, the planet nearest the Sun, is distant from him 37 millions of miles. Venus, the next, revolves at a distance of 68 millions:—the Earth, which comes next, at a distance of 95 millions:—Mars, then, at a distance of 144 millions. Now come the eight Asteroids (Ceres, Juno, Vesta, Pallas, Astræa, Flora, Iris, and Hebe) at an average distance of about 250 millions. Then we have Jupiter, distant 490 millions; then Saturn, 900 millions; then Uranus, 19 hundred millions; finally Neptune, lately discovered, and revolving at a distance, say of 28 hundred millions. Leaving Neptune out of the account—of which as yet we know little accurately and which is, possibly, one of a system of Asteroids—it will be seen that, within certain limits, there exists an _order of interval_ among the planets. Speaking loosely, we may say that each outer planet is twice as far from the Sun as is the next inner one. May not the _order_ here mentioned—_may not the law of Bode—be deduced from consideration of the analogy suggested by me as having place between the solar discharge of rings and the mode of the atomic irradiation_?

The numbers hurriedly mentioned in this summary of distance, it is folly to attempt comprehending, unless in the light of abstract arithmetical facts. They are not practically tangible ones. They convey no precise ideas. I have stated that Neptune, the planet farthest from the Sun, revolves about him at a distance of 28 hundred millions of miles. So far good:—I have stated a mathematical fact; and, without comprehending it in the least, we may put it to use—mathematically. But in mentioning, even, that the Moon revolves about the Earth at the comparatively trifling distance of 237,000 miles, I entertained no expectation of giving any one to understand—to know—to feel—how far from the Earth the Moon actually _is_. 237,000 _miles_! There are, perhaps, few of my readers who have not crossed the Atlantic ocean; yet how many of them have a distinct idea of even the 3,000 miles intervening between shore and shore? I doubt, indeed, whether the man lives who can force into his brain the most remote conception of the interval between one milestone and its next neighbor upon the turnpike. We are in some measure aided, however, in our consideration of distance, by combining this consideration with the kindred one of velocity. Sound passes through 1100 feet of space in a second of time. Now were it possible for an inhabitant of the Earth to see the flash of a cannon discharged in the Moon, and to hear the report, he would have to wait, after perceiving the former, more than 13 entire days and nights before getting any intimation of the latter.

However feeble be the impression, even thus conveyed, of the Moon’s real distance from the Earth, it will, nevertheless, effect a good object in enabling us more clearly to see the futility of attempting to grasp such intervals as that of the 28 hundred millions of miles between our Sun and Neptune; or even that of the 95 millions between the Sun and the Earth we inhabit. A cannon-ball, flying at the greatest velocity with which such a ball has ever been known to fly, could not traverse the latter interval in less than 20 years; while for the former it would require 590.

Our Moon’s real diameter is 2160 miles; yet she is comparatively so trifling an object that it would take nearly 50 such orbs to compose one as great as the Earth.

The diameter of our own globe is 7912 miles—but from the enunciation of these numbers what positive idea do we derive?

If we ascend an ordinary mountain and look around us from its summit, we behold a landscape stretching, say 40 miles, in every direction; forming a circle 250 miles in circumference; and including an area of 5000 square miles. The extent of such a prospect, on account of the _successiveness_ with which its portions necessarily present themselves to view, can be only very feebly and very partially appreciated:—yet the entire panorama would comprehend no more than one 40,000th part of the mere _surface_ of our globe. Were this panorama, then, to be succeeded, after the lapse of an hour, by another of equal extent; this again by a third, after the lapse of another hour; this again by a fourth after lapse of another hour—and so on, until the scenery of the whole Earth were exhausted; and were we to be engaged in examining these various panoramas for twelve hours of every day; we should nevertheless, be 9 years and 48 days in completing the general survey.

But if the mere surface of the Earth eludes the grasp of the imagination, what are we to think of its cubical contents? It embraces a mass of matter equal in weight to at least 2 sextillions, 200 quintillions of tons. Let us suppose it in a state of quiescence; and now let us endeavor to conceive a mechanical force sufficient to set it in motion! Not the strength of all the myriads of beings whom we may conclude to inhabit the planetary worlds of our system—not the combined physical strength of _all_ these beings—even admitting all to be more powerful than man—would avail to stir the ponderous mass _a single inch_ from its position.

What are we to understand, then, of the force, which under similar circumstances, would be required to move the _largest_ of our planets, Jupiter? This is 86,000 miles in diameter, and would include within its periphery more than a thousand orbs of the magnitude of our own. Yet this stupendous body is actually flying around the Sun at the rate of 29,000 miles an hour—that is to say, with a velocity 40 times greater than that of a cannon-ball! The thought of such a phænomenon cannot well be said to _startle_ the mind:—it palsies and appals it. Not unfrequently we task our imagination in picturing the capacities of an angel. Let us fancy such a being at a distance of some hundred miles from Jupiter—a close eye-witness of this planet as it speeds on its annual revolution. Now _can_ we, I demand, fashion for ourselves any conception so distinct of this ideal being’s spiritual exaltation, as _that_ involved in the supposition that, even by this immeasurable mass of matter, whirled immediately before his eyes, with a velocity so unutterable, he—an angel—angelic though he be—is not at once struck into nothingness and overwhelmed?

At this point, however, it seems proper to suggest that, in fact, we have been speaking of comparative trifles. Our Sun, the central and controlling orb of the system to which Jupiter belongs, is not only greater than Jupiter, but greater by far than all the planets of the system taken together. This fact is an essential condition, indeed, of the stability of the system itself. The diameter of Jupiter has been

mentioned:—it is 86,000 miles:—that of the Sun is 882,000 miles. An inhabitant of the latter, travelling 90 miles a day, would be more than 80 years in going round a great circle of its circumference. It occupies a cubical space of 681 quadrillions, 472 trillions of miles. The Moon, as has been stated, revolves about the Earth at a distance of 237,000 miles—in an orbit, consequently, of nearly a million and a half. Now, were the Sun placed upon the Earth, centre over centre, the body of the former would extend, in every direction, not only to the line of the Moon's orbit, but beyond it, a distance of 200,000 miles.

And here, once again, let me suggest that, in fact, we have _still_ been speaking of comparative trifles. The distance of the planet Neptune from the Sun has been stated:—it is 28 hundred millions of miles; the circumference of its orbit, therefore, is about 17 billions. Let this be borne in mind while we glance at some one of the brightest stars. Between this and the star of _our_ system, (the Sun,) there is a gulf of space, to convey any idea of which we should need the tongue of an archangel. From _our_ system, then, and from _our_ Sun, or star, the star at which we suppose ourselves glancing is a thing altogether apart:—still, for the moment, let us imagine it placed upon our Sun, centre over centre, as we just now imagined this Sun itself placed upon the Earth. Let us now conceive the particular star we have in mind, extending, in every direction, beyond the orbit of Mercury—of Venus—of the Earth:—still _on_, beyond the orbit of Mars—of Jupiter—of Uranus—until, finally, we fancy it filling the circle—17 _billions of miles in circumference_—which is described by the revolution of Leverrier's planet. When we have conceived all this, we shall have entertained no extravagant conception. There is the very best reason for believing that many of the stars are even far larger than the one we have imagined. I mean to say that we have the very best _empirical_ basis for such belief:—and, in looking back at the original, atomic arrangements for _diversity_, which have been assumed as a part of the Divine plan in the constitution of the Universe, we shall be enabled easily to understand, and to credit, the existence of even far vaster disproportions in stellar size than any to which I have hitherto alluded. The largest orbs, of course, we must expect to find rolling through the widest vacancies of Space.

I remarked, just now, that to convey an idea of the interval between our Sun and any one of the other stars, we should require the eloquence of an archangel. In so saying, I should not be accused of exaggeration; for, in simple truth, these are topics on which it is scarcely possible to exaggerate. But let us bring the matter more distinctly before the eye of the mind.

In the first place, we may get a general, _relative_ conception of the interval referred to, by comparing it with the inter-planetary spaces. If, for example, we suppose the Earth, which is, in reality, 95 millions of miles from the Sun, to be only _one foot_ from that luminary; then Neptune would be 40 feet distant; _and the star Alpha Lyræ, at the very least_, 159.

Now I presume that, in the termination of my last sentence, few of my readers have noticed anything especially objectionable—particularly wrong. I said that the distance of the Earth from the Sun being taken at _one foot_, the distance of Neptune would be 40 feet, and that of Alpha Lyræ, 159. The proportion between one foot and

159 has appeared, perhaps, to convey a sufficiently definite impression of the proportion between the two intervals—that of the Earth from the Sun and that of Alpha Lyræ from the same luminary. But my account of the matter should, in reality, have run thus:—The distance of the Earth from the Sun being taken at one foot, the distance of Neptune would be 40 feet, and that of Alpha Lyræ, 159——_miles_:—that is to say, I had assigned to Alpha Lyræ, in my first statement of the case, only the 5280_th_ _part_ of that distance which is the _least distance possible_ at which it can actually lie.

To proceed:—However distant a mere _planet_ is, yet when we look at it through a telescope, we see it under a certain form—of a certain appreciable size. Now I have already hinted at the probable bulk of many of the stars; nevertheless, when we view any one of them, even through the most powerful telescope, it is found to present us with _no form_, and consequently with _no magnitude_ whatever. We see it as a point and nothing more.

Again;—Let us suppose ourselves walking, at night, on a highway. In a field on one side of the road, is a line of tall objects, say trees, the figures of which are distinctly defined against the background of the sky. This line of objects extends at right angles to the road, and from the road to the horizon. Now, as we proceed along the road, we see these objects changing their positions, respectively, in relation to a certain fixed point in that portion of the firmament which forms the background of the view. Let us suppose this fixed point—sufficiently fixed for our purpose—to be the rising moon. We become aware, at once, that while the tree nearest us so far alters its position in respect to the moon, as to seem flying behind us, the tree in the extreme distance has scarcely changed at all its relative position with the satellite. We then go on to perceive that the farther the objects are from us, the less they alter their positions; and the converse. Then we begin, unwittingly, to estimate the distances of individual trees by the degrees in which they evince the relative alteration. Finally, we come to understand how it might be possible to ascertain the actual distance of any given tree in the line, by using the amount of relative alteration as a basis in a simple geometrical problem. Now this relative alteration is what we call “parallax;” and by parallax we calculate the distances of the heavenly bodies. Applying the principle to the trees in question, we should, of course, be very much at a loss to comprehend the distance of _that_ tree, which, however far we proceeded along the road, should evince _no_ parallax at all. This, in the case described, is a thing impossible; but impossible only because all distances on our Earth are trivial indeed:—in comparison with the vast cosmical quantities, we may speak of them as absolutely nothing.

Now, let us suppose the star Alpha Lyræ directly overhead; and let us imagine that, instead of standing on the Earth, we stand at one end of a straight road stretching through Space to a distance equalling the diameter of the Earth’s orbit—that is to say, to a distance of 190 _millions of miles_. Having observed, by means of the most delicate micrometrical instruments, the exact position of the star, let us now pass along this inconceivable road, until we reach its other extremity. Now, once again, let us look at the star. It is _precisely_ where we left it. Our instruments, however delicate, assure us that its relative position is absolutely—is

identically the same as at the commencement of our unutterable journey. _No_ parallax—none whatever—has been found.

The fact is, that, in regard to the distance of the fixed stars—of any one of the myriads of suns glistening on the farther side of that awful chasm which separates our system from its brothers in the cluster to which it belongs—astronomical science, until very lately, could speak only with a negative certainty. Assuming the brightest as the nearest, we could say, even of _them_, only that there is a certain incomprehensible distance on the _hither_ side of which they cannot be:—how far they are beyond it we had in no case been able to ascertain. We perceived, for example, that Alpha Lyræ cannot be nearer to us than 19 trillions, 200 billions of miles; but, for all we knew, and indeed for all we now know, it may be distant from us the square, or the cube, or any other power of the number mentioned. By dint, however, of wonderfully minute and cautious observations, continued, with novel instruments, for many laborious years, _Bessel_, not long ago deceased, has lately succeeded in determining the distance of six or seven stars; among others, that of the star numbered 61 in the constellation of the Swan. The distance in this latter instance ascertained, is 670,000 times that of the Sun; which last it will be remembered, is 95 millions of miles. The star 61 Cygni, then, is nearly 64 trillions of miles from us—or more than three times the distance assigned, _as the least possible_, for Alpha Lyræ.

In attempting to appreciate this interval by the aid of any considerations of _velocity_, as we did in endeavoring to estimate the distance of the moon, we must leave out of sight, altogether, such nothings as the speed of a cannon-ball, or of sound. Light, however, according to the latest calculations of Struve, proceeds at the rate of 167,000 miles in a second. Thought itself cannot pass through this interval more speedily—if, indeed, thought can traverse it at all. Yet, in coming from 61 Cygni to us, even at this inconceivable rate, light occupies more than _ten years_; and, consequently, were the star this moment blotted out from the Universe, still, _for ten years_, would it continue to sparkle on, undimmed in its paradoxical glory.

Keeping now in mind whatever feeble conception we may have attained of the interval between our Sun and 61 Cygni, let us remember that this interval, however unutterably vast, we are permitted to consider as but the _average_ interval among the countless host of stars composing that cluster, or “nebula,” to which our system, as well as that of 61 Cygni, belongs. I have, in fact, stated the case with great moderation:—we have excellent reason for believing 61 Cygni to be one of the _nearest_ stars, and thus for concluding, at least for the present, that its distance from us is _less_ than the average distance between star and star in the magnificent cluster of the Milky Way.

And here, once again and finally, it seems proper to suggest that even as yet we have been speaking of trifles. Ceasing to wonder at the space between star and star in our own or in any particular cluster, let us rather turn our thoughts to the intervals between cluster and cluster, in the all comprehensive cluster of the Universe.

I have already said that light proceeds at the rate of 167,000 miles in a second—that is, about 10 millions of miles in a minute, or about 600 millions of miles

in an hour:—yet so far removed from us are some of the “nebulæ” that even light, speeding with this velocity, could not and does not reach us, from those mysterious regions, in less than 3 _millions of years_. This calculation, moreover, is made by the elder Herschell, and in reference merely to those comparatively proximate clusters within the scope of his own telescope. There _are_ “nebulæ,” however, which, through the magical tube of Lord Rosse, are this instant whispering in our ears the secrets of _a million of ages_ by-gone. In a word, the events which we behold now—at this moment—in those worlds—are the identical events which interested their inhabitants _ten hundred thousand centuries ago_. In intervals—in distances such as this suggestion forces upon the _soul_—rather than upon the mind—we find, at length, a fitting climax to all hitherto frivolous considerations of _quantity_.

Our fancies thus occupied with the cosmical distances, let us take the opportunity of referring to the difficulty which we have so often experienced, while pursuing _the beaten path_ of astronomical reflection, _in accounting_ for the immeasurable voids alluded to—in comprehending why chasms so totally unoccupied and therefore apparently so needless, have been made to intervene between star and star—between cluster and cluster—in understanding, to be brief, a sufficient reason for the Titanic scale, in respect of mere _Space_, on which the Universe is seen to be constructed. A rational cause for the phænomenon, I maintain that Astronomy has palpably failed to assign:—but the considerations through which, in this Essay, we have proceeded step by step, enable us clearly and immediately to perceive that _Space and Duration are one_. That the Universe might _endure_ throughout an æra at all commensurate with the grandeur of its component material portions and with the high majesty of its spiritual purposes, it was necessary that the original atomic diffusion be made to so inconceivable an extent as to be only not infinite. It was required, in a word, that the stars should be gathered into visibility from invisible nebulosity—proceed from nebulosity to consolidation—and so grow grey in giving birth and death to unspeakably numerous and complex variations of vitalic development:—it was required that the stars should do all this—should have time thoroughly to accomplish all these Divine purposes—_during the period_ in which all things were effecting their return into Unity with a velocity accumulating in the inverse proportion of the squares of the distances at which lay the inevitable End.

Throughout all this we have no difficulty in understanding the absolute accuracy of the Divine _adaptation_. The density of the stars, respectively, proceeds, of course, as their condensation diminishes; condensation and heterogeneity keep pace with each other; through the latter, which is the index of the former, we estimate the vitalic and spiritual development. Thus, in the density of the globes, we have the measure in which their purposes are fulfilled. _As_ density proceeds—_as_ the divine intentions _are_ accomplished—_as_ less and still less remains _to be_ accomplished—so—in the same ratio—should we expect to find an acceleration of _the End_:—and thus the philosophical mind will easily comprehend that the Divine designs in constituting the stars, advance _mathematically_ to their fulfilment:—and more; it will readily give the advance a mathematical expression; it will decide that this advance is inversely proportional with the squares of the distances of all created things from the starting-point and goal of their creation.

Not only is this Divine adaptation, however, mathematically accurate, but there is that about it which stamps it _as divine_, in distinction from that which is merely the work of human constructiveness. I allude to the complete _mutuality_ of adaptation. For example; in human constructions a particular cause has a particular effect; a particular intention brings to pass a particular object; but this is all; we see no reciprocity. The effect does not re-act upon the cause; the intention does not change relations with the object. In Divine constructions the object is either design or object as we choose to regard it—and we may take at any time a cause for an effect, or the converse—so that we can never absolutely decide which is which.

To give an instance:—In polar climates the human frame, to maintain its animal heat, requires, for combustion in the capillary system, an abundant supply of highly azotized food, such as train-oil. But again:—in polar climates nearly the sole food afforded man is the oil of abundant seals and whales. Now, whether is oil at hand because imperatively demanded, or the only thing demanded because the only thing to be obtained? It is impossible to decide. There is an absolute _reciprocity of adaptation_.

The pleasure which we derive from any display of human ingenuity is in the ratio of _the approach_ to this species of reciprocity. In the construction of _plot_, for example, in fictitious literature, we should aim at so arranging the incidents that we shall not be able to determine, of any one of them, whether it depends from any one other or upholds it. In this sense, of course, _perfection_ of _plot_ is really, or practically, unattainable—but only because it is a finite intelligence that constructs. The plots of God are perfect. The Universe is a plot of God.

And now we have reached a point at which the intellect is forced, again, to struggle against its propensity for analogical inference—against its monomaniac grasping at the infinite. Moons have been seen _revolving_ about planets; planets about stars; and the poetical instinct of humanity—its instinct of the symmetrical, if the symmetry be but a symmetry of surface:—this _instinct_, which the Soul, not only of Man but of all created beings, took up, in the beginning, from the _geometrical_ basis of the Universal irradiation—impels us to the fancy of an endless extension of this system of _cycles_. Closing our eyes equally to _de_duction and _in_duction, we insist upon imagining a _revolution_ of all the orbs of the Galaxy about some gigantic globe which we take to be the central pivot of the whole. Each cluster in the great cluster of clusters is imagined, of course, to be similarly supplied and constructed; while, that the “analogy” may be wanting at no point, we go on to conceive these clusters themselves, again, as _revolving_ about some still more august sphere;—this latter, still again, _with_ its encircling clusters, as but one of a yet more magnificent series of agglomerations, _gyrating_ about yet another orb central _to them_—some orb still more unspeakably sublime—some orb, let us rather say, of infinite sublimity endlessly multiplied by the infinitely sublime. Such are the conditions, continued in perpetuity, which the voice of what some people term “analogy” calls upon the Fancy to depict and the Reason to contemplate, if possible, without becoming dissatisfied with the picture. Such, _in general_, are the interminable gyrations beyond gyration which we have been instructed by Philosophy to comprehend and to account for, at

least in the best manner we can. Now and then, however, a philosopher proper—one whose phrenzy takes a very determinate turn—whose genius, to speak more reverentially, has a strongly-pronounced washerwomanish bias, doing every thing up by the dozen—enables us to see _precisely_ that point out of sight, at which the revolutionary processes in question do, and of right ought to, come to an end.

It is hardly worth while, perhaps, even to sneer at the reveries of Fourrier:—but much has been said, latterly, of the hypothesis of Mädler—that there exists, in the centre of the Galaxy, a stupendous globe about which all the systems of the cluster revolve. The _period_ of our own, indeed, has been stated—117 millions of years.

That our Sun has a motion in space, independently of its rotation, and revolution about the system's centre of gravity, has long been suspected. This motion, granting it to exist, would be manifested perspectively. The stars in that firmamental region which we were leaving behind us, would, in a very long series of years, become crowded; those in the opposite quarter, scattered. Now, by means of astronomical History, we ascertain, cloudily, that some such phænomena have occurred. On this ground it has been declared that our system is moving to a point in the heavens diametrically opposite the star Zeta Herculis:—but this inference is, perhaps, the maximum to which we have any logical right. Mädler, however, has gone so far as to designate a particular star, Alcyone in the Pleiades, as being at or about the very spot around which a general _revolution_ is performed.

Now, since by "analogy" we are led, in the first instance, to these dreams, it is no more than proper that we should abide by analogy, at least in some measure, during their development; and that analogy which suggests the revolution, suggests at the same time a central orb about which it should be performed:—so far the astronomer was consistent. This central orb, however, should, dynamically, be greater than all the orbs, taken together, which surround it. Of these there are about 100 millions. "Why, then," it was of course demanded, "do we not _see_ this vast central sun—_at least equal_ in mass to 100 millions of such suns as ours—why do we not _see_ it—_we_, especially, who occupy the mid region of the cluster—the very locality _near_ which, at all events, must be situated this incomparable star?" The reply was ready—"It must be non-luminous, as are our planets." Here, then, to suit a purpose, analogy is suddenly let fall. "Not so," it may be said—"we know that non-luminous suns actually exist." It is true that we have reason at least for supposing so; but we have certainly no reason whatever for supposing that the non-luminous suns in question are encircled by _luminous_ suns, while these again are surrounded by non-luminous planets:—and it is precisely all this with which Mädler is called upon to find any thing analogous in the heavens—for it is precisely all this which he imagines in the case of the Galaxy. Admitting the thing to be so, we cannot help here picturing to ourselves how sad a puzzle the _why it is so_ must prove to all _à priori_ philosophers.

But granting, in the very teeth of analogy and of every thing else, the non-luminosity of the vast central orb, we may still inquire how this orb, so enormous, could fail of being rendered visible by the flood of light thrown upon it from the 100 millions of glorious suns glaring in all directions about it. Upon the urging of this

question, the idea of an actually solid central sun appears, in some measure, to have been abandoned; and speculation proceeded to assert that the systems of the cluster perform their revolutions merely about an immaterial centre of gravity common to all. Here again then, to suit a purpose, analogy is let fall. The planets of our system revolve, it is true, about a common centre of gravity; but they do this in connexion with, and in consequence of, a material sun whose mass more than counterbalances the rest of the system.

The mathematical circle is a curve composed of an infinity of straight lines. But this idea of the circle—an idea which, in view of all ordinary geometry, is merely the mathematical, as contradistinguished from the practical, idea—is, in sober fact, the _practical_ conception which alone we have any right to entertain in regard to the majestic circle with which we have to deal, at least in fancy, when we suppose our system revolving about a point in the centre of the Galaxy. Let the most vigorous of human imaginations attempt but to take a single step towards the comprehension of a sweep so ineffable! It would scarcely be paradoxical to say that a flash of lightning itself, travelling _forever_ upon the circumference of this unutterable circle, would still, _forever_, be travelling in a straight line. That the path of our Sun in such an orbit would, to any human perception, deviate in the slightest degree from a straight line, even in a million of years, is a proposition not to be entertained:—yet we are required to believe that a curvature has become apparent during the brief period of our astronomical history—during a mere point—during the utter nothingness of two or three thousand years.

It may be said that Mädler _has_ really ascertained a curvature in the direction of our system’s now well-established progress through Space. Admitting, if necessary, this fact to be in reality such, I maintain that nothing is thereby shown except the reality of this fact—the fact of a curvature. For its _thorough_ determination, ages will be required; and, when determined, it will be found indicative of some binary or other multiple relation between our Sun and some one or more of the proximate stars. I hazard nothing however, in predicting, that, after the lapse of many centuries, all efforts at determining the path of our Sun through Space, will be abandoned as fruitless. This is easily conceivable when we look at the infinity of perturbation it must experience, from its perpetually-shifting relations with other orbs, in the common approach of all to the nucleus of the Galaxy.

But in examining other “nebulæ” than that of the Milky Way—in surveying, generally, the clusters which overspread the heavens—do we or do we not find confirmation of Mädler’s hypothesis? We do _not_. The forms of the clusters are exceedingly diverse when casually viewed; but on close inspection, through powerful telescopes, we recognize the sphere, very distinctly, as at least the proximate form of all:—their constitution, in general, being at variance with the idea of revolution about a common centre.

“It is difficult,” says Sir John Herschell, “to form any conception of the dynamical state of such systems. On one hand, without a rotary motion and a centrifugal force, it is hardly possible not to regard them as in a state of _progressive collapse_. On the other, granting such a motion and such a force, we find it no less difficult to reconcile

their forms with the rotation of the whole system [meaning cluster] around any single axis, without which internal collision would appear to be inevitable."

Some remarks lately made about the "nebulæ" by Dr. Nichol, in taking quite a different view of the cosmical conditions from any taken in this Discourse—have a very peculiar applicability to the point now at issue. He says:

"When our greatest telescopes are brought to bear upon them, we find that those which were thought to be irregular, are not so; they approach nearer to a globe. Here is one that looked oval; but Lord Rosse's telescope brought it into a circle.... Now there occurs a very remarkable circumstance in reference to these comparatively sweeping circular masses of nebulæ. We find they are not entirely circular, but the reverse; and that all around them, on every side, there are volumes of stars, _stretching out apparently as if they were rushing towards a great central mass in consequence of the action of some great power_."[12]

[12] I must be understood as denying, _especially_, only the _revolutionary_ portion of Mädler's hypothesis. Of course, if no great central orb exists _now_ in our cluster, such will exist hereafter. Whenever existing, it will be merely the _nucleus_ of the consolidation.

Were I to describe, in my own words, what must necessarily be the existing condition of each nebula on the hypothesis that all matter is, as I suggest, now returning to its original Unity, I should simply be going over, nearly verbatim, the language here employed by Dr. Nichol, without the faintest suspicion of that stupendous truth which is the key to these nebular phænomena.

And here let me fortify my position still farther, by the voice of a greater than Mädler—of one, moreover, to whom all the data of Mädler have long been familiar things, carefully and thoroughly considered. Referring to the elaborate calculations of Argelander—the very researches which form Mädler's basis—_Humboldt_, whose generalizing powers have never, perhaps been equalled, has the following observation:

"When we regard the real, proper, or non-perspective motions of the stars, we find _many groups of them moving in opposite directions_; and the data as yet in hand render it not necessary, at least, to conceive that the systems composing the Milky Way, or the clusters, generally, composing the Universe, are revolving about any particular centre unknown, whether luminous or non-luminous. It is but Man's longing for a fundamental First Cause, that impels both his intellect and his fancy to the adoption of such an hypothesis."[13]

[13] Betrachtet man die nicht perspectivischen eigenen Bewegungen der Sterne, so scheinen viele gruppenweise in ihrer Richtung entgegengesetzt; und die bisher gesammelten Thatsachen machen es auf's wenigste nicht nothwendig, anzunehmen, dass alle Theile unserer Sternenschicht oder gar der gesammten Sterneninseln, welche den Weltraum füllen, sich um einen grossen, unbekannten, leuchtenden oder dunkeln Centralkörper bewegen. Das Streben nach den letzten und höchsten Grundursachen macht freilich die reflectirende Thätigkeit des Menschen, wie seine Phantasie, zu einer solchen Annahme geneigt.

The phænomenon here alluded to—that of “many groups moving in opposite directions”—is quite inexplicable by Mädler’s idea; but arises, as a necessary consequence, from that which forms the basis of this Discourse. While the _merely general direction_ of each atom—of each moon, planet, star, or cluster—would, on my hypothesis, be, of course, absolutely rectilinear; while the _general_ path of all bodies would be a right line leading to the centre of all; it is clear, nevertheless, that this general rectilinearity would be compounded of what, with scarcely any exaggeration, we may term an infinity of particular curves—an infinity of local deviations from rectilinearity—the result of continuous differences of relative position among the multitudinous masses, as each proceeded on its own proper journey to the End.

I quoted, just now, from Sir John Herschell, the following words, used in reference to the clusters:—“On one hand, without a rotary motion and a centrifugal force, it is hardly possible not to regard them as in a state of _progressive collapse_.” The fact is, that, in surveying the “nebulæ” with a telescope of high power, we shall find it quite impossible, having once conceived this idea of “collapse,” not to gather, at all points, corroboration of the idea. A nucleus is always apparent, in the direction of which the stars seem to be precipitating themselves; nor can these nuclei be mistaken for merely perspective phænomena:—the clusters are _really_ denser near the centre—sparser in the regions more remote from it. In a word, we see every thing as we _should_ see it were a collapse taking place; but, in general, it may be said of these clusters, that we can fairly entertain, while looking at them, the idea of _orbitual movement about a centre_, only by admitting the _possible_ existence, in the distant domains of space, of dynamical laws with which _we_ are unacquainted.

On the part of Herschell, however, there is evidently _a reluctance_ to regard the nebulæ as in “a state of progressive collapse.” But if facts—if even appearances justify the supposition of their being in this state, _why_, it may well be demanded, is he disinclined to admit it? Simply on account of a prejudice;—merely because the supposition is at war with a preconceived and utterly baseless notion—that of the endlessness—that of the eternal stability of the Universe.

If the propositions of this Discourse are tenable, the “state of progressive collapse” is _precisely_ that state in which alone we are warranted in considering All Things; and, with due humility, let me here confess that, for my part, I am at a loss to conceive how any _other_ understanding of the existing condition of affairs, could ever have made its way into the human brain. “The tendency to collapse” and “the attraction of gravitation” are convertible phrases. In using either, we speak of the rëaction of the First Act. Never was necessity less obvious than that of supposing Matter imbued with an ineradicable _quality_ forming part of its material nature—a quality, or instinct, _forever_ inseparable from it, and by dint of which inalienable principle every atom is _perpetually_ impelled to seek its fellow-atom. Never was necessity less obvious than that of entertaining this unphilosophical idea. Going boldly behind the vulgar thought, we have to conceive, metaphysically, that the gravitating principle appertains to Matter _temporarily_—only while diffused—only while existing as Many instead of as One—appertains to it by virtue of its state of irradiation alone—appertains, in a word, altogether to its _condition_, and not in the

slightest degree to _itself_. In this view, when the irradiation shall have returned into its source—when the rëaction shall be completed—the gravitating principle will no longer exist. And, in fact, astronomers, without at any time reaching the idea here suggested, seem to have been approximating it, in the assertion that “if there were but one body in the Universe, it would be impossible to understand how the principle, Gravity, could obtain:”—that is to say, from a consideration of Matter as they find it, they reach a conclusion at which I deductively arrive. That so pregnant a suggestion as the one just quoted should have been permitted to remain so long unfruitful, is, nevertheless, a mystery which I find it difficult to fathom.

It is, perhaps, in no little degree, however, our propensity for the continuous—for the analogical—in the present case more particularly for the symmetrical—which has been leading us astray. And, in fact, the sense of the symmetrical is an instinct which may be depended upon with an almost blindfold reliance. It is the poetical essence of the Universe—_of the Universe_ which, in the supremeness of its symmetry, is but the most sublime of poems. Now symmetry and consistency are convertible terms:—thus Poetry and Truth are one. A thing is consistent in the ratio of its truth—true in the ratio of its consistency. _A perfect consistency, I repeat, can be nothing but an absolute truth._ We may take it for granted, then, that Man cannot long or widely err, if he suffer himself to be guided by his poetical, which I have maintained to be his truthful, in being his symmetrical, instinct. He must have a care, however, lest, in pursuing too heedlessly the superficial symmetry of forms and motions, he leave out of sight the really essential symmetry of the principles which determine and control them.

That the stellar bodies would finally be merged in one—that, at last, all would be drawn into the substance of _one stupendous central orb already existing_—is an idea which, for some time past, seems, vaguely and indeterminately, to have held possession of the fancy of mankind. It is an idea, in fact, which belongs to the class of the _excessively obvious_. It springs, instantly, from a superficial observation of the cyclic and seemingly _gyrating_, or _vorticial_ movements of those individual portions of the Universe which come most immediately and most closely under our observation. There is not, perhaps, a human being, of ordinary education and of average reflective capacity, to whom, at some period, the fancy in question has not occurred, as if spontaneously, or intuitively, and wearing all the character of a very profound and very original conception. This conception, however, so commonly entertained, has never, within my knowledge, arisen out of any abstract considerations. Being, on the contrary, always suggested, as I say, by the vorticial movements about centres, a reason for it, also,—a _cause_ for the ingathering of all the orbs into one, _imagined to be already existing_, was naturally sought in the same direction—among these cyclic movements themselves.

Thus it happened that, on announcement of the gradual and perfectly regular decrease observed in the orbit of Enck’s comet, at every successive revolution about our Sun, astronomers were nearly unanimous in the opinion that the cause in question was found—that a principle was discovered sufficient to account, physically, for that final, universal agglomeration which, I repeat, the analogical, symmetrical or

poetical instinct of Man had predetermined to understand as something more than a simple hypothesis.

This cause—this sufficient reason for the final ingathering—was declared to exist in an exceedingly rare but still material medium pervading space; which medium, by retarding, in some degree, the progress of the comet, perpetually weakened its tangential force; thus giving a predominance to the centripetal; which, of course, drew the comet nearer and nearer at each revolution, and would eventually precipitate it upon the Sun.

All this was strictly logical—admitting the medium or ether; but this ether was assumed, most illogically, on the ground that no _other_ mode than the one spoken of could be discovered, of accounting for the observed decrease in the orbit of the comet:—as if from the fact that we could _discover_ no other mode of accounting for it, it followed, in any respect, that no other mode of accounting for it existed. It is clear that innumerable causes might operate, in combination, to diminish the orbit, without even a possibility of our ever becoming acquainted with one of them. In the meantime, it has never been fairly shown, perhaps, why the retardation occasioned by the skirts of the Sun’s atmosphere, through which the comet passes at perihelion, is not enough to account for the phænomenon. That Enck’s comet will be absorbed into the Sun, is probable; that all the comets of the system will be absorbed, is more than merely possible; but, in such case, the principle of absorption must be referred to eccentricity of orbit—to the close approximation to the Sun, of the comets at their perihelia; and is a principle not affecting, in any degree, the ponderous _spheres_, which are to be regarded as the true material constituents of the Universe.—Touching comets, in general, let me here suggest, in passing, that we cannot be far wrong in looking upon them as the _lightning-flashes of the cosmical Heaven_.

The idea of a retarding ether and, through it, of a final agglomeration of all things, seemed at one time, however, to be confirmed by the observation of a positive decrease in the orbit of the solid moon. By reference to eclipses recorded 2500 years ago, it was found that the velocity of the satellite’s revolution _then_ was considerably less than it is _now_; that on the hypothesis that its motions in its orbit is uniformly in accordance with Kepler’s law, and was accurately determined _then_—2500 years ago—it is now in advance of the position it _should_ occupy, by nearly 9000 miles. The increase of velocity proved, of course, a diminution of orbit; and astronomers were fast yielding to a belief in an ether, as the sole mode of accounting for the phænomenon, when Lagrange came to the rescue. He showed that, owing to the configurations of the spheroids, the shorter axes of their ellipses are subject to variation in length; the longer axes being permanent; and that this variation is continuous and vibratory—so that every orbit is in a state of transition, either from circle to ellipse, or from ellipse to circle. In the case of the moon, where the shorter axis is _de_creasing, the orbit is passing from circle to ellipse and, consequently, is _de_creasing too; but, after a long series of ages, the ultimate eccentricity will be attained; then the shorter axis will proceed to _in_crease, until the orbit becomes a circle; when the process of shortening will again take place;—and so on forever. In

the case of the Earth, the orbit is passing from ellipse to circle. The facts thus demonstrated do away, of course, with all necessity for supposing an ether, and with all apprehension of the system's instability—on the ether's account.

It will be remembered that I have myself assumed what we may term _an ether_. I have spoken of a subtle _influence_ which we know to be ever in attendance upon matter, although becoming manifest only through matter's heterogeneity. To this _influence_—without daring to touch it at all in any effort at explaining its awful _nature_—I have referred the various phænomena of electricity, heat, light, magnetism; and more—of vitality, consciousness, and thought—in a word, of spirituality. It will be seen, at once, then, that the ether thus conceived is radically distinct from the ether of the astronomers; inasmuch as theirs is _matter_ and mine _not_.

With the idea of a material ether, seems, thus, to have departed altogether the thought of that universal agglomeration so long predetermined by the poetical fancy of mankind:—an agglomeration in which a sound Philosophy might have been warranted in putting faith, at least to a certain extent, if for no other reason than that by this poetical fancy it _had_ been so predetermined. But so far as Astronomy—so far as mere Physics have yet spoken, the cycles of the Universe are perpetual—the Universe has no conceivable end. Had an end been demonstrated, however, from so purely collateral a cause as an ether, Man's instinct of the Divine _capacity to adapt_, would have rebelled against the demonstration. We should have been forced to regard the Universe with some such sense of dissatisfaction as we experience in contemplating an unnecessarily complex work of human art. Creation would have affected us as an imperfect _plot_ in a romance, where the _dénoûment_ is awkwardly brought about by interposed incidents external and foreign to the main subject; instead of springing out of the bosom of the thesis—out of the heart of the ruling idea—instead of arising as a result of the primary proposition—as inseparable and inevitable part and parcel of the fundamental conception of the book.

What I mean by the symmetry of mere surface will now be more clearly understood. It is simply by the blandishment of this symmetry that we have been beguiled into the general idea of which Mädler's hypothesis is but a part—the idea of the vorticial indrawing of the orbs. Dismissing this nakedly physical conception, the symmetry of principle sees the end of all things metaphysically involved in the thought of a beginning; seeks and finds in this origin of all things the _rudiment_ of this end; and perceives the impiety of supposing this end likely to be brought about less simply—less directly—less obviously—less artistically—than through _the rëaction of the originating Act_.

Recurring, then, to a previous suggestion, let us understand the systems—let us understand each star, with its attendant planets—as but a Titanic atom existing in space with precisely the same inclination for Unity which characterized, in the beginning, the actual atoms after their irradiation throughout the Universal sphere. As these original atoms rushed towards each other in generally straight lines, so let us conceive as at least generally rectilinear, the paths of the system-atoms towards their respective centres of aggregation:—and in this direct drawing together of the

systems into clusters, with a similar and simultaneous drawing together of the clusters themselves while undergoing consolidation, we have at length attained the great _Now_—the awful Present—the Existing Condition of the Universe.

Of the still more awful Future a not irrational analogy may guide us in framing an hypothesis. The equilibrium between the centripetal and centrifugal forces of each system, being necessarily destroyed upon attainment of a certain proximity to the nucleus of the cluster to which it belongs, there must occur, at once, a chaotic or seemingly chaotic precipitation, of the moons upon the planets, of the planets upon the suns, and of the suns upon the nuclei; and the general result of this precipitation must be the gathering of the myriad now-existing stars of the firmament into an almost infinitely less number of almost infinitely superior spheres. In being immeasurably fewer, the worlds of that day will be immeasurably greater than our own. Then, indeed, amid unfathomable abysses, will be glaring unimaginable suns. But all this will be merely a climacic magnificence foreboding the great End. Of this End the new genesis described, can be but a very partial postponement. While undergoing consolidation, the clusters themselves, with a speed prodigiously accumulative, have been rushing towards their own general centre—and now, with a thousand-fold electric velocity, commensurate only with their material grandeur and with the spiritual passion of their appetite for oneness, the majestic remnants of the tribe of Stars flash, at length, into a common embrace. The inevitable catastrophe is at hand.

But this catastrophe—what is it? We have seen accomplished the ingathering of the orbs. Henceforward, are we not to understand _one material globe of globes_ as constituting and comprehending the Universe? Such a fancy would be altogether at war with every assumption and consideration of this Discourse.

I have already alluded to that absolute _reciprocity of adaptation_ which is the idiosyncrasy of the divine Art—stamping it divine. Up to this point of our reflections, we have been regarding the electrical influence as a something by dint of whose repulsion alone Matter is enabled to exist in that state of diffusion demanded for the fulfilment of its purposes:—so far, in a word, we have been considering the influence in question as ordained for Matter’s sake—to subserve the objects of matter. With a perfectly legitimate reciprocity, we are now permitted to look at Matter, as created _solely for the sake of this influence_—solely to serve the objects of this spiritual Ether. Through the aid—by the means—through the agency of Matter, and by dint of its heterogeneity—is this Ether manifested—is _Spirit individualized_. It is merely in the development of this Ether, through heterogeneity, that particular masses of Matter become animate—sensitive—and in the ratio of their heterogeneity;—some reaching a degree of sensitiveness involving what we call _Thought_ and thus attaining Conscious Intelligence.

In this view, we are enabled to perceive Matter as a Means—not as an End. Its purposes are thus seen to have been comprehended in its diffusion; and with the return into Unity these purposes cease. The absolutely consolidated globe of globes would be _objectless_:—therefore not for a moment could it continue to exist. Matter, created for an end, would unquestionably, on fulfilment of that end, be Matter no

longer. Let us endeavor to understand that it would disappear, and that God would remain all in all.

That every work of Divine conception must cöexist and cöexpire with its particular design, seems to me especially obvious; and I make no doubt that, on perceiving the final globe of globes to be _objectless_, the majority of my readers will be satisfied with my "_therefore_ it cannot continue to exist." Nevertheless, as the startling thought of its instantaneous disappearance is one which the most powerful intellect cannot be expected readily to entertain on grounds so decidedly abstract, let us endeavor to look at the idea from some other and more ordinary point of view:—let us see how thoroughly and beautifully it is corroborated in an _à posteriori_ consideration of Matter as we actually find it.

I have before said that "Attraction and Repulsion being undeniably the sole properties by which Matter is manifested to Mind, we are justified in assuming that Matter _exists_ only as Attraction and Repulsion—in other words that Attraction and Repulsion _are_ Matter; there being no conceivable case in which we may not employ the term Matter and the terms 'Attraction' and 'Repulsion' taken together, as equivalent, and therefore convertible, expressions in Logic."[14]

[14] Page 37.

Now the very definition of Attraction implies particularity—the existence of parts, particles, or atoms; for we define it as the tendency of "each atom &c. to every other atom" &c. according to a certain law. Of course where there are _no_ parts—where there is absolute Unity—where the tendency to oneness is satisfied—there can be no Attraction:—this has been fully shown, and all Philosophy admits it. When, on fulfilment of its purposes, then, Matter shall have returned into its original condition of _One_—a condition which presupposes the expulsion of the separative ether, whose province and whose capacity are limited to keeping the atoms apart until that great day when, this ether being no longer needed, the overwhelming pressure of the finally collective Attraction shall at length just sufficiently predominate[15] and expel it:—when, I say, Matter, finally, expelling the Ether, shall have returned into absolute Unity,—it will then (to speak paradoxically for the moment) be Matter without Attraction and without Repulsion—in other words, Matter without Matter—in other words, again, _Matter no more_. In sinking into Unity, it will sink at once into that Nothingness which, to all Finite Perception, Unity must be—into that Material Nihility from which alone we can conceive it to have been evoked—to have been _created_ by the Volition of God.

[15] "Gravity, therefore, must be the strongest of forces."—See page 39.

I repeat then—Let us endeavor to comprehend that the final globe of globes will instantaneously disappear, and that God will remain all in all.

But are we here to pause? Not so. On the Universal agglomeration and dissolution, we can readily conceive that a new and perhaps totally different series of conditions may ensue—another creation and irradiation, returning into itself—another action and rëaction of the Divine Will. Guiding our imaginations by that omniprevalent law of laws, the law of periodicity, are we not, indeed, more than justified in entertaining

a belief—let us say, rather, in indulging a hope—that the processes we have here ventured to contemplate will be renewed forever, and forever, and forever; a novel Universe swelling into existence, and then subsiding into nothingness, at every throb of the Heart Divine?

And now—this Heart Divine—what is it? _It is our own._

Let not the merely seeming irreverence of this idea frighten our souls from that cool exercise of consciousness—from that deep tranquillity of self-inspection—through which alone we can hope to attain the presence of this, the most sublime of truths, and look it leisurely in the face.

The _phænomena_ on which our conclusions must at this point depend, are merely spiritual shadows, but not the less thoroughly substantial.

We walk about, amid the destinies of our world-existence, encompassed by dim but ever present _Memories_ of a Destiny more vast—very distant in the by-gone time, and infinitely awful.

We live out a Youth peculiarly haunted by such dreams; yet never mistaking them for dreams. As Memories we _know_ them. _During our Youth_ the distinction is too clear to deceive us even for a moment.

So long as this Youth endures, the feeling _that we exist_, is the most natural of all feelings. We understand it _thoroughly_. That there was a period at which we did _not_ exist—or, that it might so have happened that we never had existed at all—are the considerations, indeed, which _during this youth_, we find difficulty in understanding. Why we should _not_ exist, is, _up to the epoch of our Manhood_, of all queries the most unanswerable. Existence—self-existence—existence from all Time and to all Eternity—seems, up to the epoch of Manhood, a normal and unquestionable condition:—_seems, because it is_.

But now comes the period at which a conventional World-Reason awakens us from the truth of our dream. Doubt, Surprise and Incomprehensibility arrive at the same moment. They say:—“You live and the time was when you lived not. You have been created. An Intelligence exists greater than your own; and it is only through this Intelligence you live at all.” These things we struggle to comprehend and cannot:—_cannot_, because these things, being untrue, are thus, of necessity, incomprehensible.

No thinking being lives who, at some luminous point of his life of thought, has not felt himself lost amid the surges of futile efforts at understanding, or believing, that anything exists _greater than his own soul_. The utter impossibility of any one’s soul feeling itself inferior to another; the intense, overwhelming dissatisfaction and rebellion at the thought;—these, with the omniprevalent aspirations at perfection, are but the spiritual, coincident with the material, struggles towards the original Unity—are, to my mind at least, a species of proof far surpassing what Man terms demonstration, that no one soul _is_ inferior to another—that nothing is, or can be, superior to any one soul—that each soul is, in part, its own God—its own Creator:—in a word, that God—the material _and_ spiritual God—_now_ exists solely in the

diffused Matter and Spirit of the Universe; and that the regathering of this diffused Matter and Spirit will be but the re-constitution of the _purely_ Spiritual and Individual God.

In this view, and in this view alone, we comprehend the riddles of Divine Injustice—of Inexorable Fate. In this view alone the existence of Evil becomes intelligible; but in this view it becomes more—it becomes endurable. Our souls no longer rebel at a _Sorrow_ which we ourselves have imposed upon ourselves, in furtherance of our own purposes—with a view—if even with a futile view—to the extension of our own _Joy_.

I have spoken of _Memories_ that haunt us during our youth. They sometimes pursue us even in our Manhood:—assume gradually less and less indefinite shapes:—now and then speak to us with low voices, saying:

"There was an epoch in the Night of Time, when a still-existent Being existed—one of an absolutely infinite number of similar Beings that people the absolutely infinite domains of the absolutely infinite space.[16] It was not and is not in the power of this Being—any more than it is in your own—to extend, by actual increase, the joy of his Existence; but just as it _is_ in your power to expand or to concentrate your pleasures (the absolute amount of happiness remaining always the same) so did and does a similar capability appertain to this Divine Being, who thus passes his Eternity in perpetual variation of Concentrated Self and almost Infinite Self-Diffusion. What you call The Universe is but his present expansive existence. He now feels his life through an infinity of imperfect pleasures—the partial and pain-intertangled pleasures of those inconceivably numerous things which you designate as his creatures, but which are really but infinite individualizations of Himself. All these creatures—_all_—those which you term animate, as well as those to whom you deny life for no better reason than that you do not behold it in operation—_all_ these creatures have, in a greater or less degree, a capacity for pleasure and for pain:—_but the general sum of their sensations is precisely that amount of Happiness which appertains by right to the Divine Being when concentrated within Himself_. These creatures are all, too, more or less conscious Intelligences; conscious, first, of a proper identity; conscious, secondly and by faint indeterminate glimpses, of an identity with the Divine Being of whom we speak—of an identity with God. Of the two classes of consciousness, fancy that the former will grow weaker, the latter stronger, during the long succession of ages which must elapse before these myriads of individual Intelligences become blended—when the bright stars become blended—into One. Think that the sense of individual identity will be gradually merged in the general consciousness—that Man, for example, ceasing imperceptibly to feel himself Man, will at length attain that awfully triumphant epoch when he shall recognize his existence as that of Jehovah. In the meantime bear in mind that all is Life—Life—Life within Life—the less within the greater, and all within the _Spirit Divine_."

[16] See pages 102-103—Paragraph commencing "I reply that the right," and ending "proper and particular God."

THE END.

155 Broadway, NEW YORK. 142 Strand, LONDON.

Of late firm of WILEY & PUTNAM.

New Works in Press,

Or recently published, by

GEORGE P. PUTNAM,

155 Broadway, New York.

G. P. PUTNAM has the pleasure of announcing that, agreeably to his contract with the distinguished author, he has now in the course of publication

A new, uniform, and complete edition

OF THE

Works of Washington Irving,

Revised and enlarged by the Author,

In Twelve Elegant Duodecimo Volumes,

Beautifully printed with new type, and on superior paper, made expressly for the purpose.

The first volume of the Series will be

The Sketch-Book,

complete in one volume,

which will be ready on the first day of September.

Knickerbocker’s History of New York,

with revisions and copious additions,

will be published on the 1st of October.

The Life and Voyages of Columbus,

Vol. I. on the 1st of November,

and the succeeding volumes will be issued on the first day of each month until completed;—as follows:

The Sketch-Book, in one volume. Knickerbocker’s New York, in one volume. Tales of a Traveller, in one volume. Bracebridge Hall, in one volume. The Conquest of Grenada, in one volume. The Alhambra, in one volume. The Spanish Legends, in one vol. The Crayon Miscellany, in one vol.—Abbotsford, Newstead, The Prairies, &c. Life and Voyages of Columbus, and The Companions of Columbus, 2 vols. Adventures of Captain Bonneville, one vol. Astoria, one volume.

The Illustrated Sketch-Book.

In October will be published,

The Sketch-Book.

BY WASHINGTON IRVING.

One volume, square octavo.

Illustrated with a series of highly-finished Engravings on wood, from Designs by Darley and others, engraved in the best style by Childs, Herrick, &c. This edition will be printed on paper of the finest quality, similar in size and style to the new edition of "Halleck's Poems." It is intended that the illustrations shall be superior to any engravings on wood yet produced in this country, and that the mechanical execution of the volume, altogether, shall be worthy of the author's reputation. It will form an elegant and appropriate gift-book for all seasons.

The Illustrated Knickerbocker,

With a series of Original Designs, in one vol., octavo, is also in preparation.

Mr. Putnam has also the honor to announce that he will publish at intervals (in connexion, and uniform with the other collected writings),

Mr. Irving's New Works,

now nearly ready for the press: including

The Life of Mohammed; The Life of Washington; new volumes of Miscellanies, Biographies, &c.

⁂ This being the first uniform and complete edition of Mr. Irving's works, either in this country or in Europe, the publisher confidently believes that the undertaking will meet with a prompt and cordial response. To say this, is perhaps superfluous and impertinent; for it is a truism that no _American_ book-case (not to say _library_) can be well filled without the works of Washington Irving; while the English language itself comprises no purer models of composition.

G. P. Putnam has also made arrangements for the early commencement of new works or new editions of the works of

Miss C. M. Sedgwick, Prof. A. Gray, Leigh Hunt, Chas. Fenno Hoffman, Mrs. E. Oakes Smith, Thomas Carlyle, George H. Calvert, Mrs. C. M. Kirkland, R. Monckton Milnes, J. Bayard Taylor, Mary Howitt, Mrs. Jameson, S. Wells Williams, W. M. Thackeray, Charles Lamb, A. J. Downing, Thos. Hood, Elliot Warburton.

The following new works are now ready, or will be published this season:

I.

Sophisms of the Protective Policy.

Translated from the French of F. Bastiat. With an introduction by Francis Lieber, LL.D. Professor in South Carolina College, Editor of the Encyclopedia Americana, &c. 12mo. 75 cents.

“It is a book not for the million but for millions, and we believe if a copy could be put into the hands of every school-boy in the Union, the next generation would be inconceivably wiser, richer, and happier than the present.”—_Mirror._

II.

Grecian and Roman Mythology:

With original illustrations. Adapted for the use of Universities and High Schools, and for popular reading. By M. A. Dwight. With an introduction by Tayler Lewis, Professor of Greek, University of New York. 12mo. (On 1st September.)

Also a fine edition in octavo, with illustrations.

⁂ This work has been prepared with great care, illustrated with 20 effective outline drawings, and is designed to treat the subject in an original, comprehensive, and unexceptionable manner, so as to fill the place as a text book which is yet unsupplied; while it will also be an attractive and readable table book for general use. It will be at once introduced as a text book in the University of New York and other colleges and schools.

III.

Eureka: a Prose Poem.

Or the Physical and Metaphysical Universe.

By Edgar A. Poe, Esq. Handsomely printed, 12mo. Cloth, 75 cents.

“A most extraordinary Essay. We shall be greatly surprised if this work does not create a most profound sensation among the literary and scientific classes.”—_New York Express._

IV.

Oriental Life Illustrated.

Being a new edition of Eöthen, or Traces of Travel in the East. With fine illustrations on Steel. 12mo. elegantly bound, $1 50.

⁂ This new and unique volume, superbly illuminated by Mapleson, and comprising original articles by distinguished writers, will be the most elegant and recherché book of the kind ever produced in this country. It will be ready in October.

A new and superior edition of the PEARLS OF AMERICAN POETRY will also be published this season.

V.

The Book of Dainty Devices.

In an elegant small folio volume.

Lays of the Western World.

VI.

Dr. Klipstein’s Anglo-Saxon Course of Study.

In uniform 12mo. volumes.

I.

A Grammar of the Anglo-Saxon Language. By Louis F. Klipstein, AA.LL.M. and PH.D., of the University of Giessen.

⁂ This work recommends itself particularly to the attention of every American student who “glories in his Anglo-Saxon descent” or Teutonic lineage, as well as of all who desire an acquaintance with a language which lies as the foundation of the English, and throws a light upon its elements and structure, derivable from no other source. Of the importance and interesting nature of the study there can be no doubt, and we agree with those who think that the time is coming when it will be considered “utterly disgraceful for any well-bred Englishman or American” to have neglected it. With regard to the merits of Dr. Klipstein’s Grammar, we will only say, that it has been already adopted as a text-book in some of the leading Institutions of our country.

[The following are also in press.]

II.

Analecta Anglo-Saxonica, with an Introductory Ethnographical Essay, Copious Notes, Critical and Explanatory, and a Glossary in which are shown the Indo-Germanic and other Affinities of the Language. _By the same._

In this work appear the fruits of considerable research, and, we may add, learning. The Ethnology of Europe is succinctly, but clearly illustrated, the Anglo-Saxon language completely analysed, revealing the utmost harmony of combination from its elements, its forms and roots compared with those in kindred dialects and cognate tongues, its position in the Teutonic family and Indo-Germanic range established, and the genuine relation of the English to its great parent properly set forth. To those who are fond of the comparative study of language, the Glossary will prove an invaluable aid, apart from its particular object.

III.

Natale Sancti Gregorii Papæ.—Ælfric’s Homily on the Birth-day of St. Gregory, and Collateral Extracts from King Alfred’s version of Bede’s Ecclesiastical History and the Saxon Chronicle, with a full rendering into English, Notes Critical and Explanatory,

and an Index of Words. _By the same._

IV.

Extracts from the Anglo-Saxon-Gospels, a Portion of the Anglo-Saxon Paraphrase of the Book of Psalms, and other Selections of a Sacred Order in the same Language, with a Translation into English, and Notes Critical and Explanatory. _By the same._

These two works are prepared in such a way as in themselves, with the aid of the Grammar, to afford every facility to the Anglo-Saxon Student. Ælfric's Homily is remarkable for beauty of composition, and interesting as setting forth Augustine's Mission to the "Land of the Angles."

V.

Tha Halgan Godspel on Englisc—the Anglo-Saxon Version of the Holy Gospels. Edited by Benjamin Thorpe, F.S.A. _Reprinted by the same. Now ready._

This, the earliest "English" version of the Four Gospels, will be found interesting to the antiquarian and theologian, as well as serviceable to the student in his investigations of the language. The Text, besides the usual but unbroken division, appears, with the Rubrics, as read in the early Anglican Church.

Nearly Ready.

Dr. Bosworth's Compendious Anglo-Saxon Dictionary. Small 8vo.

VII.

Study of Modern Languages.

Part First; French, Italian, Spanish, Portuguese, German, and English.

By L. F. Klipstein, AA.LL.M. and Ph.D. One Vol. Imperial 8vo. 75 cents paper; $1 00 cloth.

This work, which is intended equally for the simultaneous and the separate study of the languages that it sets forth, and which is adapted as well for the native of Germany, France, Italy, Spain, or Portugal, as for him to whom English is vernacular, in the acquirement of any one of the other tongues besides his own, will be found an acceptable manual not only to the tyro, but to the more advanced scholar. The reading portion of the matter is interesting, and the text in every case remarkably correct, while the Elementary Phrases, forms of Cards, Letters, Bills of Exchange, Promissory Notes, Receipts, &c., in the six languages, constitute what has long been a desideratum from the American press. For the comparative study of the _Romanic_ tongues the work affords unusual facilities.

VIII.

Pedestrian Tour in Europe.

Views a-Foot; or Europe seen with Knapsack and Staff.

By J. Bayard Taylor.

A new edition with an additional chapter, and a sketch of the author in pedestrian costume, from a drawing by T. Buchanan Read. 12mo. Cloth.

IX.

A New Edition of

Clarke’s Shakspeare Concordance.

A Complete Concordance to Shakspeare: being a Verbal Index to ALL the PASSAGES in the Dramatic Works of the Poet. By Mrs. Cowden Clarke.

“Order gave each thing view.”

One large Vol. comprising 2560 closely printed columns,—(indicating _every word and passage_ in Shakspeare’s Works). Price $6. Cloth.

“The result of sixteen years of untiring labor. The different editions of Shakspeare have been carefully collated by the compiler, and every possible means taken to insure the correctness of the work. As it now stands, a person can find a particular passage in Shakspeare by simply remembering one word of it, and is also referred to the act and scene of the play in which it occurs. As a mere dictionary of Shakspearian language and phrases, it is of great value; but it is also a dictionary of his thoughts and imaginations. It altogether supersedes the volumes of Twiss and Ayscough, and should be on every student’s shelves”—_Boston Courier._

⁂ This extraordinary work is printed in London and the price there _at present_ is £2. 5s. 0d. or about $12. A large part of the edition having been purchased for this market, it is furnished here for the very low price of $6, bound in cloth.

Also—By same Author.

The Book of Shakspeare Proverbs.

18mo. 75 cts.

Dr. Lieber’s Poetical Address to the American Republic.

16mo. 25 cents.

The West:

A Metrical Epistle.

BY FRANCIS LIEBER.

⁂ Dr. Lieber, the distinguished Professor of Political Economy in South Carolina College, Author of “Political Ethics,” &c., has just sailed for his native

country—Germany—with the view of aiding in the great cause of Constitutional and Rational Freedom. This little volume proves that he has well studied that subject during his long residence in this his adopted country—and his able and valuable opinions on American Society and Progress, carry with them a peculiar interest at this time.

RECENT PUBLICATIONS.

Alexander.—Commentary on the Earlier Prophecies of Isaiah. By Prof. J. A. Alexander. Royal 8vo. cloth, $3.

Alexander.—Commentary on the Later Prophecies of Isaiah. By Prof. J. A. Alexander. Royal 8vo. cloth, $2 50.

Ancient Moral Tales, from the Gesta Romanorum, &c. 1 vol. 12mo. green cloth.

"A quiet humor, a quaintness and terseness of style, will strongly recommend them."—_English Churchman._

Architecture.—Hints on Public Architecture; issued under the Direction of the "Smithsonian Institution." Imperial 4to. with Illustrations. (In preparation.)

This work will contain numerous and valuable illustrations, including two perspective views of the buildings of the Smithsonian Institution. The Appendix will contain the results of a research under the auspices of the Institution to test the properties of the most important building materials throughout the United States.

Bastiat.—Sophisms of the Protective Policy. Translated from the French of F. Bastiat. With an Introduction, by Francis Lieber, LL.D., Professor in South Carolina College, Editor of the Encyclopædia Americana, &c., &c. 12mo. 75 cts.

Bibliotheca Sacra and Theological Review. Conducted by B. B. Edwards and E. A. Park, Professors at Andover, with the Special Aid of Dr. Robinson and Professor Stuart. Published quarterly in February, May, August, and November $4 per annum. Vols. 1, 2, 3, and 4, 8vo. cloth, each $4.

"This is, perhaps, the most ambitious journal in the United States. We use the word in a good sense, as meaning that there is no journal among us which seems more laudably desirous to take the lead in literary and theological science. Its handsome type and paper give it a pleasing exterior; its typographical errors, though sufficiently numerous, are so comparatively few, as to show that it has the advantage of the best American proof-reading; while for thoroughness of execution in the departments of history and criticism, it aims to be pre-eminent."—_N. Y. Churchman._

Burton.—The Anatomy of Melancholy. By Burton. New and beautiful edition, with Engravings. 1 vol. royal 8vo. cloth, $2 50.

⁂ This is one of those sterling old works which were written for "all time," full of learning, humor, and quaint conceits. No library can be complete without it.

Calvert.—Scenes and Thoughts in Europe. By an American. 1 vol. 12mo. green cloth, 50 cents.

“His descriptions of scenery, his remarks on art, his accounts of the different people among whom he sojourned, are all good.”—_Cincinnati Gazette._

Carlyle.—The French Revolution: a History. By Thomas Carlyle. 2 vols. 12mo. green cloth, $2.

“His French Revolution is considered one of the most remarkable works of the age—as at once the poetry and philosophy of history.”—_Hunt’s Merchants’ Mag._

Carlyle.—Letters and Speeches of Oliver Cromwell. By Thos. Carlyle. 2 vols. 12mo. green cloth, $2 50.

“A work more valuable as a guide to the study of the singular and complex character of our pious revolutionist, our religious demagogue, our preaching and praying warrior, has not been produced.”—_Blackwood’s Magazine._

Carlyle.—Past and Present: Chartism. By Thomas Carlyle. 1 vol. 12mo. green cloth, $1

“To say that the book is replete with instruction, thought, and quaint fancy, is unnecessary: but we may mention it as one, _par excellence_, which should be read at the present juncture.”-_Tribune._

Chaucer and Spenser.—Selections from the Poetical Works of Geoffrey Chaucer. By Charles D. Deshler. Spenser, and the Faery Queen. By Mrs. C. M. Kirkland. 1 vol. 12mo. $1 13.

—— The same, extra gilt, $1 50.

“A portion of their writings are presented in a beautiful and convenient form, and with the requisite notes and modifications.”—_Home Journal._

Coe.—Studies in Drawing, in a Progressive Series of Lessons on Cards; beginning with the most Elementary Studies, and Adapted for Use at Home and Schools. By Benjamin H. Coe, Teacher of Drawing. In Ten Series—marked 1 and 10—each containing about eighteen Studies. 25 cents each.

The design is:

I.—To make the exercises in drawing highly interesting to the pupil.

II.—To make drawings so simple, and so gradually progressive, as to enable any teacher, whether acquainted with drawing or not, to instruct his pupils to advantage.

III.—To take the place of one-half of the writing lessons, with confidence that the learner will acquire a knowledge of writing in less than time is usually required.

IV.—To give the pupils a bold, rapid, and artist-like style of drawing.

Coleridge.—Biographia Literaria; or, Biographical Sketches of my Literary Life and Opinions. By Samuel Taylor Coleridge. From the 2d London edition, Edited by H. N. Coleridge. 2 vols. 12mo. green cloth, $2.

Cortez.—Letters and Despatches of Hernando Cortez. Translated by Hon. George Folsom. 1 vol. 8vo. $1 25.

Dana.—A System of Mineralogy, comprising the most Recent Discoveries. By James D. Dana. Woodcuts and copperplates, 8vo. cloth, $3 50.

Downing.—Cottage Residences; or, a Series of Designs for Rural Cottages and Cottage Villas, and their Gardens and Grounds; adapted to North America. By A. J. Downing. Numerous plates, 3d edition, 8vo. cloth, $2.

Downing.—A Treatise on the Theory and Practice of Landscape Gardening adapted to North America; with Remarks on Rural Architecture. By A. J. Downing. Plates, 2d edition, thick 8vo. cloth, $3 50.

Downing.—The Fruits and Fruit Trees of America; or, the Culture, Propagation, and Management, in the Garden and Orchard, of Fruit Trees generally. By A. J. Downing. Plates, 9th edition, revised, 12mo. cloth, $1 50.

—— The same, 8vo. cloth, $2 50.

—— The same, with 80 superb Illustrations, drawn and beautifully colored by Paris Artists, royal 8vo. half morocco, top edge gilt. New edition shortly.

Dwight.—Grecian and Roman Mythology; with original Illustrations. Adapted for the Use of Universities and High Schools, and for Popular Reading. By M. A. Dwight. With an Introduction by Tayler Lewis, Professor of Greek, University of New York. 12mo. [In September.

—— Also a fine edition in octavo, with Illustrations.

⁂ This work has been prepared with great care, illustrated with twenty effective outline drawings, and is designed to treat the subject in an original, comprehensive, and unexceptionable manner, so as to fill the place as a text-book which is yet unsupplied; while it will also be an attractive and readable table-book for general use. It will be at once introduced as a text-book in the University of New York, and other colleges and schools.

Ford.—The Spaniards and their Country. By Richard Ford. 1 vol. 12mo. green cloth, 87 cents.

“The best description of national character and manners of Spain that has ever appeared.”—_Quarterly Review._

“The volumes appear to treat of almost everything save the graver questions of religion and politics, which may possibly be taken up hereafter. In one respect it has the advantage over more directly historical works—it portrays the Spanish character, as well as country, with fidelity.”—_Commercial Advertiser._

Fouqué.—Undine, a Tale; and Sintram and his Companions, a Tale. From the German of La Motte Fouqué. 1 vol. 12mo. green cloth. 50 cts.

“The style and execution of this delightful romance are very graceful.”—_Hawkins’s Germany._

“Fouqué’s romances I always recommend—especially the wild, graceful, and touching Undine.”—_Sarah Austin._

French.—Historical Collections of Louisiana. By B. F. French. 8vo. cloth, $1 50.

Goldsmith.—The Vicar of Wakefield. By Oliver Goldsmith. 1 vol. 12mo. neatly printed, cloth, 50 cents.

—— The same, with Illustrated Designs by Mulready, elegantly bound, gilt edges, $1.

Gray.—Botanical Text-Book. By Prof. Asa Gray. Many hundred cuts, 2d edition, large 12mo. cloth, $1 75.

Green.—A Treatise on Diseases of the Air Passages; comprising an Inquiry into the History, Pathology, Causes, and Treatment of those Affections of the Throat called Bronchitis, &c. By Horace Green, M.D. Colored plates, 8vo. cloth. $2 50.

“A new and eminently successful treatment of lung complaints.”

Hackley.—Elements of Trigonometry, Plane and Spherical. By Rev. C. W. Hackley, Professor of Mathematics, Columbia College, New York. 8vo. cloth, $1 25.

Hamilton Papers.—The Official Papers of the late Major-General Alexander Hamilton. Compiled from the Originals in the Possession of Mrs. Hamilton. 1 vol. 8vo. cloth, $2 50.

Hahn’s Hebrew Bible.—New and complete stereotype edition, being a fac-simile of the Leipsic edition. In 1 vol. 8vo. In press.

Hazlitt’s (William) Miscellaneous Works. 4 vols. 12mo. cloth, $5.

Hazlitt’s Life of Napoleon. 3 vols. 12mo. cloth.

—— Spirit of the Age. 12mo., 50 cents.

—— Table Talk, both series, in 2 vols. cloth, $2 25.

—— Characters of Shakspeare, 12mo. 50 cts.

—— Literature of the Age of Queen Elizabeth, 12mo. 50 cts.

—— English Comic Writers, 50 cts.

—— Lectures on English Poets, 50 cts.

Head.—Bubbles from the Brunnen. By Sir Francis Head. 12mo. green cloth.

“At once an instructive and amusing book. It contains a great deal of information.”—_London Times._

Hervey.—The Book of Christmas; descriptive of the Customs, Ceremonies, Traditions, Superstitions, Fun, Feeling, and Festivities of the Christmas Season. By Thomas K. Hervey. 12mo. green cloth, 63 cents.

—— The same, gilt extra. $1.

“Every leaf of this book affords a feast worthy of the season.”—_Dr. Hawks’s Church Record._

Hood.—Prose and Verse. By Thomas Hood. 12mo. green cloth. 87 cents.

—— The same, gilt extra, $1 25.

“A very judicious selection, designed to embrace Hood’s more earnest writings, those which were written most directly from the heart, which reflect most faithfully his life and opinions.”—_Broadway Journal._

Howitt.—Ballads and other Poems. By Mary Howitt. 1 vol. 12mo. green cloth, 63 cents.

—— The same, with fine Portrait, gilt extra, $1 25.

“Her poems are always graceful and beautiful.”—_Mrs. S. C. Hall._

“We cannot commend too highly the present publication, and only hope that the reading public will relish ‘Mary Howitt’s Ballads and other Poems,’ now for the first time put forth in a collected form.”—_Albion._

Hunt.—Imagination and Fancy. By Leigh Hunt. 1 vol. 12mo. green cloth, 62 cents.

—— The same, gilt extra, $1.

Hunt.—Stories from the Italian Poets: being a Summary in Prose of the Poems of Dante, Pulci, Boiardo, Aristo, and Tasso; with Comments throughout, occasional passages Versified, and Critical Notices of the Lives and Genius of the Authors. By Leigh Hunt. 12mo. cloth, $1 25.

—— The same, fancy gilt. $1 75.

“Mr. Hunt’s book has been aptly styled, a series of exquisite engravings of the magnificent pictures painted by these great Italian masters.”—_Journal of Commerce._

Irving.—Works of Washington Irving; Revised and Enlarged by the Author. In twelve elegant duodecimo volumes, beautifully printed with new type, and on superior paper, made expressly for the purpose, and bound in cloth.

As follows:—

The Sketch-Book, in one volume. _Knickerbocker’s New York_, in one volume. _Tales of a Traveller_, in one vol. _Bracebridge Hall_, in one volume. _The Conquest of Grenada_, in one volume. _The Alhambra_, in one volume. _Astoria_, in one volume. _The Crayon Miscellany_, in one volume. Abbotsford, Newstead, The Prairies, &c. _The Spanish Legends_, in one vol. _The Life and Voyages of Columbus_, and _The Companions of Columbus_, in two volumes. _Adventures of Capt. Bonneville_, in one volume.

(Now publishing.)

Irving.—The Sketch-Book. By Washington Irving. Complete in one volume, 12mo. cloth. In September.

Irving.—The Illustrated Sketch-Book. By Washington Irving. In October will be published, THE SKETCH-BOOK, by Washington Irving, one vol. square octavo, Illustrated with a series of highly-finished Engravings on Wood, from Designs by DARLEY and others, engraved in the best style by CHILDS, HERRICK, &c. This

edition will be printed on paper of the finest quality, similar in size and style to the new edition of “Halleck’s Poems.” It is intended that the illustrations shall be superior to any engravings on wood yet produced in this country, and that the mechanical execution of the volume, altogether, shall be worthy of the author’s reputation. It will form an elegant and appropriate gift-book for all seasons.

Irving.—Knickerbocker’s History of New York. By Washington Irving. With Revisions and copious Additions. Will be published on the 1st of October.

Irving.—The Illustrated Knickerbocker; with a series of original Designs, in one volume, octavo, uniform with the “Sketch-Book,” is also in preparation.

Irving.—The Life and Voyages of Columbus. By Washington Irving. Vol. I. on the 1st of November.

The succeeding volumes will be issued on the first day of each month until completed.

Keats.—The Poetical Works of John Keats. 1 vol. 12mo. cloth.

—— The same, gilt extra.

“They are flushed all over with the rich lights of fancy; and so colored and bestrewn with the flowers of poetry that, even while perplexed and bewildered in their labyrinths, it is impossible to resist the intoxication of their sweetness, or to shut our hearts to the enchantment they so lavishingly present.”—_Francis Jeffrey._

Kinglake.—Eöthen; or, Traces of Travel brought from the East. 12mo. green cloth. 50 cts.

“Eöthen is a book with which everybody, fond of eloquent prose and racy description, should be well acquainted.”—_U. S. Gazette._

Klipstein’s Anglo-Saxon Course of Study. In uniform 12mo. volumes, as follows:

I.

Klipstein.—A Grammar of the Anglo-Saxon Language. By Louis F. Klipstein, AA.LL.M. and PH.D., of the University of Giessen. 12mo. cloth, $1 25.

II.

Klipstein.—Analecta Anglo-Saxonica, with an Introductory Ethnographical Essay, Copious Notes, Critical and Explanatory, and a Glossary in which are shown the Indo-Germanic and other Affinities of the Language. By Louis F. Klipstein, AA.LL.M. and PH.D., of the University of Giessen.

III.

Klipstein.—Natale Sancti Gregorii Papæ.—Ælfric’s Homily on the Birth-day of St. Gregory, and Collateral Extracts from King Alfred’s Version of Bede’s Ecclesiastical

History and the Saxon Chronicle, with a full Rendering into English, Notes Critical and Explanatory, and an Index of Words. By Louis F. Klipstein, AA.LL.M. and PH.D., of the University of Giessen.

IV.

Klipstein.—Extracts from the Anglo-Saxon Gospels, a Portion of the Anglo-Saxon Paraphrase of the Book of Psalms, and other Selections of a Sacred Order in the same Language, with a Translation into English, and Notes Critical and Explanatory. By Louis F. Klipstein, AA.LL.M. and PH.D., of the University of Giessen.

V.

Klipstein.—Tha Halgan Godspel on Englisc—the Anglo-Saxon Version of the Holy Gospels. Edited by Benjamin Thorpe, F.S.A. _Reprinted by the same. Now ready._ 12mo. cloth, $1 25.

Klipstein.—Study of Modern Languages.—Part First; French, Italian, Spanish, Portuguese, German, and English. By L. F. Klipstein, AA.LL.M. and PH.D. One vol. Imperial 8vo. Cloth, $1; paper 75 cents.

Lamb.—Essays of Elia. By Charles Lamb. 1 vol. 12mo., cloth. $1.

—— The same, gilt extra, $1 25.

“Shakspeare himself might have read them, and Hamlet have quoted them: for truly was our excellent friend of the genuine line of Yorick.”—_Leigh Hunt’s London Journal._

Lamb.—Specimens of the English Dramatic Poets. By Charles Lamb. 1 vol. 12mo., green cloth, $1 13.

—— The same, gilt extra, $1 50.

“Nowhere are the resources of the English tongue in power, in sweetness, terror, pathos; in description and dialogue, so well displayed.”—_Broadway Journal._

Mahan.—On Advanced Guards, Outposts, and Military Duty. By D. H. Mahan, M.A. 18mo. cloth, 75 cents

Mahan’s Course of Civil Engineering. Third edition, 8vo. Illustrated. $3 50.

Milton.—The Prose Works of John Milton. Edited by Rev. Rufus Wilmott Griswold. 2 vols. 8vo., cloth, $4.

Modern Painters. By a Graduate of Oxford. 12mo. cloth, $1 25.

—— The same. Second vol. 12mo.

Montagu.—Selections from the Works of Taylor, Latimer, Hall, Milton, Barrow, Lowth, Brown, Fuller, and Bacon. By Basil Montagu. 1 vol. 12mo., green cloth, 50 cents; cloth gilt, $1.

“This volume contains choice extracts from some of the noblest of the old English writers.”—_Cincinnati Atlas._

Nordheimer.—A Critical Grammar of the Hebrew Language. By Isaac Nordheimer, Phil. Doctor. 8vo. cloth, $3 50.

Oriental Life Illustrated. Being a new edition of Eöthen, or Traces of Travel in the East. With fine Illustrations on Steel.

Parsons.—The Rose; its History, Poetry, and Culture. By S. B. Parsons. With colored Plates. Royal 8vo. cloth, $1 50.

Patrick, Lowth, Arnold, and Whitby.—Commentary on the Bible, by Bishops Patrick, Lowth, Arnold, Whitby, and Lowman. 4 vols. imperial 8vo. cloth, $16.

Peacock.—Headlong Hall and Nightmare Abbey. 1 vol. 12mo., green cloth, 50 cents.

“This is a witty and amusing book.”—_Tribune._

Poe.—Eureka, A Prose Poem: Or the Physical and Metaphysical Universe. By Edgar A. Poe, Esq. Handsomely printed. 12mo. cloth, 75 cents.

Pearls of American Poetry. Second edition, superbly illuminated in the manner of the ancient missals by T. W. Gwilt Mapleson, Esq. Printed in gold and colors on Bristol board. Elegantly and strongly bound in full Morocco, Antique style. One volume quarto, $12.

⁂ Of this splendid and costly work, a small number were issued for this season, but it was not ready for actual publication. It is now completed in a superior style, and is the most splendid book of the time.

“On beautiful vellum paper, are printed in colored characters and with every variety of type, some of the choicest brief poems of American writers—Bryant, Longfellow, C. F. Hoffman, and others. Each initial letter is a picture, and each page is illuminated as exquisitely as any of the choicest of antique illuminated volumes—and all from original designs. The conception of these works of art, as they richly deserve to be called, the drawing, painting, gilding, are of the highest order. The binding is in keeping with the rest—that of the olden day—solid, rich, and tasteful. Altogether this is a volume of great attraction for the rare beauty of its adorning and the discrimination of its selections.”—_Courier._

Princeton Theological Essays. First Series. Royal 8vo. cloth, $2 50.

Princeton Theological Essays. Second Series. Royal 8vo. cloth, $2 50.

St. John.—The Three Days of February, 1848: with Portrait of Lamartine. 18mo. cloth, 63 cts.

Tappan.—Elementary Logic. By Prof. H. P. Tappan. 1 vol. 12mo. cloth. $1.

Tasso.—Godfrey of Bulloigne; or, the Recovery of Jerusalem: done into English Heroical Verse. From the Italian of Tasso, by Edward Fairfax. Introductory Essay, by Leigh Hunt, and the Lives of Tasso and Fairfax, by Charles Knight. 1 vol. 12mo. $1

25.

“The completest translation, and nearest like its original of any we have seen.”—_Leigh Hunt._

Taylor.—Views a-Foot; or, Europe seen with Knapsack and Staff. By J. Bayard Taylor. New edition, with an additional Chapter, &c., and a Sketch of the Author in Pedestrian Costume, from a Drawing by T. Buchanan Read. 12mo. cloth. Nearly ready. $1 25.

“Besides being one of the most entertaining books of travel we ever read, it is written under circumstances of the most interesting; although at a first glance, seemingly the most unfavorable.”—_Boston Atlas._

Thackeray.—Journey from Cornhill to Cairo. By Michael Angelo Titmarsh. 1 vol. 12mo. green cloth. 50 cts.

“It is wonderful what a description of people and things, what numerous pictures, what innumerable remarks and allusions it contains.”—_Douglas Jerrold’s Mag._

Torrey and Gray.—Flora of North America. By Professors Torrey and Gray. 1 vol. 8vo. cloth, $6. Parts 1 and 6, each $1 50; Part 7, $1.

Tschudi.—Travels in Peru. By Dr. J. J. Von Tschudi. 1 vol. 12mo. cloth, 87 cents.

“The book contains a great deal of curious information, and will be found useful as a book of reference by all who are interested in the commerce, natural history, and general statistics of Peru.”—_Blackwood’s Magazine._

Tupper.—Proverbial Philosophy. By Martin Farquhar Tupper. 12mo. green cloth.

—— The same, gilt extra.

—— The same, morocco extra.

Walton.—The Lives of Donne, Walton, Hooker, Herbert, and Sanderson. By Izaak Walton. New edition, 1 vol. 12mo. green cloth, 75 cents.

“The Lives are the most delightful kind of reading. Walton possesses an inimitable simplicity and vivacity of style.”—_Mrs. C. M. Kirkland._

Warburton.—The Crescent and the Cross; or, the Romance and Reality of Eastern Travel. By Elliot Warburton. 1 vol. 12mo. green cloth, $1.

“This delightful work is, from first to last, a splendid panorama of Eastern scenery, in the full blaze of its magnificence.”—_London Morning Post._

A valuable Work for Libraries.

Now Ready. 8vo. $1 in paper, or $1 25 half bound.

An Alphabetical Index to Subjects treated in the Reviews, and other Periodicals, to which no Indexes have been Published.

⁂ This volume comprises an Index to all articles in 560 volumes of the most important periodical works.

POPULAR VOLUMES FOR PRESENTATION,

Elegantly bound in extra cloth, gilt edges.

Chaucer and Spenser $1 50 Fairfax’s Tasso’s Jerusalem Delivered 1 50 Fouqué’s Undine, and Sintram 1 00 Goldsmith’s Vicar of Wakefield, with plates 1 00 Hervey’s Book of Christmas 1 00 Howitt’s (Mary) Ballads and Poems 1 00 Hood’s Prose and Verse 1 25 Hunt’s Imagination and Fancy 1 00 —— Italian Poets 1 75 Keats’s Poems 1 25 Lamb’s Dramatic Specimens 1 50 Lamb’s Essays of Elia 1 25 THE SYBIL; or, New Oracles from the Poets. By Mrs. Gilman. An elegant and attractive book 1 50

ILLUSTRATED JUVENILES.

FACTS AND FANCIES. By Miss Sedgwick. 16mo. with cuts, cloth .50

GLIMPSES OF THE WONDERFUL.—An entertaining Account of Curiosities of Nature and Art. First, Second, and Third Series, with numerous fine Illustrations engraved in London. Square 16mo. cloth, each .75

HOME TREASURY, THE; Comprising new versions of Cinderella, Beauty and the Beast, Grumble and Cheery, The Eagle’s Verdict, The Sleeping Beauty. Revised and illustrated. Small 4to .50

MORALS AND MANNERS; or, Hints for our Young People. By Miss Sedgwick. 16mo .25

YOUNG NATURALIST’S RAMBLES—through many Lands, with an account of the principal Animals and Birds of the Old and New Continents. Cloth .50

George P. Putnam

(OF THE LATE FIRM OF WILEY AND PUTNAM),

Has taken the new and commodious premises,

155 BROADWAY, NEW YORK

(_Next building to that of the late Firm_),

And continues the business of

PUBLISHING,

AND THE

IMPORTATION OF FOREIGN BOOKS,

AS ABOVE, AND AT

PUTNAM’S AMERICAN LITERARY AGENCY, 142 _Strand_,

LONDON

[ESTABLISHED IN 1838].

* * * * *

Arrangements have been made to secure at the London Agency the services of an experienced and competent Bibliographer, so that the business of executing SPECIAL ORDERS FOR THE TRADE AND PUBLIC INSTITUTIONS may be thoroughly regulated, and all parties giving such orders, may be fully satisfied both with regard to _expedition_ and _economy_.

The interests of Public Institutions, and those ordering _Books in quantities_ will receive special attention, while it is also intended that any one ordering _a single volume_ from Europe, may receive it promptly (if procurable), without disappointment or unnecessary expense.

Mr. PUTNAM believes that his _twelve years' experience_ abroad in purchasing Books for the American market, will be of service to those who may favor him with orders.

⁂ Correspondence established with PARIS, ROME, LEIPSIC, BRUSSELS, and all the principal cities on the Continent. All American Publications on the best terms, by the quantity or singly.

N. B.—CATALOGUES of extensive collections of _Foreign and American Books_, on all subjects may be had on application.